Hanwha Investment&Securities

October 26, 2026

SECTOR INITIATION

Defense Sector (Positive)

Beyond Infinity

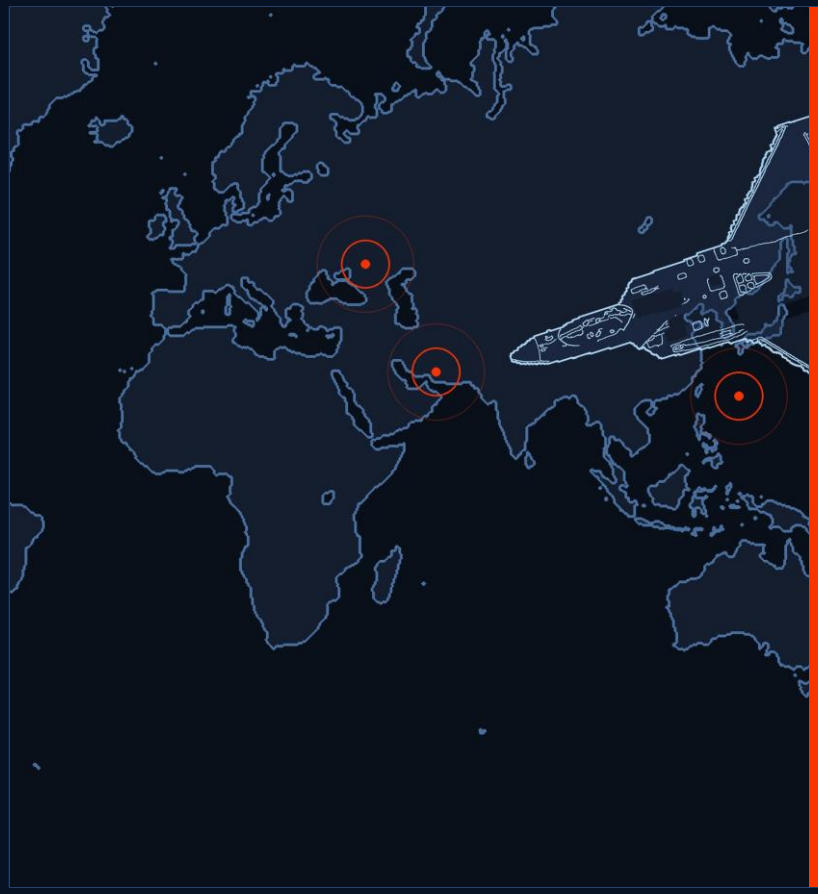
Defense Chae Un-Sam unsam1@hanwha.com 3772-8419

Hanwha Aerospace | Hanwha Systems | LIG D&A | KAI | Hyundai Rotem

Summary

Summary

We initiate coverage of the defense industry with a Positive opinion and present Hanwha Aerospace as our Top Pick. We expect global weapons demand to remain on an uptrend irrespective of a ceasefire, as the market shifts from emergency force replenishment to structural rearmament, while supply expansion is unlikely to cause oversupply. The order backlog of more than KRW 100 trillion secured by Korean defense companies and the decline in valuations make the current level an attractive medium- to long-term buying opportunity.



I

Executive Summary

Defense Sector (Positive)

Defense Sector: Beyond Infinity

Executive Summary (1)

- Defense sector share prices tend to move with visibility of new orders. New orders are an important investment point in that they increase visibility on future earnings. Because the defense industry has a high share of government-to-government transactions and is affected by variables such as defense budgets and government policy, it is regrettably difficult to anticipate the timing of individual order contracts. In this report, we analyzed global weapons demand and supply in order to gauge the medium- to long-term earnings growth potential of the defense sector.

- The biggest concern for the defense sector is likely to be a decline in weapons demand following an end of war/ceasefire. However, historical data show that military spending tended to increase after the end of wars in both the belligerent countries and neighboring countries. The key variables determining military spending after the end of a war are security threats and fiscal capacity. Even if the Russia-Ukraine and US-Iran wars end, we expect military spending in the US, Russia and neighboring countries to increase, while Ukraine and Iran are likely to see declines due to insufficient fiscal capacity. Irrespective of a ceasefire/end of war, the uptrend in global military spending is expected to continue.

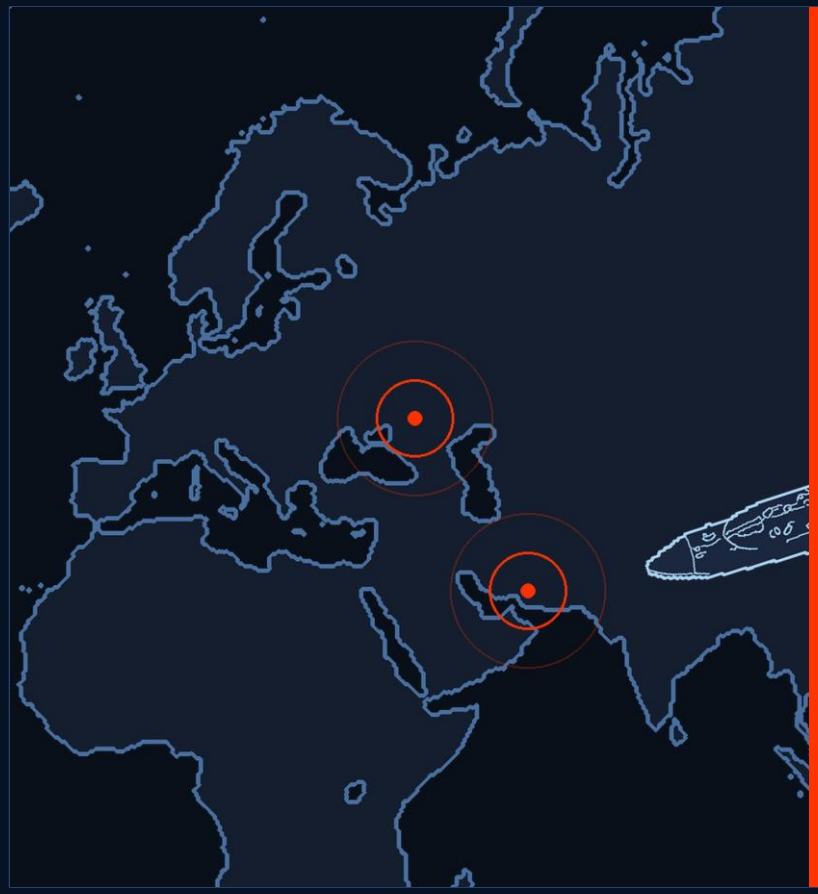
- Although interest in the defense sector has risen since the Russia-Ukraine war, total global arms imports and exports have not increased as much as expected. The net increase in demand was concentrated in Ukraine and Poland, and the military capabilities of most countries have not changed significantly from before the war. We judge that global weapons demand is currently in a phase of transition from urgent force replenishment to structural rearmament. There are three factors supporting the demand increase. First, the prolongation of geopolitical competition. Second, the need to replace accumulated aging forces. Third, demand for new weapons procurement driven by defense budget expansion.

- Defense companies are focusing on expanding production capacity. We note that European defense companies are actively expanding. The key is whether intra-European demand can sufficiently absorb the increasing supply. Based on our analysis, we judge that concerns over weapons oversupply are limited.

Executive Summary (2)

- Looking at self-propelled howitzers divided into tracked and wheeled types, production capacity expansion is limited for tracked types, while supply increases are expected for wheeled types. Because potential demand for wheeled types is abundant, we judge the likelihood of oversupply to be low. For multiple rocket launchers, the likelihood of capacity expansion by companies is high, but long delivery lead times are expected to persist, and we judge there is ample room for Chunmoo to expand its market share. In air defense, US and European companies are expanding production capacity, but the supply shortage of air defense systems capable of intercepting ballistic missiles is expected to persist. We expect ample export opportunities for Cheongung-II, Cheongung-III and L-SAM. In fighters, Western companies are pushing for increased production, but the pace of production expansion is falling short of targets. We judge that new order wins for the KF-21 are ultimately a matter of time. In tanks, competitors are preparing for next-generation tank development and production transitions, so aggressive production increases of existing tanks are not appearing. With the competitive structure centered on the US, Germany and Korea maintained, the number of K2 tank operators is expected to expand.

- We initiate coverage of the defense industry with a Positive opinion and present Hanwha Aerospace as our Top Pick. The weapons demand growth cycle is accompanied by force buildups in most weapon systems, not just specific weapons. Hanwha Aerospace has secured a land, naval and air business portfolio, and structural earnings growth is expected over the medium to long term in the global force buildup cycle.



II

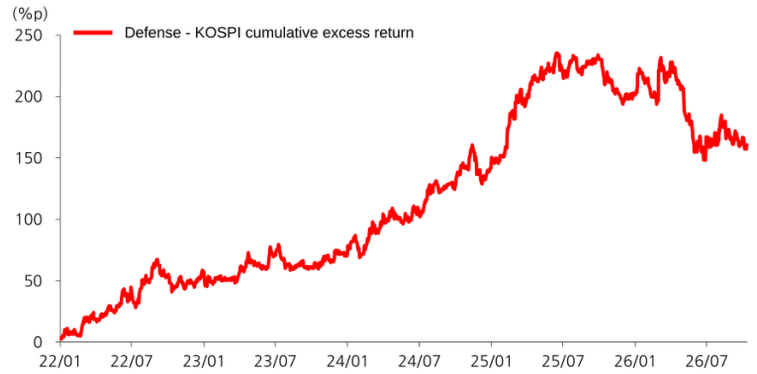
Key Charts

Figures 1–12 · Tables 1–3

Defense Sector: Beyond Infinity

Key Charts (1)

[Figure 1] Korean defense stocks underperforming the KOSPI index since 2H25



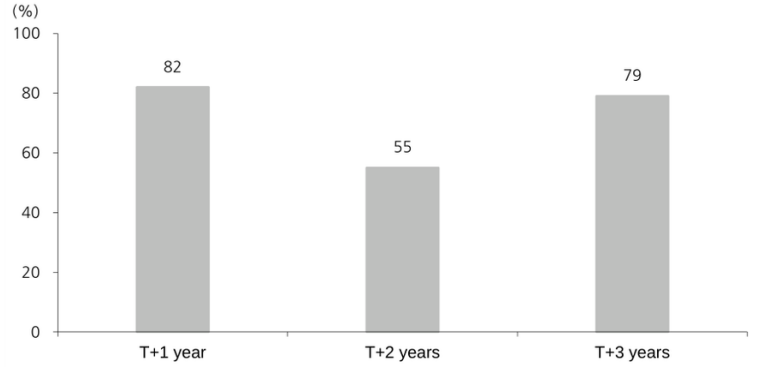
Source: Hanwha Research Center

[Figure 2] The biggest concern in defense industry investment is the decline in weapons demand following an end of war/ceasefire



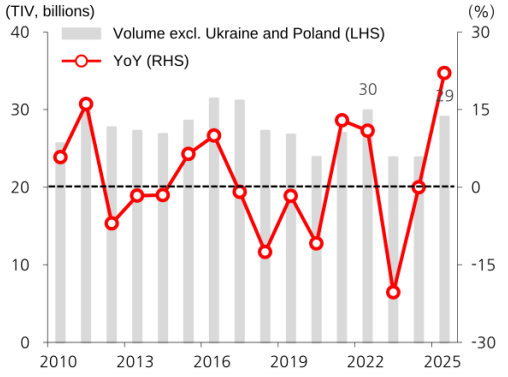
Source: Hanwha Research Center

[Figure 3] Share of countries with increased military spending after past ends of war/ceasefire. Military spending increased in many cases



Source: SIPRI, Hanwha Research Center

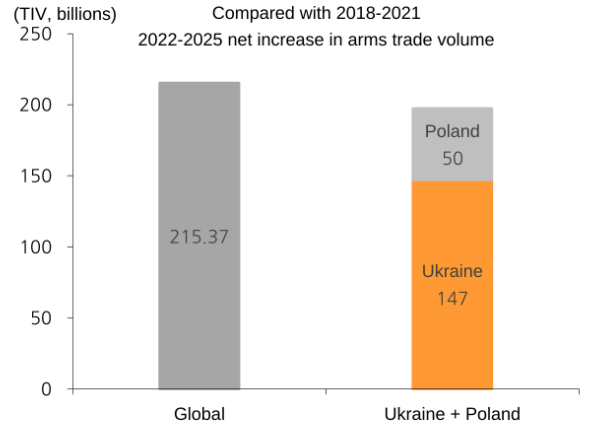
[Figure 4] Limited increase in global arms trade volume since the Russia-Ukraine war



Source: SIPRI, Hanwha Investment & Securities

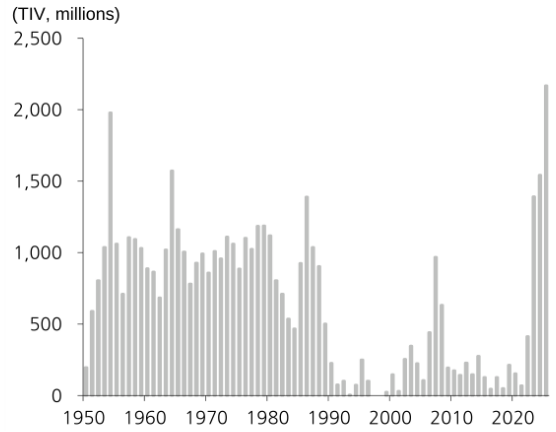
Key Charts (2)

[Figure 5] Most of the net increase in trade volume is attributable to Ukraine and Poland



Source: SIPRI, Hanwha Investment & Securities

[Figure 6] Poland's arms import volume. A representative case of structural force buildup



Source: SIPRI, Hanwha Investment & Securities

Key Charts (3)

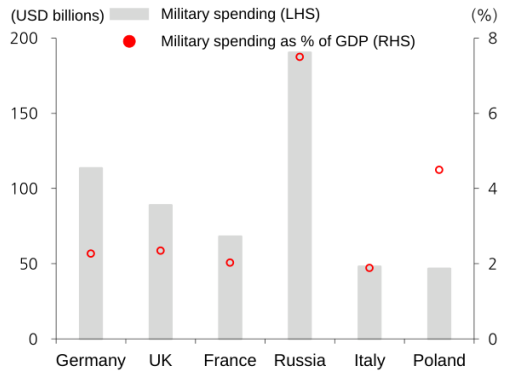
[Table 1] Poland: Force Levels and Modernization Progressing Simultaneously

Category	End-2021 Force Level (Share of New Units Introduced Within Last 10 Yrs)	End-2025 Force Level (Share of New Units Introduced Within Last 10 Yrs)	Change (%)
MBT	797 (14.9%)	897 (46.0%)	12.5%
IFV+APC	1,611 (3.2%)	1,516 (14.0%)	-5.9%
Self-Propelled Howitzers	410 (17.6%)	558 (43.2%)	36.1%
Multiple Launch Rocket Systems	179 (0.0%)	299 (56.2%)	67.0%
Fighter/Attack Aircraft	94 (0.0%)	77 (20.8%)	-18.1%
Attack Helicopters	28 (0.0%)	24 (33.3%)	-14.3%
Submarines	1 (0.0%)	1 (0.0%)	-
Surface Combatants	2 (0.0%)	2 (0.0%)	-

Source: IISS, Hanwha Investment & Securities

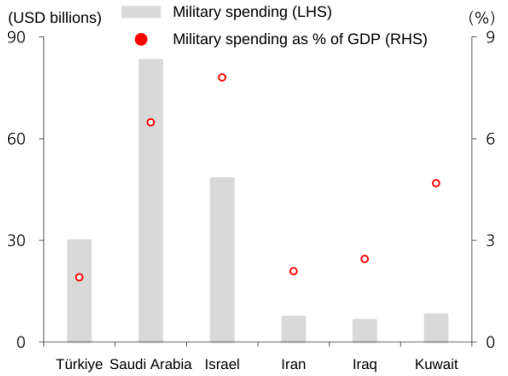
Key Charts (4)

[Figure 7] Military spending and military spending as a share of GDP in major European countries



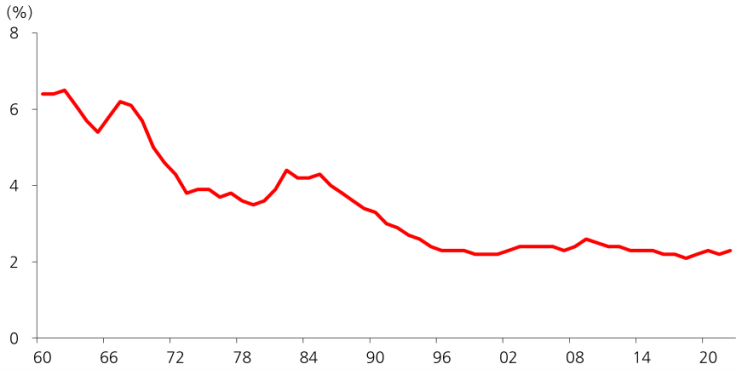
Source: SIPRI, Hanwha Investment & Securities

[Figure 8] Military spending and military spending as a share of GDP in major Middle Eastern countries



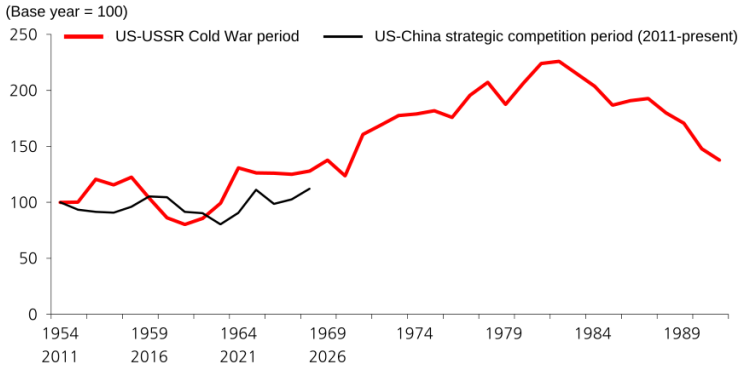
Source: SIPRI, Hanwha Investment & Securities

[Figure 9] Trend in global military spending as a share of GDP



Source: SIPRI, Hanwha Research Center

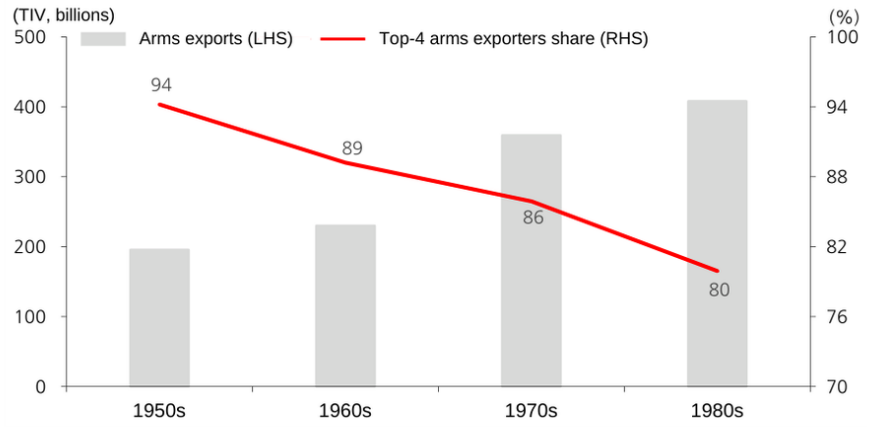
[Figure 10] Global arms trade volume in the US-Soviet Cold War and the US-China strategic competition period. Not a cycle that ends in a short period



Source: SIPRI, Hanwha Investment & Securities

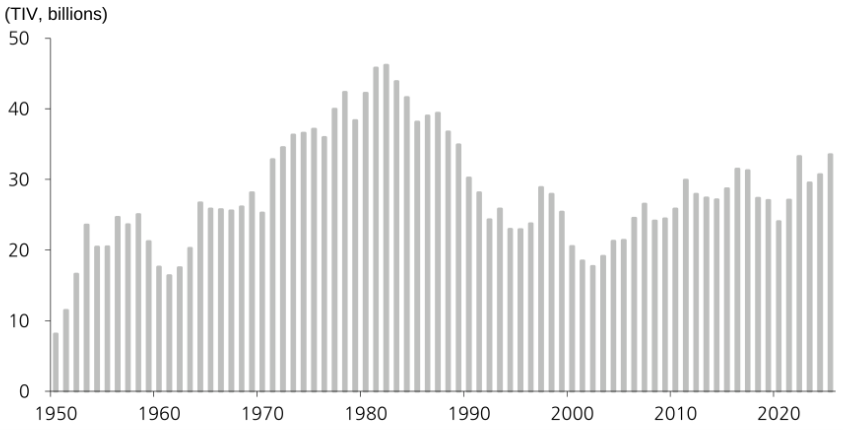
Key Charts (5)

[Figure 11] Share of the top four arms exporters in global arms exports during the US-Soviet Cold War. Supplier oligopoly position persisted for a long period



Source: SIPRI, Hanwha Investment & Securities

[Figure 12] Global arms trade volume trend. Replacement pressure is ample given typical operating lives of 30-40 years



Source: SIPRI, Hanwha Investment & Securities

Key Charts (6)

[Table 2] Key Competing Weapon Systems of Korea and Europe

Category	Korea	European Competing Products	Key Companies	Capacity Direction
Self-Propelled Howitzers	K9	CAESAR	KNDS France	Expanding
		RCH 155	KNDS Deutschland	Expanding
		Archer	BAE Systems	Expanding
		PzH 2000	KNDS Deutschland	Unknown
Multiple Launch Rocket Systems	Chunmoo	PULS	Elbit Systems	Expanding
		EuroPULS	KNDS / Elbit	Expansion planned
Tanks	K2	Leopard 2	KNDS / Rheinmetall	Expanding
Air Defense	Cheongung II	SAMP/T NG	MBDA	Expanding
		IRIS-T SLM	Diehl Defence	Expanding
Fighters	KF-21	Rafale	Dassault Aviation	Expanding
		Eurofighter	Airbus / BAE / Leonardo	Expanding
		Gripen	Saab	Expanding

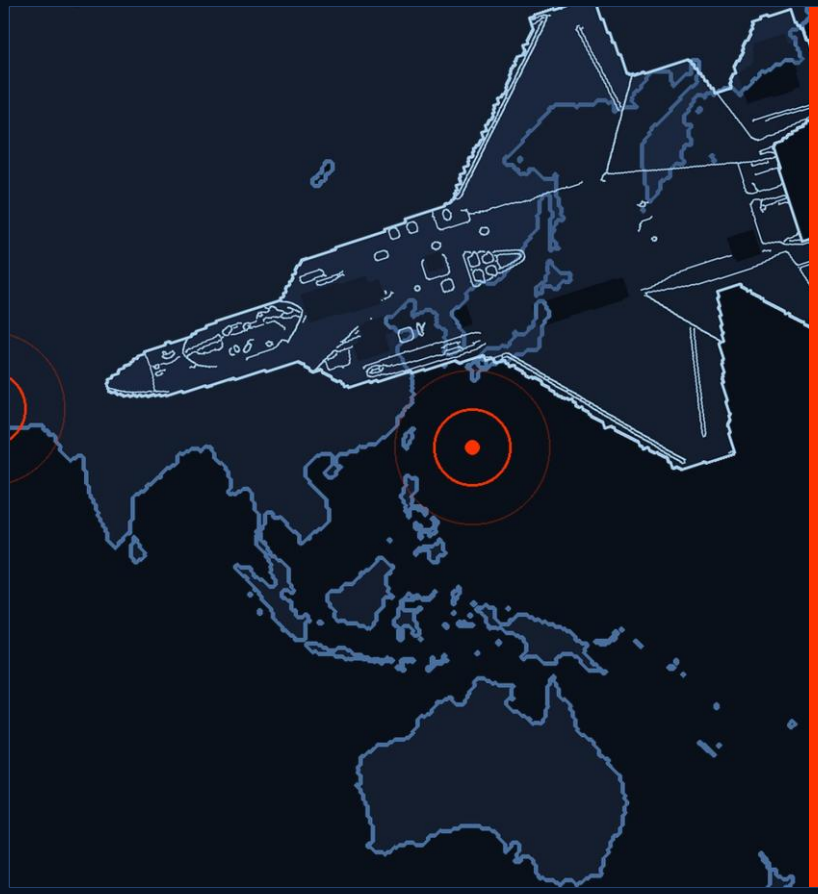
Source: Companies, Hanwha Research Center

Key Charts (7)

[Table 3] Production Capacity and Expansion Status of Weapon Systems Competing with Korean Defense

Category	Weapon System	Annual CAPA Before the Russia-Ukraine War	Current Annual CAPA	Target CAPA	Remarks
Self-Propelled Howitzers	M109A7	60 guns (est.)	60 guns (est.)	Not disclosed	Low likelihood of expansion
	PzH2000	5 guns (est.)	10-15 guns (est.)	Not disclosed	Low likelihood of expansion
	RCH155	0 guns (before mass production)	25-35 guns (est.)	Expanding	Figures undisclosed
	CAESAR	24 guns	120 guns	144 guns	Low likelihood of further expansion
	ARCHER	10-15 guns (est.)	15 guns (est.)	Expanding	Low likelihood of further expansion
Multiple Launch Rocket Systems	HIMARS	48 units	96 units	Not disclosed	High likelihood of expansion
	PULS	Unknown	Unknown	Expanding (est.)	High likelihood of further expansion
Air Defense	Patriot	Unknown	12 batteries	Expanding	Figures undisclosed
	SAMP/T	1-3 batteries (est.)	4-6 batteries (est.)	Expanding	Figures undisclosed
	IRIS-T	Unknown	10 batteries	16 batteries	Long-term target of increasing production to 30 batteries
	NASAMS	1-3 batteries (est.)	3-5 batteries (est.)	Not disclosed	High likelihood of expansion
Fighters	F-16	Mass production started after plant relocation	20-30 units (est.)	48 units	Limited likelihood of further expansion
	F-15	10-15 units (est.)	12-17 units (est.)	24 units	Possibility of further expansion exists
	Rafale	12-20 units (est.)	20-30 units (est.)	48 units	Possibility of further expansion exists
	Eurofighter	10-12 units (est.)	12-14 units (est.)	20-30 units	Limited likelihood of further expansion
	Gripen	6-10 units (est.)	15 units (est.)	20-30 units	Limited likelihood of further expansion
Tanks	Leopard 2	Unknown	20-30 units (est.)	40-50 units (est.)	Low likelihood of further expansion
	M1A2	180 units (est.)	180 units (est.)	Expanding	Investing in facility modernization (est.)

Source: Companies, press reports, Hanwha Research Center



III

Investment Strategy and Valuation

Tables 4–5

Defense Sector: Beyond Infinity

Investment Strategy and Valuation (1)

- **Economy, international affairs and earnings: defense becomes more favorable over time** We initiate coverage of the defense industry with a Positive opinion and present Hanwha Aerospace as our Top Pick. We expect an increasingly favorable investment environment for the defense sector over time. The key point is that three factors - the economy, international affairs and earnings - are highly likely to act favorably on the defense sector. The recent rise in global government bond yields and inflation burden, and the resulting possibility that the US Federal Reserve maintains a high-interest-rate stance, are factors that raise downside risk rather than upside potential for the economy going forward. Typically, when economic uncertainty expands, investors pay attention to sectors with high visibility on earnings growth. Weapons demand in the defense sector is affected more by geopolitical conflicts and national security policy than by economic fluctuations. The sector stands out in that it has secured visibility on earnings growth, based on its order backlog, regardless of the economic cycle. We judge that the relative appeal of defense will stand out as economic concerns grow.

- **Self-reliant defense is the trend of the times** The current international situation, which is converging on self-reliant defense, is also unlikely to change in the short term. More than two years remain until the end of President Trump's term. We believe the stance of self-reliant defense is likely to continue even after the Trump administration. Within the US, public opinion supporting active engagement in international affairs is weaker than in the past, and the perception that domestic issues should be prioritized is spreading. These changes suggest that US foreign security policy may not be determined solely by the inclinations of a particular administration. Considering China's steep military buildup, we judge that the US will need to concentrate considerable military capabilities on containing China going forward. Even if the next administration takes office, it is unlikely to be easy for the US to provide its allies with the same level of security as in the past. We judge that the expansion of global self-reliant defense is not the policy of a particular administration, but a medium- to long-term trend that will continue as long as US-China competition persists.

- **Order backlog and earnings to be re-rated** The order backlog of more than KRW 100 trillion secured by Korean defense companies is expected to be re-rated over time. Major domestic defense companies already have a foundation to continue earnings growth through at least 2028 on the order backlog secured to date alone. In 2026, share prices of major global defense companies, including Korean defense, were relatively neglected due to the steep earnings growth of AI-related companies and the concentration of investor interest. This gap is expected to narrow gradually. Based on current market consensus, the 2027 earnings growth rate of major domestic tech companies remains high, but the difference from the defense sector is expected to narrow significantly. In 2028, the earnings growth rate of the Korean defense sector may even exceed that of major tech companies. If new overseas orders are also secured, visibility on earnings growth could extend into the 2030s.

Investment Strategy and Valuation (2)

▪ **An attractive buying opportunity from a medium- to long-term perspective** We judge that the valuation of the defense sector does not sufficiently reflect its medium- to long-term growth potential. The valuation burden has eased as share prices were relatively neglected in 2026. The average 2027E PER based on expected net profit of the five major domestic defense companies is currently 21.3x, which is 37.9% and 54.5% below the highest PER of 2025 and 2026, respectively. The defense sector still simultaneously secures relative defensiveness against an economic slowdown, structural demand from the expansion of self-reliant defense, and high earnings visibility. We judge that the relative advantage of the Korean defense sector will widen over time in the three aspects of the economy, international affairs and earnings, and we expect the current valuation to be an attractive buying opportunity from a medium- to long-term perspective.

[Table 4] Target Prices and Valuation of Five Domestic Defense Companies

Item	Hanwha Aerospace	Hanwha Systems	LIG D&A	Korea Aerospace Industries	Hyundai Rotem
Investment Rating	BUY (Initiated)	BUY (Initiated)	BUY (Initiated)	BUY (Initiated)	BUY (Initiated)
Target Price (KRW)	1,300,000	100,000	670,000	140,000	110,000
Current Price (KRW, 10/23)	792,000	66,300	475,500	102,200	82,700
Market Cap (KRW bn, 10/23)	40,838	12,525	10,461	9,962	9,026
Upside (%)	▲64.1	▲50.8	▲40.9	▲37.0	▲33.0
2026E P/E	17.2	100.2	25.9	39.1	11.3
2026E P/B	3.1	2.6	5.9	4.8	2.4
2026E P/S	1.3	2.9	2.1	1.9	1.4
2026E ROE	20.8	2.5	25.0	13.0	23.2
2027E P/E	12.5	36.7	18.3	20.5	12.7
2027E P/B	2.6	2.4	4.7	4.0	2.0
2027E P/S	1.2	2.4	1.8	1.5	1.4
2027E ROE	22.4	6.7	28.3	21.2	17.2

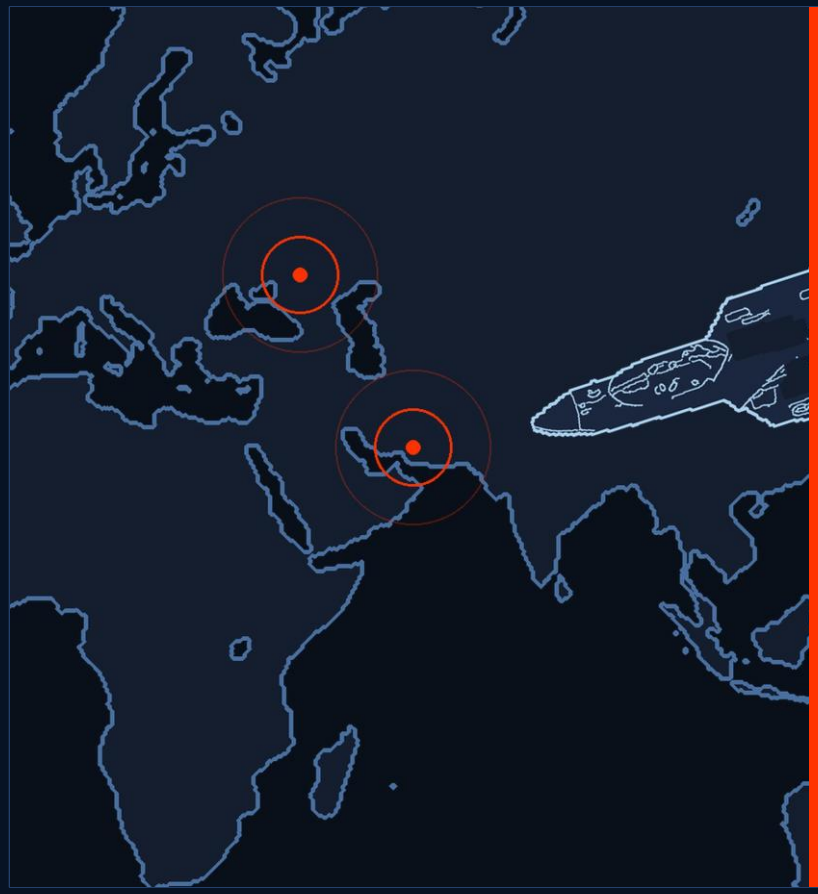
Source: Hanwha Research Center

Investment Strategy and Valuation (3)

[Table 5] Global Defense Companies Valuation Table

Country	Company	Market Cap (USD bn)	Relative Return 1M	Relative Return 6M	Relative Return 1Y	P/E 26	P/E 27	P/E 28	P/B 26	P/B 27	P/B 28	P/S 26	P/S 27	P/S 28	ROE 26	ROE 27	ROE 28
Korea	Hanwha Aerospace	31.3	-4.6	-31.4	-66.1	18.0	14.1	12.6	3.4	2.8	2.4	1.3	1.2	1.1	21.0	22.0	21.5
	Hanwha Systems	10.5	14.0	-33.6	-12.7	125.4	37.9	28.2	2.8	2.5	2.3	3.1	2.9	2.5	2.5	7.0	8.6
	LIG D&A	8.0	-12.7	-39.6	-39.2	29.5	20.8	15.3	6.1	4.9	3.9	2.1	1.8	1.5	22.9	26.0	28.3
	Korea Aerospace Industries	8.0	1.3	-30.8	-39.5	47.4	25.2	18.8	5.3	4.5	3.7	2.0	1.6	1.4	11.6	19.2	21.7
	Hyundai Rotem	7.1	-7.6	-52.1	-109.6	12.0	10.7	8.3	2.5	2.1	1.7	1.5	1.3	1.1	23.2	21.2	22.3
US	RTX	258.6	19.4	17.4	-38.6	25.5	23.9	21.7	3.7	3.6	3.4	2.7	2.5	2.3	14.8	14.9	15.6
	Boeing	152.0	16.3	-5.3	-57.5	-	53.5	24.4	24.1	15.2	9.2	1.6	1.4	1.2	-8.1	35.5	44.8
	Lockheed Martin	127.3	24.7	10.8	-33.1	18.2	16.9	15.6	11.8	10.0	8.7	1.6	1.5	1.4	81.4	65.6	59.6
	General Dynamics	93.3	19.5	19.0	-44.4	20.3	18.5	17.0	3.3	3.1	3.0	1.7	1.6	1.5	17.3	17.5	17.7
	Northrop Grumman	76.6	24.9	3.0	-55.9	18.4	18.0	16.6	4.0	3.6	3.3	1.7	1.6	1.5	23.4	21.1	20.8
Europe	Safran	147.5	13.6	26.7	-43.1	30.0	24.1	21.2	7.9	6.8	6.0	3.6	3.3	3.0	27.1	28.6	28.3
	BAE System	77.7	16.1	3.0	-43.0	22.8	20.1	17.8	4.6	4.2	3.9	1.7	1.6	1.5	19.9	21.1	22.0
	Rheinmetall	53.6	22.0	-16.8	-88.0	28.0	19.1	13.9	7.5	5.9	4.5	3.4	2.5	1.9	28.4	33.4	36.2
	Hensoldt	10.9	28.1	19.8	-59.7	46.0	35.1	27.0	8.5	7.4	6.2	3.4	2.9	2.5	17.2	20.4	23.8
	Leonardo	31.3	15.7	-0.6	-50.8	19.8	16.2	13.7	2.6	2.4	2.1	1.3	1.1	1.0	13.3	14.5	15.1
	Saab AB	35.2	23.5	25.0	-16.3	42.6	33.2	26.9	6.8	5.8	5.0	3.6	2.9	2.5	17.2	18.6	19.9
	Thales	53.2	16.5	6.6	-57.5	21.5	18.6	16.4	5.4	4.7	4.1	2.0	1.8	1.7	24.7	25.2	24.7
	Dassault Avia	25.9	23.1	9.5	-40.1	19.6	16.1	13.6	3.3	2.9	2.6	2.6	2.4	2.0	16.7	18.2	19.4

Note: As of October 21, 2026, based on Bloomberg
Source: Bloomberg, Hanwha Research Center



IV

The Unending Weapons Demand Cycle

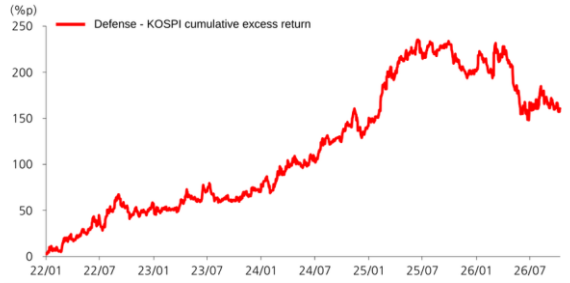
Weapons Demand Continuing After the End of the War · Force Buildup Slower Than Expected · From Urgent Force Replenishment to the Start of Structural Force Rebuilding

Defense Sector: Beyond Infinity

Gap Between Concerns Over an End of War and Past Military Spending Cases

- **Sharp fall in defense stocks driven by concerns over weapons demand** Recently, the price-to-earnings ratios (PER) of defense companies in the US and Europe, including Korea, have declined significantly. The 2027E-earnings-based PERs of Korea's Hanwha Aerospace and Germany's Rheinmetall, representatives of the global land defense sector, have fallen to about 13x and 19x, respectively. Given that the PER of a typical manufacturing company is about 10x, it is clear that the defense sector still commands a premium. Considering that the defense sector is likely to continue earnings growth into the 2030s on a long-secured order backlog, we judge that current valuations are undervalued.
- One of the biggest concerns in defense investment at present is whether the current uptrend in weapons demand can be maintained even after an end of war or ceasefire. This is because, when a war ends, geopolitical uncertainty eases and the uptrend in global military spending could slow. This leads to concern that the earnings growth rates of Korean defense companies could peak before 2030. There is also the possibility that defense budgets will be deprioritized amid postwar reconstruction and expanded welfare and social infrastructure investment. This could lead to delays in replacing aging forces and reductions in force buildup plans, becoming a factor that slows the pace of ongoing global rearmament.
- The expectation that military spending will decline after the end of a war is quite reasonable. When a war ends, the cost of conducting large-scale military operations falls and weapons demand may also decline. However, looking at past cases of the end of wars, the results differ from expectations. There were more cases where military spending actually increased not only immediately after a war or ceasefire but also 3-5 years later.

[Figure 13] Korean defense underperforming the KOSPI



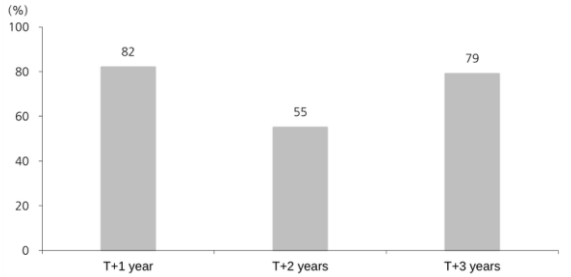
Source: Hanwha Research Center

[Figure 14] Key concerns of defense investors



Source: SIPRI, Hanwha Research Center

[Figure 15] Share of countries with increased military spending at T+1, T+2 and T+5 years after past ends of war/ceasefire



Source: SIPRI, Hanwha Research Center

In Many Past Cases, Military Spending Increased YoY Even Immediately After the End of War/Ceasefire (1)

- **In past cases, military spending increased in the majority after the war ended** Looking at past cases of the end of wars/ceasefires, military spending did not decrease in many countries. Of 67 interstate wars that have occurred since 1950, 22 involved cumulative casualties of 1,000 or more. Setting the end of war/ceasefire of those wars as T and looking at changes in military spending of the belligerent countries, at T+1 year, military spending rose YoY in 23 of the 28 countries (82%) for which data exist. At T+2 years the share of countries with rising military spending fell to 16 of 29 countries (55%), but at T+5 years it rose again to 22 of 28 countries (78%). Immediately after the end of the war, military spending continued to rise in most countries, and even though the share of countries with rising military spending temporarily declined thereafter, military spending rose again in about 78% of countries after five years. This suggests that military spending to restore forces consumed in the war, modernize the military and respond to the changed security environment may have continued despite the end of the war. Considering that military spending continued to rise in a considerable number of countries even after the end of the war, it is difficult to conclude that the end of war or ceasefire alone immediately reversed the direction of weapons demand.
- The results are similar when approached from the perspective of changes in military spending as a share of GDP. Comparing the average military spending as a share of GDP for the five years before the end of war/ceasefire with that at T+1, the share rose in 23 of 28 countries (82%). At T+2, 21 of 28 countries (75%), and at T+5, 18 of 25 countries (72%) exceeded the five-year average before the end of the war. Even after the war ended, a considerable number of countries maintained not only the scale of military spending but also military spending as a share of GDP at levels higher than before the war.

In Many Past Cases, Military Spending Increased YoY Even Immediately After the End of War/Ceasefire (2)

[Table 6] Casualties in Major Interstate Wars

No.	Interstate War/Conflict	Period	Estimated Battle-Related Casualties	Annual Average Estimate
1	North Vietnam-South Vietnam	1965~1975	1,461,050	132,823
2	North Korea-South Korea	1950~1953	995,000	248,750
3	Iran-Iraq	1980~1988	644,500	71,611
4	Russia-Ukraine	2022~2025	372,588	93,147
5	Eritrea-Ethiopia	1998~2000	98,192	32,731
6	China-Vietnam	1978~1981	45,255	11,314
7	Iraq-Kuwait	1990~1991	22,848	11,424
8	India-Pakistan	1971	11,223	11,223
9	Chad-Libya	1987	8,500	8,500
10	US/UK/Australia-Iraq	2003	7,927	7,927
11	Israel-Egypt	1973	6,450	6,450
12	Israel-Egypt	1969~1970	5,520	2,760
13	India-Pakistan	1964~1965	5,000	2,500
14	Israel-Syria	1973	4,634	4,634
15	Israel-Egypt	1967	3,337	3,337
16	Iran-Israel	2018~2025	2,863	358
17	China-India	1962	2,105	2,105
18	India-Pakistan	1996~2003	1,965	246
19	Afghanistan-US/UK	2001	1,397	1,397
20	Tunisia-France	1961	1,394	1,394
21	Israel-Jordan	1967	1,349	1,349
22	China-Taiwan	1954	1,000	1,000

Source: PRIO, UCDP, Hanwha Research Center

In Many Past Cases, Military Spending Increased YoY Even Immediately After the End of War/Ceasefire (3)



[Table 7] Changes in Military Spending and Military Spending as a Share of GDP After the End of War/Ceasefire for Belligerents (T+1 Year)

Country	War Name and Period	Military Spending YoY (%)	Military Spending as % of GDP YoY (%p)
South Korea	Korean War (1950~53)	150.8	3.5
Iran	Iran-Iraq War (1980~88)	-26.0	-1.7
Eritrea	Eritrea-Ethiopia War (1998~2000)	-	-10.5
Ethiopia	Eritrea-Ethiopia War (1998~2000)	-36.5	-3.3
Kuwait	Iraq-Kuwait War (1990~91)	-49.3	-85.6
India	India-Pakistan War (1971)	5.4	0.1
Pakistan	India-Pakistan War (1971)	12.1	0.1
Chad	Chad-Libya War (1987)	95.1	1.7
US	US/UK/Australia-Iraq War (2003)	9.0	0.2
UK	US/UK/Australia-Iraq War (2003)	1.2	-0.1
Australia	US/UK/Australia-Iraq War (2003)	4.1	-0.0
Iraq	US/UK/Australia-Iraq War (2003)	-	1.7
Israel	Israel-Egypt War (1973)	2.6	-0.2
Egypt	Israel-Egypt War (1973)	33.1	3.8
Israel	Israel-Egypt War (1969~70)	8.1	-0.6
Egypt	Israel-Egypt War (1969~70)	12.4	1.2

In Many Past Cases, Military Spending Increased YoY Even Immediately After the End of War/Ceasefire (4)

[Table 7] Changes in Military Spending and Military Spending as a Share of GDP After the End of War/Ceasefire for Belligerents (T+1 Year) (cont.)

Country	War Name and Period	Military Spending YoY (%)	Military Spending as % of GDP YoY (%p)
India	India-Pakistan War (1964~65)	-5.8	-0.3
Pakistan	India-Pakistan War (1964~65)	16.6	0.8
Israel	Israel-Syria War (1973)	2.6	-0.2
Syria	Israel-Syria War (1973)	-	-2.4
Israel	Israel-Egypt War (1967)	26.7	2.0
Egypt	Israel-Egypt War (1967)	29.8	11.1
India	China-India War (1962)	63.7	1.3
India	India-Pakistan War (1996~2003)	16.2	0.2
Pakistan	India-Pakistan War (1996~2003)	4.1	-0.1
US	Afghanistan-US/UK War (2001)	12.3	0.3
UK	Afghanistan-US/UK War (2001)	5.9	0.1
Tunisia	Tunisia-France (1961)	-23.7	-0.5
France	Tunisia-France (1961)	3.3	-0.1
Israel	Israel-Jordan (1967)	26.7	2.0
Jordan	Israel-Jordan (1967)	22.0	0.6

Source: SIPRI, Hanwha Research Center

In Many Past Cases, Military Spending Increased YoY Even Immediately After the End of War/Ceasefire (5)



[Table 8] Changes in Military Spending and Military Spending as a Share of GDP After the End of War/Ceasefire for Belligerents (T+2 Years)

Country	War Name and Period	Military Spending YoY (%)	Military Spending as % of GDP YoY (%p)
South Korea	Korean War (1950~53)	10.9	0.4
Iran	Iran-Iraq War (1980~88)	-11.2	-1.6
Eritrea	Eritrea-Ethiopia War (1998~2000)	0.0	-1.5
Ethiopia	Eritrea-Ethiopia War (1998~2000)	-16.9	-0.6
Kuwait	Iraq-Kuwait War (1990~91)	-51.6	-19.4
India	India-Pakistan War (1971)	-11.7	-0.6
Pakistan	India-Pakistan War (1971)	-6.6	-0.7
Chad	Chad-Libya War (1987)	-19.4	-0.8
US	US/UK/Australia-Iraq War (2003)	4.6	0.1
UK	US/UK/Australia-Iraq War (2003)	0.9	-0.1
Australia	US/UK/Australia-Iraq War (2003)	3.5	-0.0
Iraq	US/UK/Australia-Iraq War (2003)	35.0	0.6
Israel	Israel-Egypt War (1973)	10.5	2.8
Egypt	Israel-Egypt War (1973)	-5.2	-1.8
Israel	Israel-Egypt War (1969~70)	1.4	-2.4
Egypt	Israel-Egypt War (1969~70)	-17.5	-3.6

In Many Past Cases, Military Spending Increased YoY Even Immediately After the End of War/Ceasefire (6)

[Table 8] Changes in Military Spending and Military Spending as a Share of GDP After the End of War/Ceasefire for Belligerents (T+2 Years) (cont.)

Country	War Name and Period	Military Spending YoY (%)	Military Spending as % of GDP YoY (%p)
India	India-Pakistan War (1964~65)	-6.6	-0.4
Pakistan	India-Pakistan War (1964~65)	-18.5	-1.3
Israel	Israel-Syria War (1973)	10.5	2.8
Syria	Israel-Syria War (1973)	0.0	3.3
Israel	Israel-Egypt War (1967)	23.3	1.7
Egypt	Israel-Egypt War (1967)	32.5	3.2
India	China-India War (1962)	-2.4	-0.2
India	India-Pakistan War (1998~2003)	6.4	0.1
Pakistan	India-Pakistan War (1998~2003)	4.1	-0.1
US	Afghanistan-US/UK War (2001)	13.8	0.4
UK	Afghanistan-US/UK War (2001)	7.1	0.1
Tunisia	Tunisia-France War (1961)	0.3	-0.2
France	Tunisia-France War (1961)	-1.8	-0.4
Israel	Israel-Jordan War (1967)	23.3	1.7
Jordan	Israel-Jordan War (1967)	-8.8	-23.9

Source: SIPRI, Hanwha Research Center

In Many Past Cases, Military Spending Increased YoY Even Immediately After the End of War/Ceasefire (7)



[Table 9] Changes in Military Spending and Military Spending as a Share of GDP After the End of War/Ceasefire for Belligerents (T+5 Years)

Country	War Name and Year	Military Spending YoY (%)	Military Spending as % of GDP YoY (%p)
South Korea	Korean War (1950~53)	18.0	0.6
Iran	Iran-Iraq War (1980~88)	17.1	-0.1
Eritrea	Eritrea-Ethiopia War (1998~2000)	0.4	-0.3
Kuwait	Iraq-Kuwait War (1990~91)	-15.0	-3.3
India	India-Pakistan War (1971)	14.8	-0.1
Pakistan	India-Pakistan War (1971)	0.8	-0.4
Chad	Chad-Libya War (1987)	-	1.4
US	US/UK/Australia-Iraq War (2003)	7.3	0.4
UK	US/UK/Australia-Iraq War (2003)	4.4	0.1
Australia	US/UK/Australia-Iraq War (2003)	3.6	-0.0
Iraq	US/UK/Australia-Iraq War (2003)	32.2	0.1
Israel	Israel-Egypt War (1973)	8.7	-0.2
Egypt	Israel-Egypt War (1973)	-18.7	-3.2
Israel	Israel-Egypt War (1969~70)	10.5	2.8
Egypt	Israel-Egypt War (1969~70)	-5.2	-1.8
India	India-Pakistan War (1964~65)	3.1	0.1

In Many Past Cases, Military Spending Increased YoY Even Immediately After the End of War/Ceasefire (8)

[Table 9] Changes in Military Spending and Military Spending as a Share of GDP After the End of War/Ceasefire for Belligerents (T+5 Years) (cont.)

Country	War Name and Year	Military Spending YoY (%)	Military Spending as % of GDP YoY (%p)
Pakistan	India-Pakistan War (1964~65)	9.1	0.7
Israel	Israel-Syria War (1973)	8.7	-0.2
Syria	Israel-Syria War (1973)	-	0.2
Israel	Israel-Egypt War (1967)	1.4	-2.4
Egypt	Israel-Egypt War (1967)	-17.5	-3.6
India	China-India War (1962)	-6.6	-0.4
India	India-Pakistan War (1996~2003)	13.4	0.2
Pakistan	India-Pakistan War (1996~2003)	-5.7	-0.1
Afghanistan	Afghanistan-US/UK War (2001)	-	-0.1
US	Afghanistan-US/UK War (2001)	1.4	-0.1
UK	Afghanistan-US/UK War (2001)	0.5	-0.1
Tunisia	Tunisia-France War (1961)	4.8	-
France	Tunisia-France War (1961)	3.0	-0.1
Israel	Israel-Jordan War (1967)	1.4	-2.4
Jordan	Israel-Jordan War (1967)	10.6	1.2

Source: SIPRI, Hanwha Research Center

Security Threats and Fiscal Capacity Determine the Direction of Military Spending After the End of War/Ceasefire

- **Five cases of decline in military spending** Cases in which military spending decreased immediately after the end of war/ceasefire fall broadly into two types. First, cases where the cost of conducting the war fell and security threats eased as the war ended. Second, cases where force rebuilding was needed but military spending could not be increased owing to insufficient fiscal capacity. The wars in which military spending actually decreased one year after the end were five in total: the Iran-Iraq War, the Eritrea-Ethiopia War, the Iraq-Kuwait War, the India-Pakistan War and the Tunisia-France War.

- Iran in particular saw its economy and finances greatly damaged by the eight-year protracted war with Iraq. Although security threats remained after the war, it was difficult to commit sufficient resources to rebuilding military power. In the remaining cases, including Kuwait, we judge that the easing of military tension after the end of the war affected the decrease in military spending. Kuwait's security environment improved owing to the weakening of Iraq's military power and US security guarantees. India and Pakistan saw short-term tension eased through the ceasefire and the Tashkent Declaration, and Tunisia and France also saw developments in which the core cause of conflict was resolved through the process leading to withdrawal from the Bizerte base. Security threats did not disappear completely in these countries either, and other factors such as the decline in war-fighting costs and fiscal burden may also have had an effect.

- It is difficult to interpret the decline in military spending after the end of a war simply as being because the war ended. We judge that not only the decrease in war-fighting costs but also changes in security threats and fiscal capacity had a combined effect. Conversely, we judge that if security threats remain and fiscal capacity is supportive, military spending is highly likely to continue rising even after the end of the war.

[Table 10] Cases of Decline in Military Spending Immediately After the End of War/Ceasefire and Changes in Military Spending as a Share of GDP

Category	T+1 Military Spending YoY	T+1 Military Spending/GDP YoY	T+2 Military Spending YoY	T+2 Military Spending/GDP YoY	T+5 Military Spending YoY	T+5 Military Spending/GDP YoY
Iran (Iran-Iraq)	-26.0	4.6	-11.2	2.9	17.1	1.7
Ethiopia (Ethiopia-Eritrea)	-36.5	4.4	-16.9	3.8	0.4	2.8
Kuwait (Iraq-Kuwait)	-49.3	31.8	-51.6	12.4	-15.0	10.3
India (India-Pakistan)	-5.8	3.6	-6.6	3.2	3.1	3.2
Tunisia (Tunisia-France)	-23.7	1.7	0.3	1.5	4.8	1.3

Source: SIPRI, Hanwha Research Center

Direction of Military Spending After the End of the Russia-Ukraine and US-Iran Wars/Ceasefires in Light of Past Cases (1)

- **Military spending outlook after the end of war: Russia - limited increase; Ukraine - uncertain; neighboring European countries - increase; US - increase; Iran - uncertain; neighboring Middle Eastern countries - increase** We judge that the direction of military spending after the end of the ongoing Russia-Ukraine war and US-Iran war will also vary depending on security threats and fiscal capacity after the end of the war. Even if a ceasefire/end of war is achieved in both wars, it is hard to say the security threats of the belligerents and neighboring countries will be completely eliminated. As the demand for force rebuilding and the fiscal capacity to support it differ, the direction of military spending may differ by country.
- In the Russia-Ukraine war, Ukraine must prepare for the possibility of a renewed Russian invasion even after the war ends. The need to replenish depleted weapons stockpiles and rebuild military power is also high. As the economy and finances have been greatly damaged by the prolonged war, actual military spending may depend heavily on whether the West provides support. Russia will also see a decrease in war-fighting costs, but it needs to restore depleted weapons stockpiles and rebuild its military power. The fiscal burden from the prolonged war could constrain future expansion of military spending.
- The situation of neighboring European countries is different. Even if the war ends, the Russian threat does not disappear completely. In addition, European NATO members have already agreed to invest 5% of GDP in defense/security spending by 2035. Even if the uptrend in military spending of Russia and Ukraine slows, demand for force buildup in neighboring countries is highly likely to continue for a considerable period.
- In the case of the US-Iran war as well, differences by country are expected. The US must replenish depleted missile and ammunition stockpiles after the war ends. Above all, the expansion of US defense spending is focused on strategic competition with China. We judge that the end of the war with Iran is unlikely to change the US trend of rising defense spending. Iran needs to rebuild its military power, but insufficient fiscal capacity could constrain the expansion of military spending. We believe neighboring Middle Eastern countries will have reconfirmed through the war that securing defense capabilities against missile and drone attacks is a core task of national security. It has been highlighted that, to respond to large-scale attacks, not only the introduction of air defense systems but also sufficient interceptor missile inventories are important. Demand for building up interceptor missile stockpiles and introducing medium- and long-range air defense systems is highly likely to continue even after the end of the war.

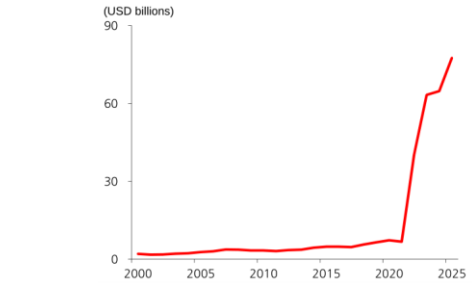
Direction of Military Spending After the End of the Russia-Ukraine and US-Iran Wars/Ceasefires in Light of Past Cases (2)

[Table 11] Military Spending Outlook of Belligerents and Neighboring Countries After the End of the Russia-Ukraine and US-Iran War/Ceasefire

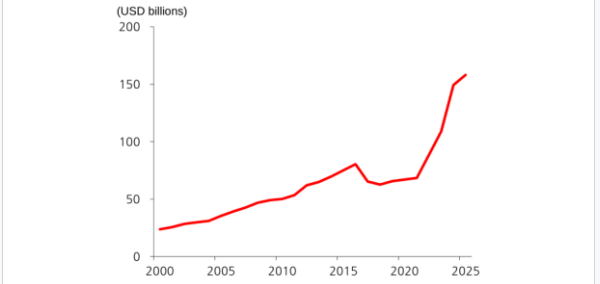
Category	Security Threat	Fiscal Capacity	Military Spending Outlook
Ukraine	Yes	Insufficient	Uncertain
Russia	Yes	Limited	Limited increase
Europe and other neighboring countries	Yes	Varies by country	Increase
US	Yes	Relatively sufficient	Increase
Iran	Yes	Insufficient	Uncertain
Middle East and other neighboring countries	Yes	Varies by country	Increase

Source: Hanwha Research Center

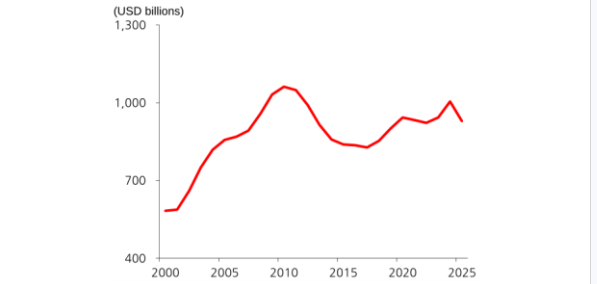
[Figure 16] Ukraine's military spending trend



[Figure 17] Russia's military spending trend



[Figure 18] US military spending trend



Direction of Military Spending After the End of the Russia-Ukraine and US-Iran Wars/Ceasefires in Light of Past Cases (3)

[Table 11] Military Spending Outlook of Belligerents and Neighboring Countries After the End of the Russia-Ukraine and US-Iran War/Ceasefire

Category	Security Threat	Fiscal Capacity	Military Spending Outlook
Ukraine	Yes	Insufficient	Uncertain
Russia	Yes	Limited	Limited increase
Europe and other neighboring countries	Yes	Varies by country	Increase
US	Yes	Relatively sufficient	Increase
Iran	Yes	Insufficient	Uncertain
Middle East and other neighboring countries	Yes	Varies by country	Increase

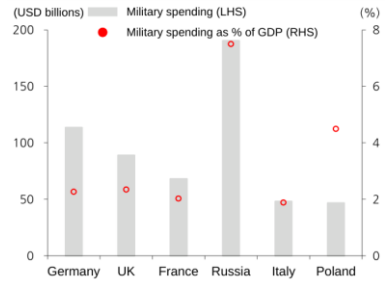
Source: Hanwha Research Center

[Figure 19] Iran's military spending trend



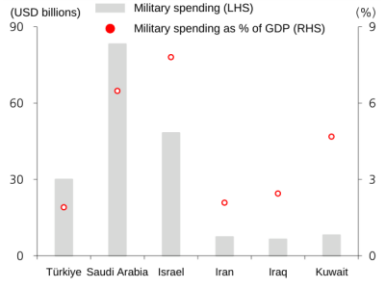
Source: SIPRI, Hanwha Research Center

[Figure 20] Military spending and military spending as a share of GDP in major European countries



Source: SIPRI, Hanwha Research Center

[Figure 21] Military spending and military spending as a share of GDP in major Middle Eastern countries



Source: SIPRI, Hanwha Research Center

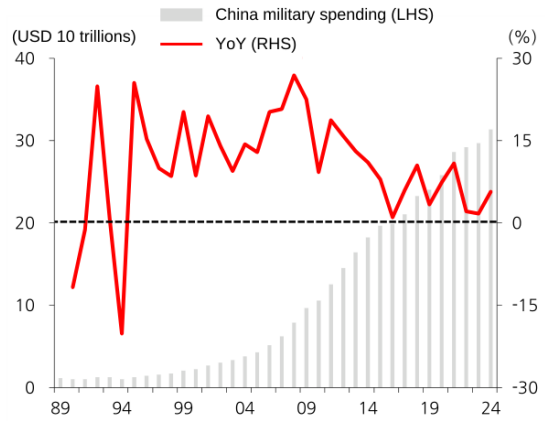
Narrowing of US Military Superiority and Weakening Capacity to Provide Security Guarantees (1)

- **The US must focus on containing China; effectiveness of security guarantees also in question** The biggest difference between the past and the present international situation is the US military intervention and allied security guarantee conditions. After the end of the Cold War, the US was able, on the strength of overwhelming military power, to intervene directly in regional conflicts or provide military support to allies. From the perspective of neighboring countries, the possibility of US military intervention alone could deter the military actions of potential adversaries. Currently, the US must concentrate its military capabilities on strategic competition with China. It must respond to the Russian threat in Europe and the need for military intervention continues in the Middle East, but it is becoming difficult to deploy the same level of military power as in the past to all regions. The US is prioritizing deterrence of China while demanding that its European and Middle Eastern allies strengthen their own defense capabilities and increase their defense spending burden. In this environment, even if the Russia-Ukraine war or Middle East war ends, it is difficult for neighboring countries to cut defense budgets on the basis of US security guarantees alone. The need to expand their own military power is increasing in preparation for situations where US military support is limited.

- Even if US security guarantees are strengthened after the end of the war, doubts remain over their effectiveness. The US possesses the world's strongest military power, but its overwhelming superiority of the past is weakening due to China's military expansion. The shortage of missile and ammunition stockpiles was highlighted through the Russia-Ukraine war and the Middle East war, and it is difficult to restore depleted forces in a short period owing to limited production capacity. This increases uncertainty over whether the US can deploy sufficient military power in a timely manner during an actual crisis even if it pledges security support. In fact, in September 2026, Saudi Arabia requested US military support to respond to attacks by Houthi rebels, but the US focused on intelligence support and withheld direct military intervention. This shows that even countries that maintain close security cooperation with the US cannot expect immediate US military intervention when a crisis occurs. We judge that even after the end of the war, US security guarantees alone are unlikely to sufficiently ease security threats, and demand for each country's own force buildup is likely to continue.

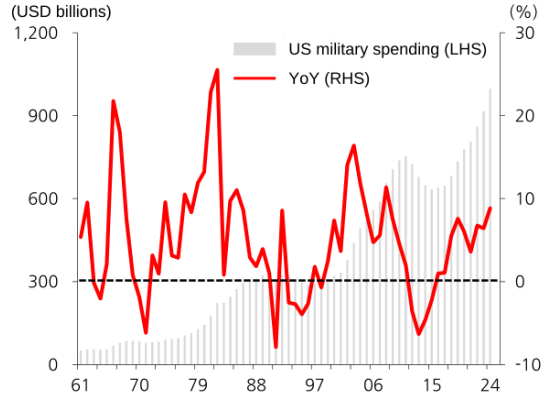
Narrowing of US Military Superiority and Weakening Capacity to Provide Security Guarantees (2)

[Figure 22] China's military spending trend



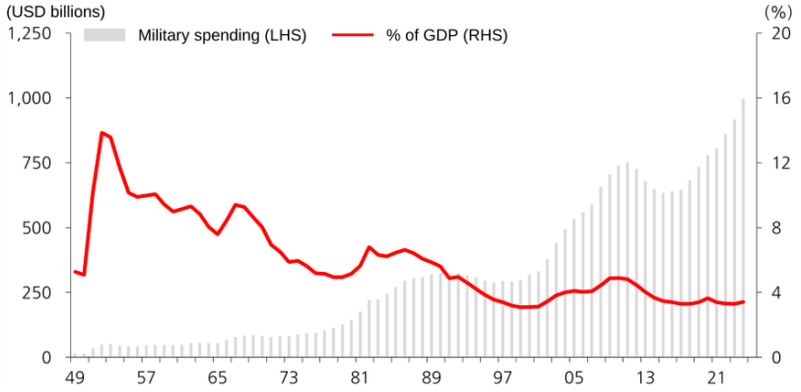
Source: SIPRI, Hanwha Research Center

[Figure 23] US military spending trend



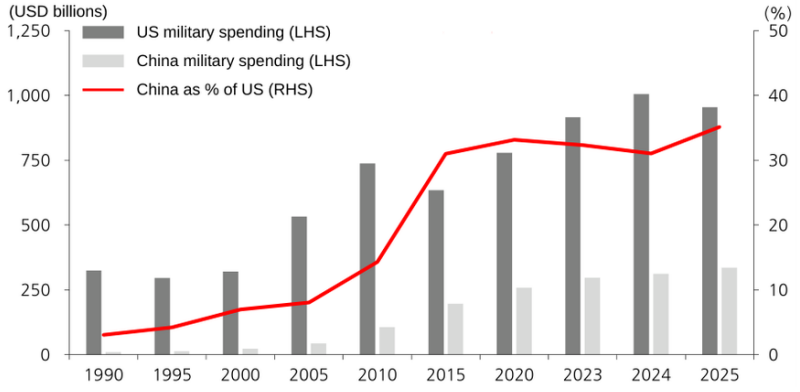
Source: SIPRI, Hanwha Research Center

[Figure 24] US military spending and its share of GDP



Source: SIPRI, Hanwha Research Center

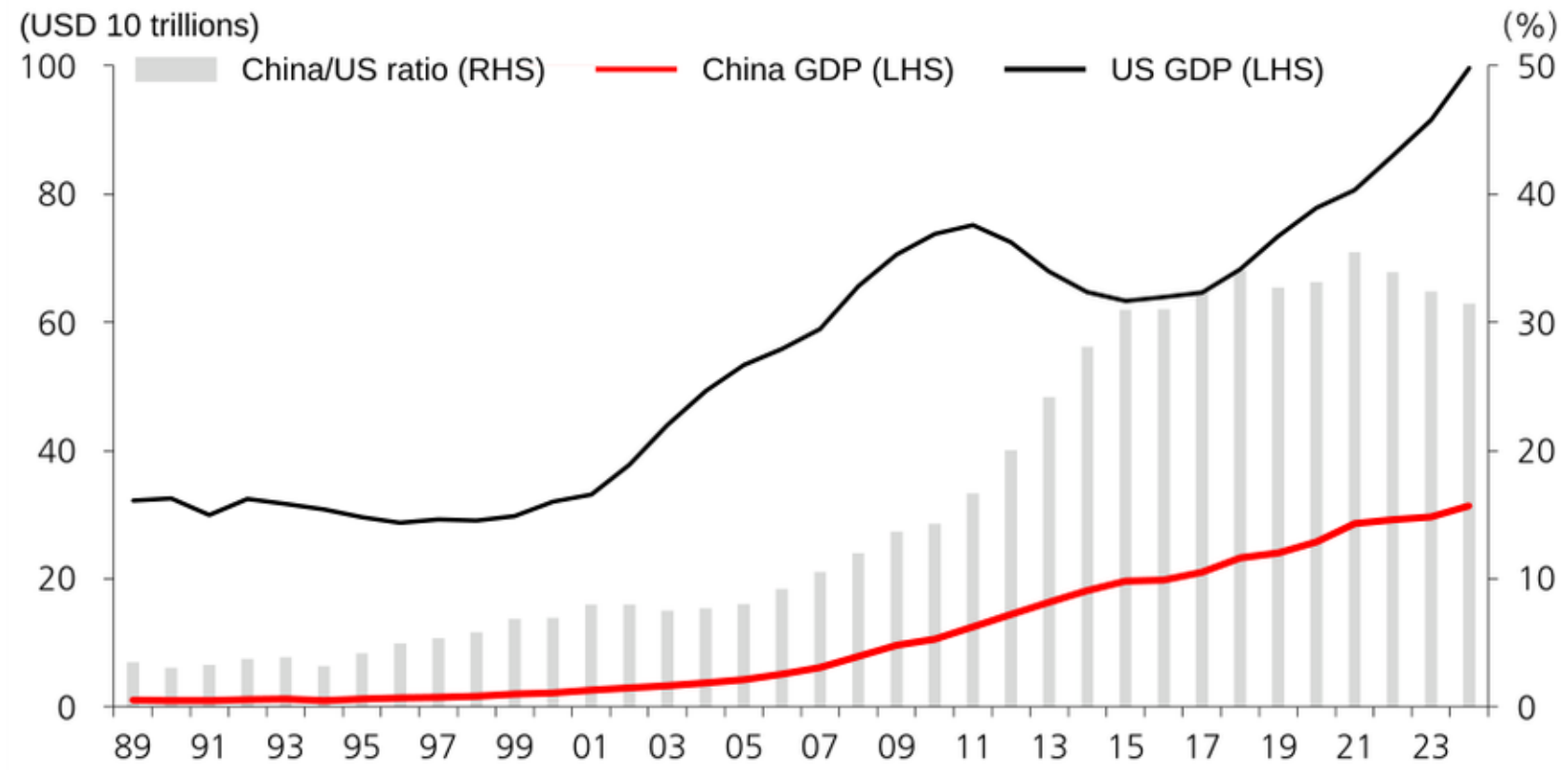
[Figure 25] Comparison of US and China military spending



Source: SIPRI, Hanwha Research Center

Narrowing of US Military Superiority and Weakening Capacity to Provide Security Guarantees (3)

[Figure 26] Comparison of US and China GDP



Source: SIPRI, Hanwha Research Center

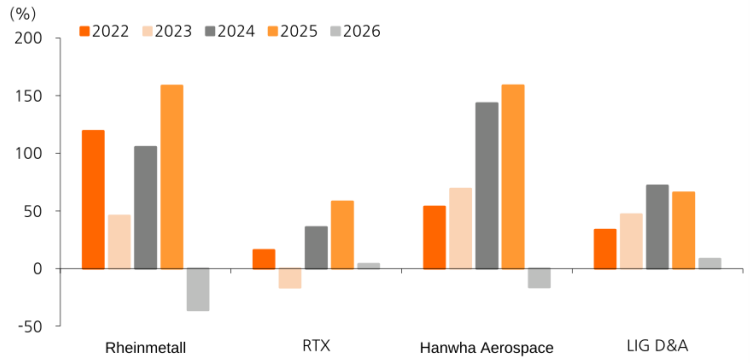
Limited Change in Total Actual Arms Trade Volume Since the Russia-Ukraine War (1)

- **Global arms trade showed no significant change after the outbreak of the Russia-Ukraine war** We judge that a full-fledged cycle of increase in global weapons demand has not yet started since the Russia-Ukraine war. Over the four years since the outbreak of the war, market interest in the global defense sector has expanded greatly, but the change in total actual arms trade volume has been limited. There was no significant difference in global arms import/export volume between 2022 and 2025, and the average annual growth rate over 2022-2025 was only 0.2%. Unlike the re-rating of the defense sector seen in the stock market over the past four years, it is hard to say that the entire global arms market has grown explosively. It is appropriate to view the period to date as one in which urgent demand to quickly replenish forces that were short immediately after the outbreak of war arose first.

- The increase in weapons demand was concentrated in a few countries. Ukraine and Poland account for about 92% of the net increase in global arms imports/exports in 2022-2025. Excluding the two countries, global arms trade volume increased by only about 3.6% compared to 2018-2021. It is still difficult to say that weapons demand has spread across the globe since the Russia-Ukraine war. We judge that the past four years were closer to the early stage before a full-fledged expansion of demand rather than the peak of global weapons demand. After urgent force replenishment is completed, a structural increase cycle in global weapons demand is expected to begin only when each country's force rebuilding, replacement of aging weapons and increased defense budgets start to translate into actual new orders.

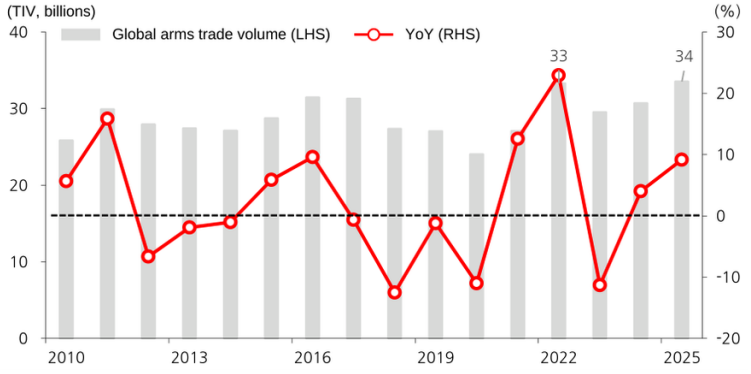
Limited Change in Total Actual Arms Trade Volume Since the Russia-Ukraine War (2)

[Figure 27] Annual share price returns of global defense companies since the outbreak of the 2022 Russia-Ukraine war



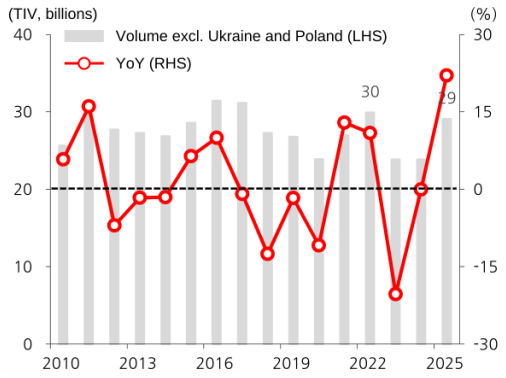
Source: Hanwha Research Center

[Figure 28] Trend in global arms import/export volume



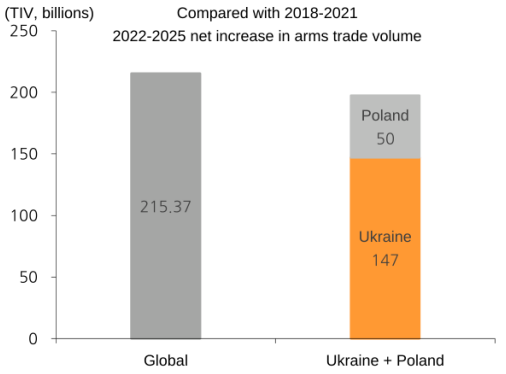
Source: SIPRI, Hanwha Research Center

[Figure 29] Trend in trade volume excluding Ukraine and Poland



Source: SIPRI, Hanwha Research Center

[Figure 30] Most of the net increase in global arms trade volume is attributable to Ukraine and Poland



Source: SIPRI, Hanwha Research Center

Limited Change in Total Actual Arms Trade Volume Since the Russia-Ukraine War (3)

[Table 12] Change in Arms Imports by Region Excluding Ukraine and Poland

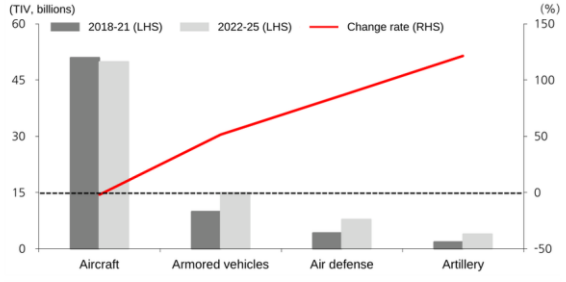
Category	2018~2021	2022~2025	Change (%)
Europe	13,199	26,322	99.4
Asia and Oceania	44,667	36,108	-19.2
Middle East	32,556	31,385	-3.6
Africa	5,697	5,598	-1.7
Americas	6,366	6,781	6.5
Total	**102,485**	**106,194**	**3.6**

Source: SIPRI, Hanwha Research Center

Demand Concentrated in Land Weapons During the Urgent Force Replenishment Phase

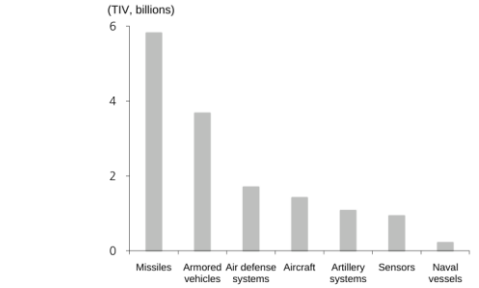
- **The current weapons demand phase is urgent replenishment centered on land weapons** Global weapons demand growth in 2022-2025 was concentrated in missiles, armored vehicles, air defense systems and artillery. Compared with 2018-2021, artillery rose 121.4%, air defense systems 86.2%, armored vehicles 51.4% and missiles 49.6%, and the four weapon systems accounted for about 85% of the net increase in global arms imports/exports. Aircraft fell 2.1%. Since the Russia-Ukraine war, the initial rearmament has proceeded in the direction of filling forces with tanks and armored vehicles, reinforcing firepower with artillery and missiles, and restoring air defense networks, rather than expanding high-value platforms such as fighters. It had a strong character of first replenishing the land forces that became depleted during the war.
- This characteristic is also confirmed in arms imports by country. Ukraine concentrated on maintaining its war-fighting capability by replenishing tanks and armored vehicles lost during the war and introducing self-propelled howitzers, multiple launch rocket systems and air defense systems. Poland expanded the introduction of tanks, self-propelled howitzers and multiple launch rocket systems to fill the force gap created by transferring equipment to Ukraine and to prepare for the Russian threat. East European countries such as Hungary and Romania also moved to replace aging land forces and strengthen air defense networks. While Ukraine's weapons introduction was force replenishment needed to wage the war, Poland and East European countries simultaneously pursued filling force gaps and strengthening medium- to long-term military power.

[Figure 31] Urgent force replenishment demand concentrated in armored vehicles, air defense and artillery



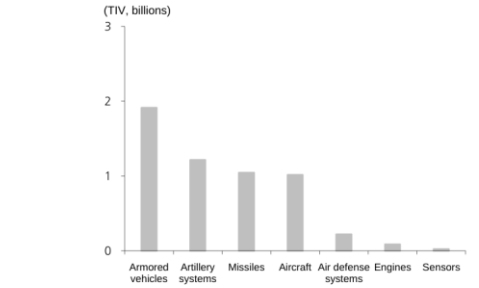
Source: SIPRI, Hanwha Research Center

[Figure 32] Ukraine's arms imports by system (2022-25)



Source: SIPRI, Hanwha Research Center

[Figure 33] Poland's arms imports by system (2022-25)



Source: SIPRI, Hanwha Research Center

Poland: A Representative Case of Structural Force Rebuilding (1)

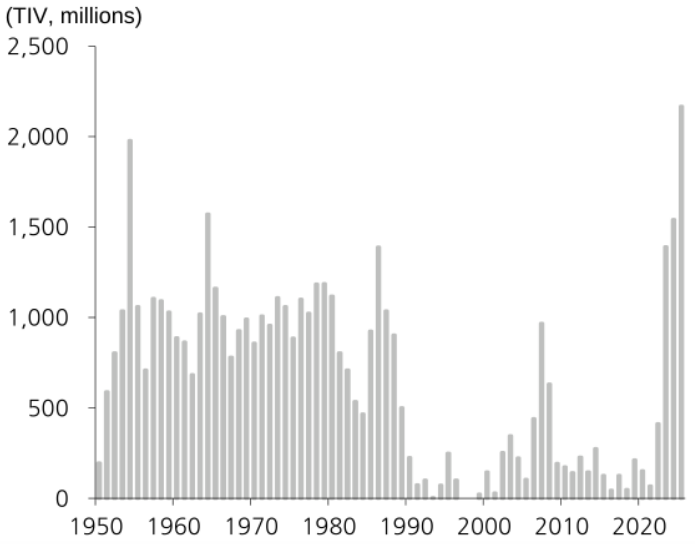
- **Poland is increasing both the quantity and performance of weapons** We judge that the case of Poland best shows the process by which urgent force replenishment transitions to structural force rebuilding. At the end of 2021, before the Russia-Ukraine war, Poland's main battle tanks were estimated at about 859, of which about 612, or 71%, were Soviet-origin tanks. The gap created by transferring old equipment to Ukraine on a large scale was quickly filled with K2 and Abrams tanks. It went beyond simple 1:1 replacement to expand absolute scale. At the end of 2021, modern tanks were virtually limited to about 247, but in 2025, with K2 and Abrams deliveries added, they are estimated to have increased to about 646. The same applies to artillery forces. Poland did not stop at replacing aging equipment with new equipment but simultaneously improved the quantity and performance of tanks, self-propelled howitzers and long-range firepower. It is the case that first shows the picture of structural rearmament.

- Although the scale of rearmament is smaller than Poland's, a similar force rebuilding trend is confirmed in Hungary. Hungary's main battle tanks numbered 48 at the end of 2021 but increased 129.2% to 110 at the end of 2025. Over the same period, the share of modern Leopard tanks rose from 8.3% to 40.9%. Infantry fighting vehicles also increased from 120 to more than 165, with newly introduced Lynx KF41 replacing existing Soviet-made armored vehicles. Changes in artillery forces are also clear. It had no self-propelled howitzers at the end of 2021 but secured 24 PzH2000 guns at the end of 2025. It also newly introduced 14 NASAMS units of medium-range air defense systems, which it previously lacked. Hungary is significant in that its military modernization program, pursued since before the Russia-Ukraine war, has led to actual force expansion since 2022. This shows that European rearmament is not limited to short-term demand arising after the war, but is a structural process in which the replacement of existing aging forces and the build-up of new forces proceed over a long period.

- Denmark, Romania and the three Baltic states are pursuing the replacement of aging forces and the expansion of long-range firepower. In Asia, Taiwan and Japan are strengthening defense capabilities in response to China's military threat, and in the Middle East, Qatar and Saudi Arabia are focusing on modernizing air and naval forces and air defense networks. Rearmament is spreading across multiple regions rather than being limited to specific countries.

Poland: A Representative Case of Structural Force Rebuilding (2)

[Figure 34] Poland's arms import volume (1950-2025)



Source: SIPRI, Hanwha Research Center

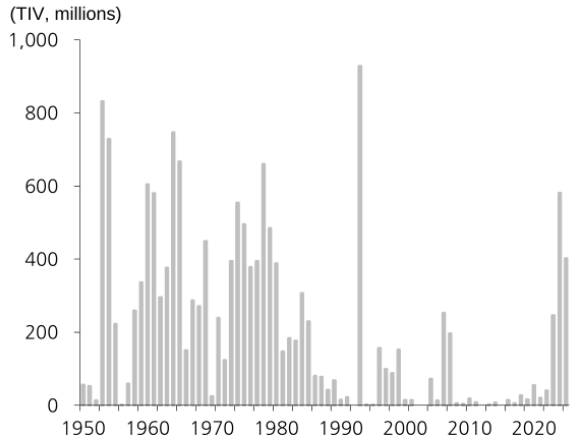
Poland: A Representative Case of Structural Force Rebuilding (3)

[Table 13] Comparison of Changes in Poland's Force Levels (2021-2025)

Category	End-2021 Force Level (Share of New Units Introduced Within Last 10 Yrs)	End-2025 Force Level (Share of New Units Introduced Within Last 10 Yrs)	Change (%)
MBT	797 (14.9%)	897 (46.0%)	12.5%
IFV+APC	1,611 (3.2%)	1,516 (14.0%)	-5.9%
Self-Propelled Howitzers	410 (17.6%)	558 (43.2%)	36.1%
Multiple Launch Rocket Systems	179 (0.0%)	299 (56.2%)	67.0%
Fighter/Attack Aircraft	94 (0.0%)	77 (20.8%)	-18.1%
Attack Helicopters	28 (0.0%)	24 (33.3%)	-14.3%
Submarines	1 (0.0%)	1 (0.0%)	-
Surface Combatants	2 (0.0%)	2 (0.0%)	-

Source: IISS, Hanwha Research Center

[Figure 35] Hungary's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

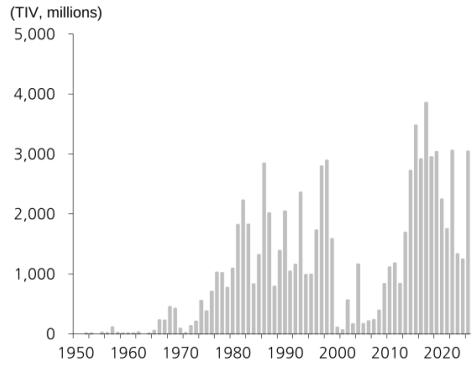
Poland: A Representative Case of Structural Force Rebuilding (4)

[Table 14] Comparison of Changes in Hungary's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	48	110	129.2%
IFV	120	165+	37.5%+
APC	282	424	50.4%
Self-Propelled Howitzers	0	24	Newly acquired
Multiple Launch Rocket Systems	0	0	-
Medium-Range Air Defense	0	14	Newly acquired
Short-Range Air Defense	16	8	-50.0%
Fighter/Attack Aircraft	14	14	-
Attack Helicopters	8	8	-
Multirole Helicopters	30	34	13.3%
Surface Combatants	0	0	-
Submarines	0	0	-

Source: IISS, Hanwha Research Center

[Figure 36] Saudi Arabia's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

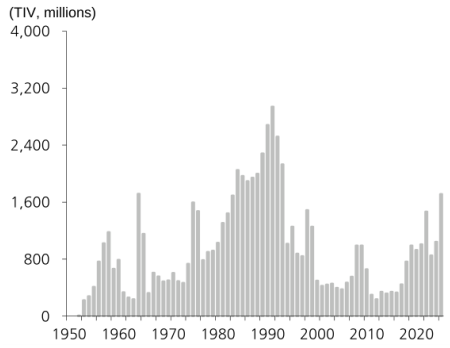
Poland: A Representative Case of Structural Force Rebuilding (5)

[Table 15] Comparison of Changes in Saudi Arabia's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	1,010	1,085	7.4%
IFV	1,975	2,145	8.6%
APC	2,253	2,253	-
Self-Propelled Howitzers	224	224	-
Multiple Launch Rocket Systems	70	88	25.7%
Medium-Range Air Defense	236	236	-
Short-Range Air Defense	186	192	3.2%
Fighter/Attack Aircraft	369	353	-4.3%
Attack Helicopters	47	59	25.5%
Multirole Helicopters	274	274	-
Surface Combatants	19	25	31.6%
Submarines	0	0	-

Note: Self-propelled howitzers are on an Army basis; APC excludes AUV
Source: IISS, Hanwha Research Center

[Figure 37] Japan's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

Poland: A Representative Case of Structural Force Rebuilding (6)

[Table 16] Comparison of Changes in Japan's Force Levels (March 2022 - March 2025)

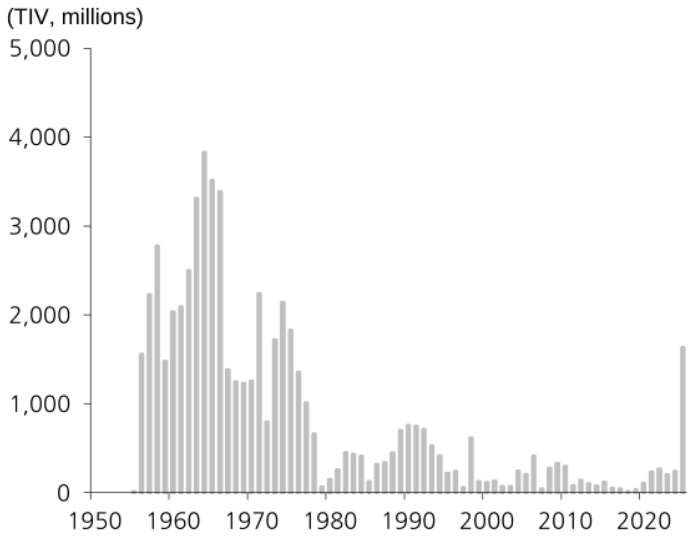
Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	579	349	-39.7%
IFV	68	68	-
APC	804	786	-2.2%
Self-Propelled Howitzers	166	150	-9.6%
Multiple Launch Rocket Systems	54	21	-61.1%
Medium-Range Air Defense	250	243	-2.8%
Short-Range Air Defense	70	48	-31.4%
Fighter/Attack Aircraft	317	330	4.1%
Attack Helicopters	99	84	-15.2%
Multirole Helicopters	282	259	-8.2%
Surface Combatants	55	60	9.1%
Submarines	22	25	13.6%

Source: IISS, Hanwha Research Center

A Long Way to Go in Force Rebuilding (1)

▪ **Many countries still need force buildup** It should be noted that there are far more countries that still need a full-fledged force buildup. Germany and the UK, which have large defense budgets in Europe, have expanded their defense budgets since the Russia-Ukraine war, but the replacement of aging forces and the replenishment of force gaps resulting from donations to Ukraine remain insufficient. This is a common problem in many European countries, and major countries around the world beyond Europe have likewise not secured sufficient independent defense capabilities. In the Middle East as well, as geopolitical risks have recently expanded, the need to expand air defense networks and military power, in addition to replenishing depleted missile and ammunition stockpiles, is growing. Force rebuilding has become visible only in some countries, and the stage has not yet been reached where defense budget increases in major countries around the world have translated into large-scale replacement and buildup of actual forces. Given that even urgent force replenishment demand, let alone structural force rebuilding, has not been sufficiently met, we judge that there is considerable room for further expansion of global weapons demand going forward.

[Figure 38] Germany's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

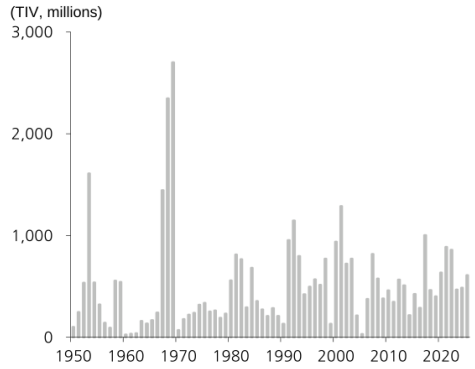
A Long Way to Go in Force Rebuilding (2)

[Table 17] Comparison of Changes in Germany's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	284	313	10.2%
IFV	674	680	0.9%
APC	728	802	10.2%
Self-Propelled Howitzers	121	109	-9.9%
Multiple Launch Rocket Systems	41	38	-7.3%
Medium-Range Air Defense	30	73	143.3%
Short-Range Air Defense	20	20	-
Fighter/Attack Aircraft	226	226	-
Attack Helicopters	51	51	-
Multirole Helicopters	189	188	-0.5%
Surface Combatants	16	16	-
Submarines	6	6	-

Source: IISS, Hanwha Research Center

[Figure 39] The UK's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

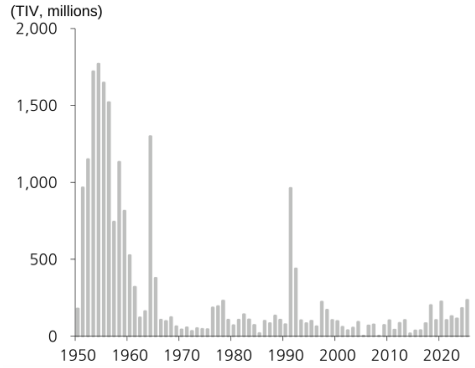
A Long Way to Go in Force Rebuilding (3)

[Table 18] Comparison of Changes in the UK's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	227	213	-6.2%
IFV	388+	388+	-
APC	805	875	8.7%
Self-Propelled Howitzers	89	53	-40.4%
Multiple Launch Rocket Systems	35	26	-25.7%
Medium-Range Air Defense	0	0	-
Short-Range Air Defense	Quantity not stated	12	Not comparable
Fighter/Attack Aircraft	167	159	-4.8%
Attack Helicopters	50	44	-12.0%
Multirole Helicopters	188	142	-24.5%
Surface Combatants	20	16	-20.0%
Submarines	10	10	-

Source: IISS, Hanwha Research Center

[Figure 40] France's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

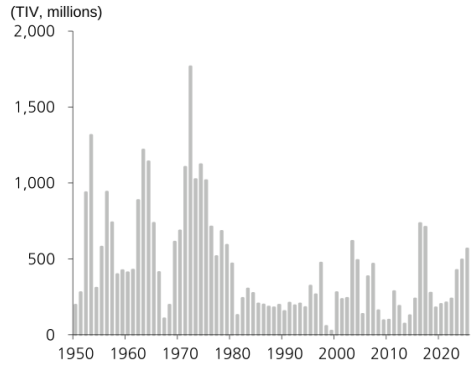
A Long Way to Go in Force Rebuilding (4)

[Table 19] Comparison of Changes in France's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	222	200	-9.9%
IFV	706	622	-11.9%
APC	2,616	2,557	-2.3%
Self-Propelled Howitzers	108	92	-14.8%
Multiple Launch Rocket Systems	13	9	-30.8%
Medium-Range Air Defense	40	40	-
Short-Range Air Defense	24	26	8.3%
Fighter/Attack Aircraft	250	235	-6.0%
Attack Helicopters	67	67	-
Multirole Helicopters	369	384	4.1%
Surface Combatants	22	22	-
Submarines	8	9	12.5%

Source: IISS, Hanwha Research Center

[Figure 41] Italy's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

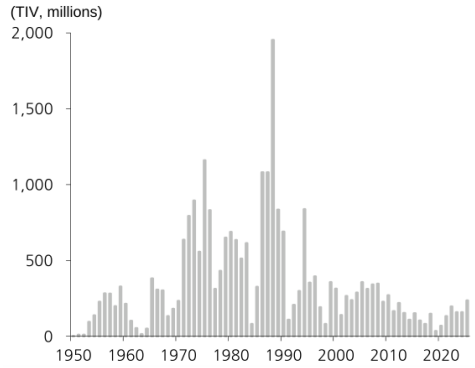
A Long Way to Go in Force Rebuilding (5)

[Table 20] Comparison of Changes in Italy's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	150	150	-
IFV	426	449	5.4%
APC	379	370	-2.4%
Self-Propelled Howitzers	70	64	-8.6%
Multiple Launch Rocket Systems	22	21	-4.5%
Medium-Range Air Defense	20	12	-40.0%
Short-Range Air Defense	32	-	-
Fighter/Attack Aircraft	238	211	-11.3%
Attack Helicopters	35	33	-5.7%
Multirole Helicopters	258	235	-8.9%
Surface Combatants	17	18	5.9%
Submarines	8	8	-

Source: IISS, Hanwha Research Center

[Figure 42] Spain's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

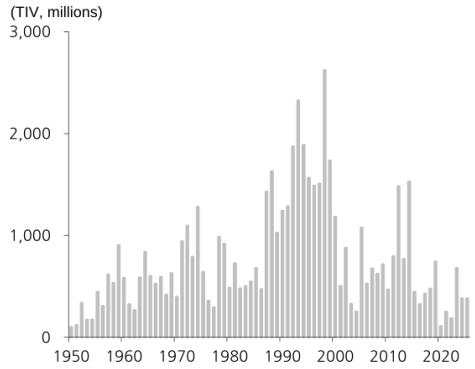
A Long Way to Go in Force Rebuilding (6)

[Table 21] Comparison of Changes in Spain's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	327	274	-16.2%
IFV	225	225	-
APC	937	949	1.3%
Self-Propelled Howitzers	101	102	1.0%
Multiple Launch Rocket Systems	-	-	-
Medium-Range Air Defense	56	46	-17.9%
Short-Range Air Defense	21	8	-61.9%
Fighter/Attack Aircraft	185	182	-1.6%
Attack Helicopters	24	18	-25.0%
Multirole Helicopters	142	134	-5.6%
Surface Combatants	11	11	-
Submarines	2	2	-

Source: IISS, Hanwha Research Center

[Figure 43] Turkey's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

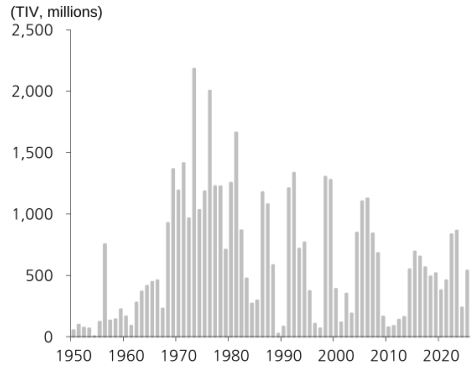
A Long Way to Go in Force Rebuilding (7)

[Table 22] Comparison of Changes in Turkey's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	2,378	2,378	-
IFV	645	645	-
APC	6,026	7,175	19.1%
Self-Propelled Howitzers	1,080	1,061	-1.8%
Multiple Launch Rocket Systems	146	98	-32.9%
Medium-Range Air Defense	32	58	81.3%
Short-Range Air Defense	148	148	-
Fighter/Attack Aircraft	279	279	-
Attack Helicopters	103	114	10.7%
Multirole Helicopters	351	364	3.7%
Surface Combatants	45	44	-2.2%
Submarines	12	13	8.3%

Source: IISS, Hanwha Research Center

[Figure 44] Israel's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

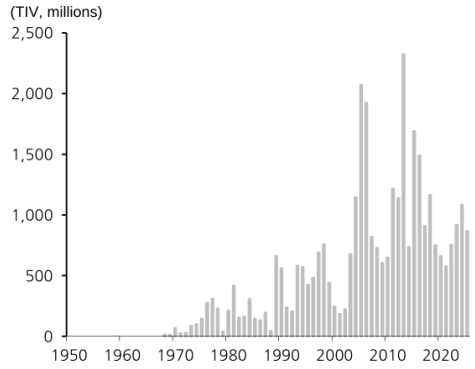
A Long Way to Go in Force Rebuilding (8)

[Table 23] Comparison of Changes in Israel's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	400	400	-
IFV	-	-	-
APC	1,190	390	-
Self-Propelled Howitzers	250	250	-
Multiple Launch Rocket Systems	30	48	60.0%
Medium-Range Air Defense	-	-	-
Short-Range Air Defense	40	40	-
Fighter/Attack Aircraft	309	310	0.3%
Attack Helicopters	43	46	7.0%
Multirole Helicopters	80	80	-
Surface Combatants	15	15	-
Submarines	5	5	-

Source: IISS, Hanwha Research Center

[Figure 45] The UAE's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

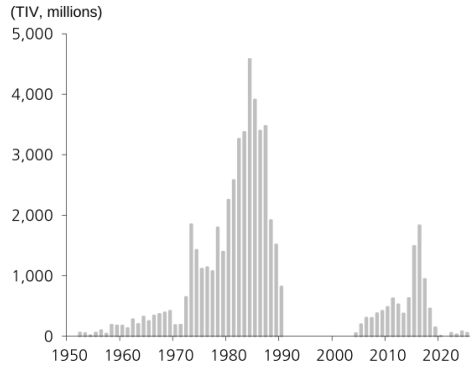
A Long Way to Go in Force Rebuilding (9)

[Table 24] Comparison of Changes in the UAE's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	353	363	2.8%
IFV	814	740	-9.1%
APC	1,656	1,656	-
Self-Propelled Howitzers	181	181	-
Multiple Launch Rocket Systems	94	94	-
Medium-Range Air Defense	-	39	-
Short-Range Air Defense	42	42	-
Fighter/Attack Aircraft	160	151	-5.6%
Attack Helicopters	28	29	3.6%
Multirole Helicopters	144	144	-
Surface Combatants	31	33	6.5%
Submarines	-	-	-

Source: IISS, Hanwha Research Center

[Figure 46] Iraq's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

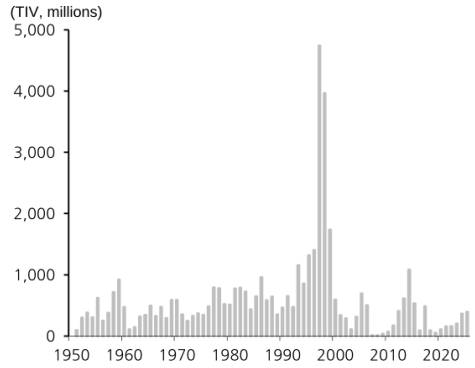
A Long Way to Go in Force Rebuilding (10)

[Table 25] Comparison of Changes in Iraq's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	391	401	2.6%
IFV	650	650	-
APC	1,592	1,592	-
Self-Propelled Howitzers	48	48	-
Multiple Launch Rocket Systems	6	6	-
Medium-Range Air Defense	-	-	-
Short-Range Air Defense	24	24	-
Fighter/Attack Aircraft	64	62	-3.1%
Attack Helicopters	35	6	-82.9%
Multirole Helicopters	107	102	-4.7%
Surface Combatants	6	6	-
Submarines	0	0	-

Source: IISS, Hanwha Research Center

[Figure 47] Taiwan's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

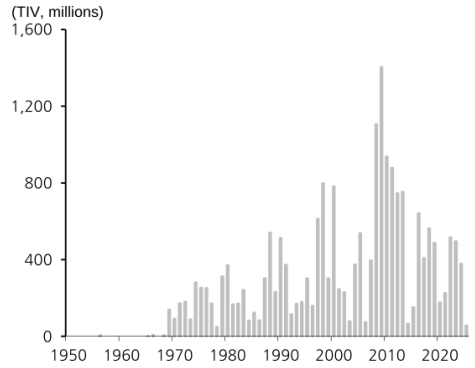
A Long Way to Go in Force Rebuilding (11)

[Table 26] Comparison of Changes in Taiwan's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	750	750	-
IFV	257	305	18.7%
APC	1,318	1,543	17.1%
Self-Propelled Howitzers	488	488	-
Multiple Launch Rocket Systems	223	234	4.9%
Medium-Range Air Defense	172	122	-29.1%
Short-Range Air Defense	30	30	-
Fighter/Attack Aircraft	407	322	-20.9%
Attack Helicopters	96	90	-6.2%
Multirole Helicopters	103	93	-9.7%
Surface Combatants	69	71	2.9%
Submarines	2	2	-

Source: IISS, Hanwha Research Center

[Figure 48] Singapore's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

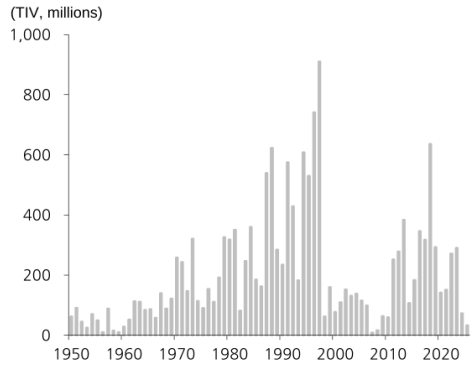
A Long Way to Go in Force Rebuilding (12)

[Table 27] Comparison of Changes in Singapore's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	96	96	-
IFV	600	420	-30.0%
APC	1,375	1,635	18.9%
Self-Propelled Howitzers	54	54	-
Multiple Launch Rocket Systems	18	18	-
Medium-Range Air Defense	4	4	-
Short-Range Air Defense	-	-	-
Fighter/Attack Aircraft	100	99	-1.0%
Attack Helicopters	19	19	-
Multirole Helicopters	56	59	5.4%
Surface Combatants	6	6	-
Submarines	4	4	-

Source: IISS, Hanwha Research Center

[Figure 49] Thailand's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

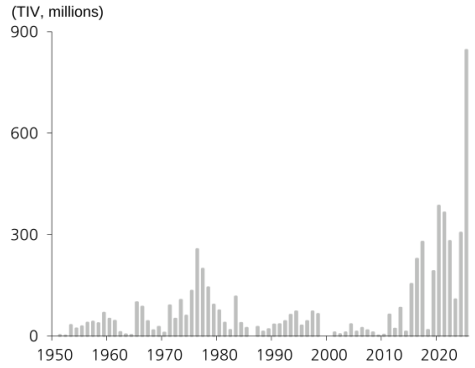
A Long Way to Go in Force Rebuilding (13)

[Table 28] Comparison of Changes in Thailand's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	370	394	6.5%
IFV	234	234	-
APC	1,214	1,289	6.2%
Self-Propelled Howitzers	48	62	29.2%
Multiple Launch Rocket Systems	68	69	1.5%
Medium-Range Air Defense	3	3	-
Short-Range Air Defense	8	10	25.0%
Fighter/Attack Aircraft	103	82	-20.4%
Attack Helicopters	7	7	-
Multirole Helicopters	205	212	3.4%
Surface Combatants	15	13	-13.3%
Submarines	0	0	-

Source: IISS, Hanwha Research Center

[Figure 50] The Philippines' arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

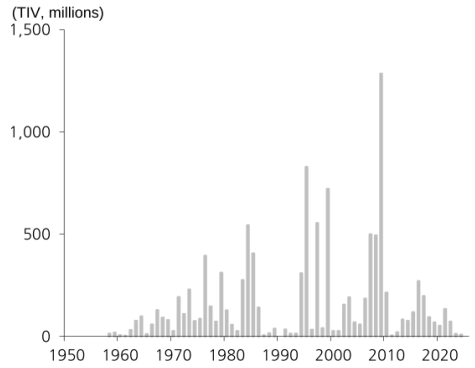
A Long Way to Go in Force Rebuilding (14)

[Table 29] Comparison of Changes in the Philippines' Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	0	0	-
IFV	54	54	-
APC	429	437	1.9%
Self-Propelled Howitzers	-	12	-
Multiple Launch Rocket Systems	-	-	-
Medium-Range Air Defense	-	6	-
Short-Range Air Defense	-	-	-
Fighter/Attack Aircraft	12	12	-
Attack Helicopters	2	8	300.0%
Multirole Helicopters	89	77	-13.5%
Surface Combatants	6	11	83.3%
Submarines	-	-	-

Source: IISS, Hanwha Research Center

[Figure 51] Malaysia's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

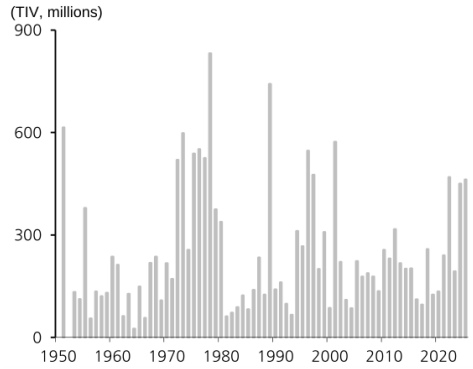
A Long Way to Go in Force Rebuilding (15)

[Table 30] Comparison of Changes in Malaysia's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	48	48	-
IFV	212	199	-6.1%
APC	629	352	-44.0%
Self-Propelled Howitzers	0	0	-
Multiple Launch Rocket Systems	36	36	-
Medium-Range Air Defense	0	0	-
Short-Range Air Defense	15	15	-
Fighter/Attack Aircraft	26	26	-
Attack Helicopters	0	0	-
Multirole Helicopters	35	41	17.1%
Surface Combatants	10	10	-
Submarines	2	2	-

Source: IISS, Hanwha Research Center

[Figure 52] Brazil's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

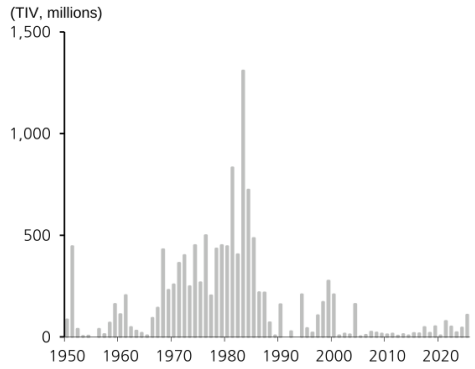
A Long Way to Go in Force Rebuilding (16)

[Table 31] Comparison of Changes in Brazil's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	296	292	-1.4%
IFV	13	13	-
APC	1,430	1,526	6.7%
Self-Propelled Howitzers	169	109	-35.5%
Multiple Launch Rocket Systems	42	44	4.8%
Medium-Range Air Defense	0	0	-
Short-Range Air Defense	-	-	-
Fighter/Attack Aircraft	76	76	-
Attack Helicopters	12	0	-100.0%
Multirole Helicopters	209	204	-2.4%
Surface Combatants	8	8	-
Submarines	5	4	-20.0%

Source: IISS, Hanwha Research Center

[Figure 53] Argentina's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

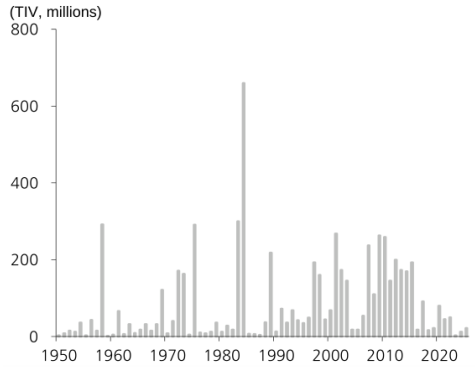
A Long Way to Go in Force Rebuilding (17)

[Table 32] Comparison of Changes in Argentina's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	231	231	-
IFV	232	232	-
APC	309	309	-
Self-Propelled Howitzers	42	42	-
Multiple Launch Rocket Systems	8	8	-
Medium-Range Air Defense	0	0	-
Short-Range Air Defense	-	-	-
Fighter/Attack Aircraft	12	11	-8.3%
Attack Helicopters	0	0	-
Multirole Helicopters	118	116	-1.7%
Surface Combatants	14	13	-7.1%
Submarines	2	2	-

Source: IISS, Hanwha Research Center

[Figure 54] Colombia's arms import trend (1950-2025)



Source: SIPRI, Hanwha Research Center

A Long Way to Go in Force Rebuilding (18)

[Table 33] Comparison of Changes in Colombia's Force Levels (End-2021 to End-2025)

Category	End-2021 Force Level	End-2025 Force Level	Change (%)
MBT	0	0	-
IFV	60	60	-
APC	114	182	59.6%
Self-Propelled Howitzers	-	-	-
Multiple Launch Rocket Systems	-	-	-
Medium-Range Air Defense	-	-	-
Short-Range Air Defense	-	-	-
Fighter/Attack Aircraft	34	26	-23.5%
Attack Helicopters	14	14	-
Multirole Helicopters	177	177	-
Surface Combatants	6	6	-
Submarines	4	4	-

Source: IISS, Hanwha Research Center

Weapons Demand Changing Into Structural Force Buildup (1)

- **Structural force rebuilding begins** If global weapons demand in 2022-2025 was concentrated on urgent force replenishment, future demand is expected to change into structural force rebuilding. Immediately after the Russia-Ukraine war, the priority was to quickly fill the force gaps created in the course of conducting the war and supporting Ukraine. Going forward, it is highly likely that the replacement of aging forces, expansion of holdings and modernization of weapon systems will proceed simultaneously. Poland is already showing this change, and we expect major countries including those in Europe to translate expanded defense budgets into actual force buildup.

- There are three grounds on which we judge that structural force rebuilding demand will continue for a long time. 1) The prolongation of geopolitical competition. During the Cold War, the US and the Soviet Union continued an arms race to secure superiority in nuclear and conventional forces. The military buildup of the two countries heightened the threat to the opposing camp, and NATO and Warsaw Pact members also embarked on force buildup to prevent military inferiority and strengthen collective defense capabilities. Currently as well, with US-China strategic competition and regional security threats continuing, demand for strengthening defense capabilities in each country is highly likely to continue for a long time. 2) The need to replace aging forces. Due to defense budget cuts and delays in introducing new weapons after the end of the Cold War, aging forces have accumulated in major countries. Because tanks, self-propelled howitzers, armored vehicles and fighters are weapon systems operated for decades, replacement demand that had been postponed is highly likely to lead to new orders over a long period in tandem with future defense budget increases. 3) Increased capacity to introduce new weapons due to defense budget expansion. In the past, a considerable portion of defense budgets went to maintaining and operating existing forces, limiting the introduction of new weapons. As major countries have recently raised defense budgets and presented medium- to long-term force buildup targets, the resources available for introducing new weapons are expanding. As force buildup programs that had remained at the planning stage are converted into actual orders, they are expected to support growth in global weapons demand.

Weapons Demand Changing Into Structural Force Buildup (2)

[Figure 55] Three grounds on which structural force rebuilding demand could continue for a long time



Source: Hanwha Research Center

① Prolongation of Geopolitical Competition: The Cold War and Today, a Changed Global Rearmament Structure (1)

▪ **Cold War and present rearmament: similar yet different** To judge how the current global rearmament cycle will unfold, it is necessary to look at the Cold War period. This is because the structure of the present is similar to that of the Cold War in that strategic competition among great powers leads to arms buildup by neighboring countries. The US-Soviet arms race during the Cold War was not confined to the two countries. As the military buildup of the two countries heightened the threat to the opposing camp, members of NATO and the Warsaw Pact also embarked on force buildup to prevent military inferiority. Weapons demand also expanded in neighboring regions such as Asia and the Middle East. There are differences between the past and present arms race structures. Whereas US-Soviet competition was the central axis of the global arms race during the Cold War, today Russia's threat to Europe and geopolitical conflict in the Middle East have been added to US-China strategic competition. In other words, security threats in multiple regions, rather than a single competitive axis, are simultaneously stimulating rearmament.

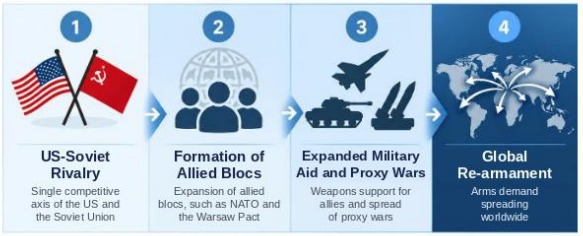
[Table 34] Comparison of the Global Rearmament Structure in the Past Cold War and Today

Category	Past US-Soviet Cold War	Current US-China Strategic Competition Period
Competitive Structure	Single competitive axis centered on the US and the Soviet Union	Multiple threat axes: US-China, Russia and the Middle East
Scope of Arms Race	US-Soviet competition spread to allies and neighboring countries	Unfolding simultaneously in the Indo-Pacific, Europe and the Middle East
Key Means of Force Buildup	Nuclear and conventional force buildup, weapons support to allies	Filling force gaps, replacing aging weapons, expanding holdings
Driver of Rearmament	Geopolitical tension between the US and the Soviet Union	Simultaneous expansion of regional security threats
Possibility of Threat Resolution	Tension could ease if US-Soviet relations improved	Threats persist in other regions even if tension eases in some regions
Key Characteristic	Global spread of a single competitive axis	Simultaneous rearmament driven by multiple competitive axes

Source: Hanwha Research Center

① Prolongation of Geopolitical Competition: The Cold War and Today, a Changed Global Rearmament Structure (2)

[Figure 56] The past Cold War: Global spread of a single competitive axis



Source: Hanwha Research Center

[Figure 57] The current US-China strategic competition period: Multiple regional threats acting simultaneously



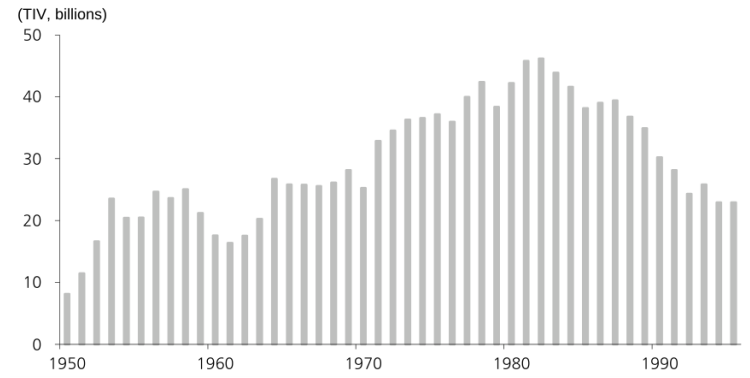
Source: Hanwha Research Center

Characteristics of Arms Trade in the US-Soviet Cold War: Long-Term Growth, Expansion of Demand Countries and Oligopoly of Supplier Countries (1)

- **Key characteristics of the US-Soviet Cold War period** Looking at global arms trade volume during the Cold War, it continued an uptrend for 32 years as US-Soviet competition intensified. Even as arms trade volume rose, there were periods of temporary decline. This suggests that even if geopolitical competition continues, weapons demand does not increase every year, and it is difficult to judge the end of a rearmament cycle from a short-term decline in trade volume alone.
- Another characteristic of Cold War weapons demand is that demand that had been concentrated in a few importing countries spread to many countries. In fact, from the early to the late Cold War, the share of the top 10 arms importers fell from 51.8% to 40.0%. The countries leading demand also changed from period to period. Even when weapons introduction in a particular country slowed, new demand arose in other countries, supporting the increase in global arms trade. We judge that Cold War weapons demand grew over a long period through the spread of demand and the turnover of leading countries rather than the continuous purchasing of specific countries.
- Weapons supply was concentrated in a few countries. During the Cold War, the global arms export market maintained an oligopolistic structure in which the top four countries held a high share. Although the number of arms-exporting countries grew toward the late Cold War, the increase was limited compared with importing countries. In other words, while demand spread to many countries, the supplier countries able to meet it remained few. It appears that an environment favorable for enjoying earnings benefits from expanded global weapons demand was created for the defense companies of the few countries that led weapons supply.

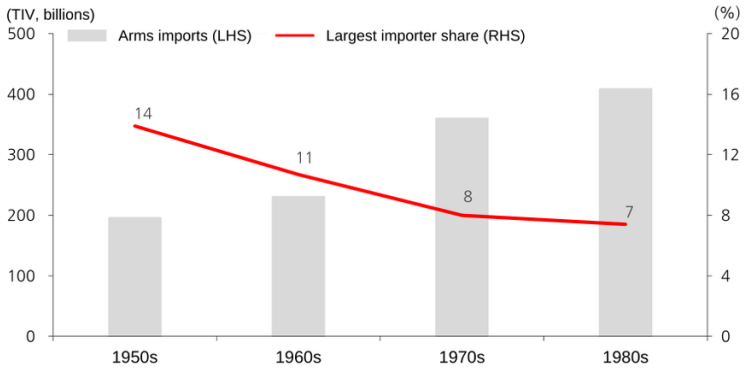
Characteristics of Arms Trade in the US-Soviet Cold War: Long-Term Growth, Expansion of Demand Countries and Oligopoly of Supplier Countries (2)

[Figure 58] Global arms trade volume during the US-Soviet Cold War (1950-1995)



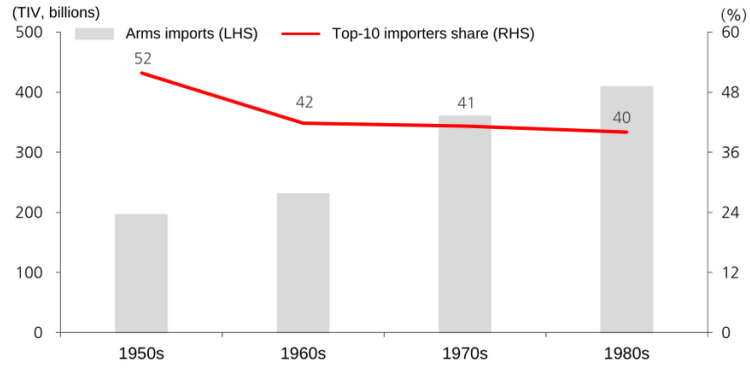
Source: SIPRI, Hanwha Research Center

[Figure 59] Share of the largest importing country during the US-Soviet Cold War



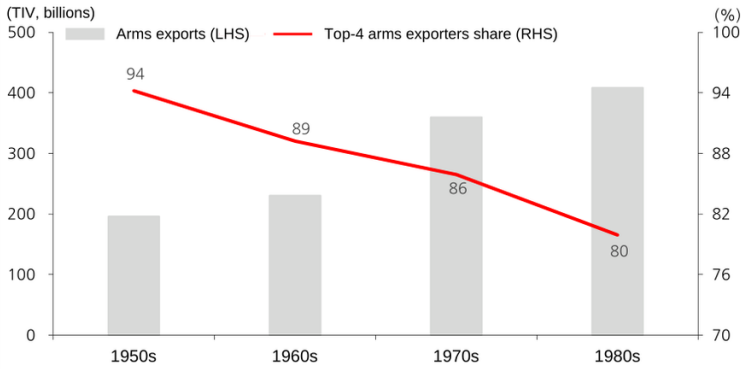
Source: SIPRI, Hanwha Research Center

[Figure 60] Share of the top 10 importing countries during the US-Soviet Cold War



Source: SIPRI, Hanwha Research Center

[Figure 61] Share of the top four arms exporting countries in global arms exports during the US-Soviet Cold War

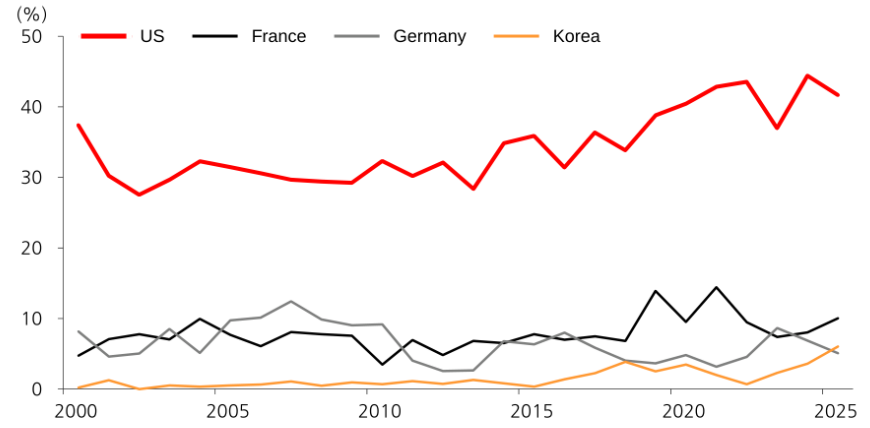


Source: SIPRI, Hanwha Research Center

US-China Strategic Competition Period: Long-Term Rearmament and Benefits for Large Defense Companies

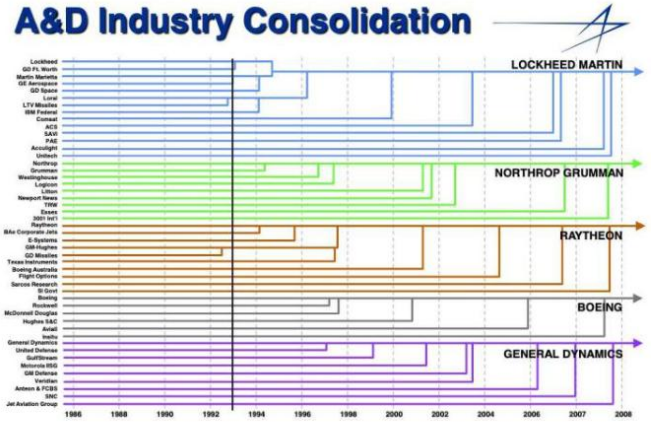
The current US-China strategic competition period The trend of arms trade seen through the US-Soviet Cold War carries significant implications for the current rearmament cycle. Considering this, global weapons demand in the current US-China strategic competition period is also likely to continue a long-term uptrend, repeating ups and downs depending on each country's order timing, rather than rising one-sidedly every year. If 2022-2025 weapons demand was concentrated in Ukraine and Poland, going forward, as force buildup in major countries including Europe gets into full swing, the number of demand countries is expected to expand gradually. We expect the countries leading weapons introduction to also differ from period to period. As in the past Cold War, weapons supply is likely to remain an oligopolistic structure centered on a few countries. As defense budgets were cut and M&A among defense companies proceeded after the end of the Cold War, the global defense market was restructured around a small number of large companies. We judge that global large defense companies, including those in Korea, are likely to enjoy earnings benefits from expanded weapons demand for a long time in a market that is more oligopolistic than during the Cold War.

[Figure 62] Share of major arms exporting countries



Source: SIPRI, Hanwha Research Center

[Figure 63] M&A and consolidation process of major US defense companies

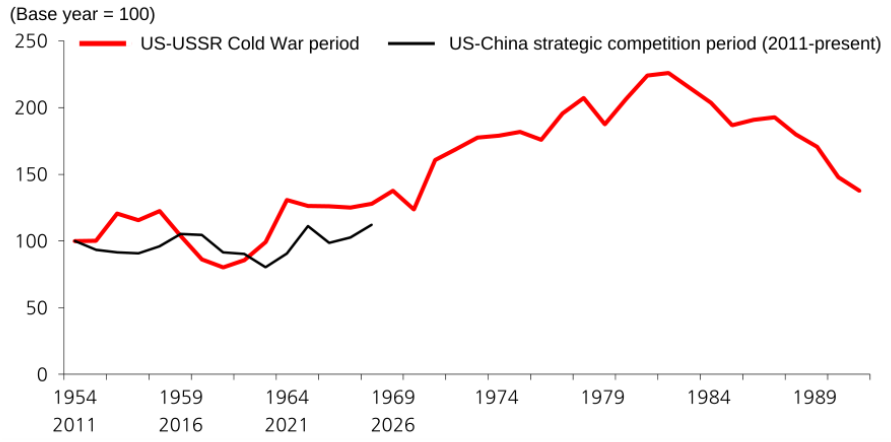


Source: PwC, Hanwha Research Center

The Global Rearmament Cycle Is Still at an Early Stage

- **The global rearmament cycle is still at an early stage** We judge that the global rearmament cycle of the current US-China strategic competition period is still at an early stage compared with the past US-Soviet Cold War. Figure 64 compares the trends of global arms trade volume in the two periods. For the US-Soviet Cold War, we set the analysis period as 1954-1991, excluding the temporary surge in arms trade volume due to the Korean War. For the US-China strategic competition period, we set it from 2011, when US strategic interest began to expand to the Asia-Pacific following the Obama administration's Pivot to Asia policy.
- During the past US-Soviet Cold War, global arms trade volume increased over a long period until the early 1980s despite short-term fluctuations. It subsequently turned to a decline, and with the dissolution of the Soviet Union in 1991, the strategic confrontation between the two countries also ended. There is no guarantee that the uptrend in arms trade volume of the past Cold War will be repeated identically today, but US-China strategic competition is still ongoing, and Russia's threat to Europe and geopolitical conflict in the Middle East are being added. Given that the structural factors triggering the arms race have not been resolved, we judge that the possibility of long-term growth in global weapons demand remains high.

[Figure 64] Comparison of global arms trade volume during the US-Soviet Cold War and the US-China strategic competition period



Note: To exclude the temporary surge in arms trade volume during the Korean War, the US-Soviet Cold War is analyzed from 1954. The US-Soviet Cold War is indexed with the 1954 arms trade volume and the US-China strategic competition period with the 2011 arms trade volume, each set to 100. The starting points of the two time series are aligned for comparison

Source: SIPRI, Hanwha Research Center

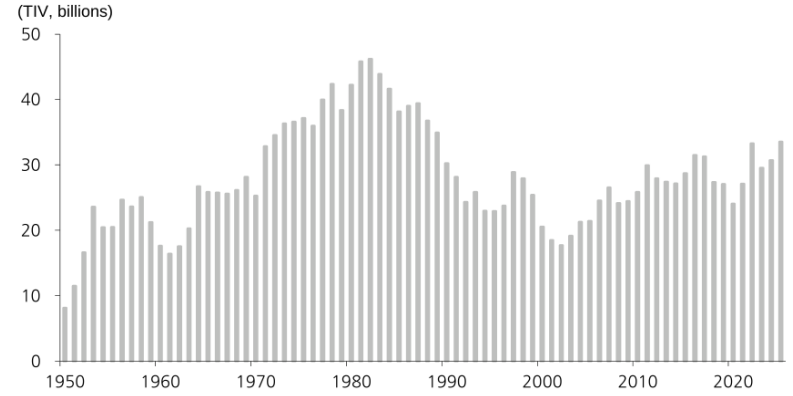
② Replacement of Aging Forces: Supporting Structural Force Rebuilding (1)

- **Weapons replacement is essential not only because of aging but also to respond to modern warfare** The aging of global weapon systems and the resulting replacement demand are also major factors supporting structural force rebuilding going forward. Major weapon systems are characterized by being operated for decades through maintenance and performance upgrades after introduction. Although there are differences by system, there are quite a few cases in which tanks, self-propelled howitzers and fighters are operated for 30-40 years or more after first introduction. In many countries, weapon systems introduced during the US-Soviet Cold War have been continuously upgraded and used as main forces. After the end of the Cold War, the approach of extending the service life of existing equipment expanded owing to defense budget cuts and the lower priority of introducing new weapons. Globally, the need to replace aging forces has accumulated over a long period.

- It should be noted that the aging of weapon systems is not simply a matter of years in service. In modern warfare, not only the performance of individual weapon systems but also real-time information sharing and organic linkage among surveillance and reconnaissance, command and control, and strike systems are becoming important. Aging weapon systems designed on the basis of past battlefield environments have structural limits in responding to such a network-centric operating environment. In the case of fighters, the ability to first detect enemy aircraft and ground threats, evade tracking by enemy radar and missiles, and engage with long-range precision-guided weapons is becoming important. Radar detection performance, electronic warfare response capability and ability to operate the latest armaments act as key factors determining the actual combat power of a fighter. Tanks and armored vehicles are also required to have, beyond simple firepower and armor protection, defense capability against drones and anti-tank missiles, information sharing with external surveillance and reconnaissance assets, and real-time target identification and response capability. Whereas in the past service life was extended by improving the engines or firepower systems of existing equipment, overall performance improvement is now required to respond to the changed battlefield environment. We judge that replacement of aging forces is not merely about maintaining existing forces, and that demand pressure to secure combat capabilities able to respond to modern warfare is expanding.

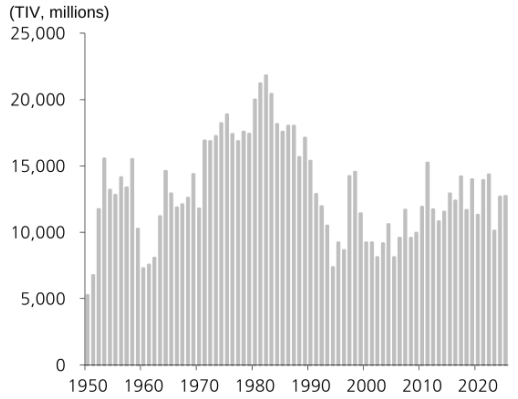
② Replacement of Aging Forces: Supporting Structural Force Rebuilding (2)

[Figure 65] Global arms trade volume trend (1950-2025)



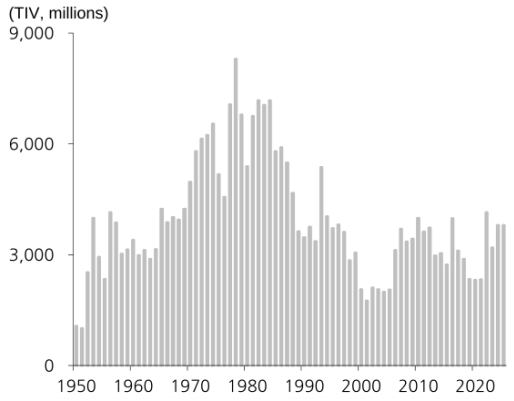
Source: SIPRI, Hanwha Research Center

[Figure 66] Global aircraft trade volume trend



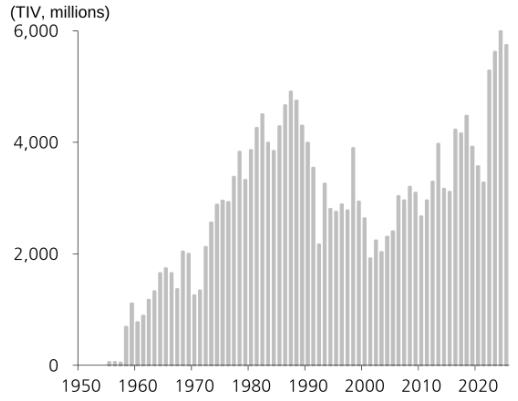
Source: Hanwha Research Center

[Figure 67] Global armored vehicle trade volume trend



Source: Hanwha Research Center

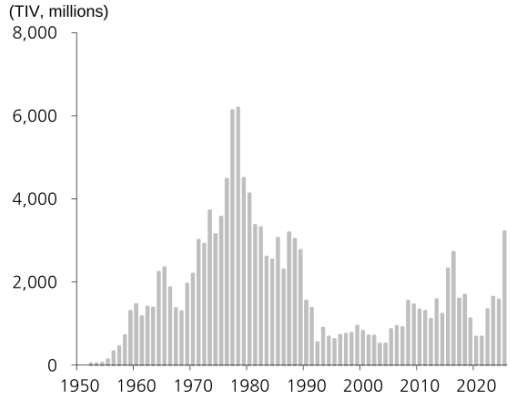
[Figure 68] Global missile trade volume trend



Source: Hanwha Research Center

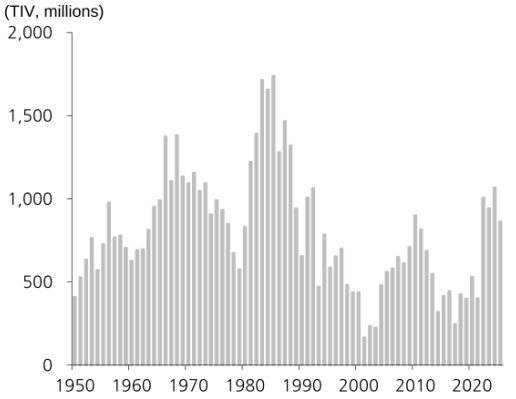
② Replacement of Aging Forces: Supporting Structural Force Rebuilding (3)

[Figure 69] Global air defense system trade volume trend



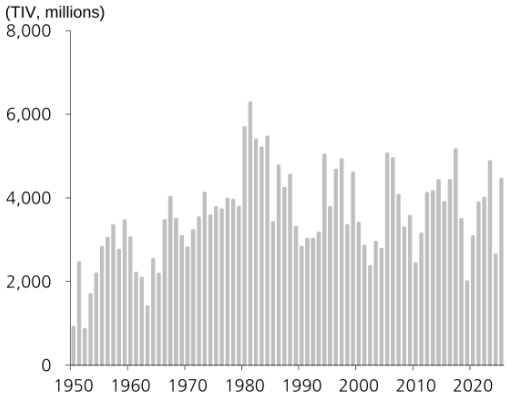
Source: Hanwha Research Center

[Figure 70] Global artillery system trade volume trend



Source: Hanwha Research Center

[Figure 71] Global naval vessel trade volume trend

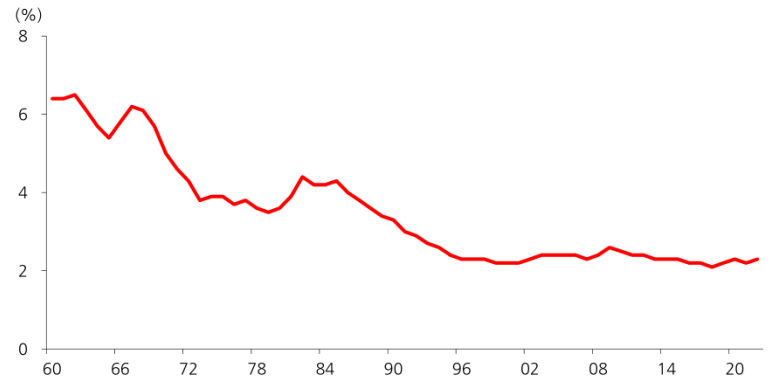


Source: Hanwha Research Center

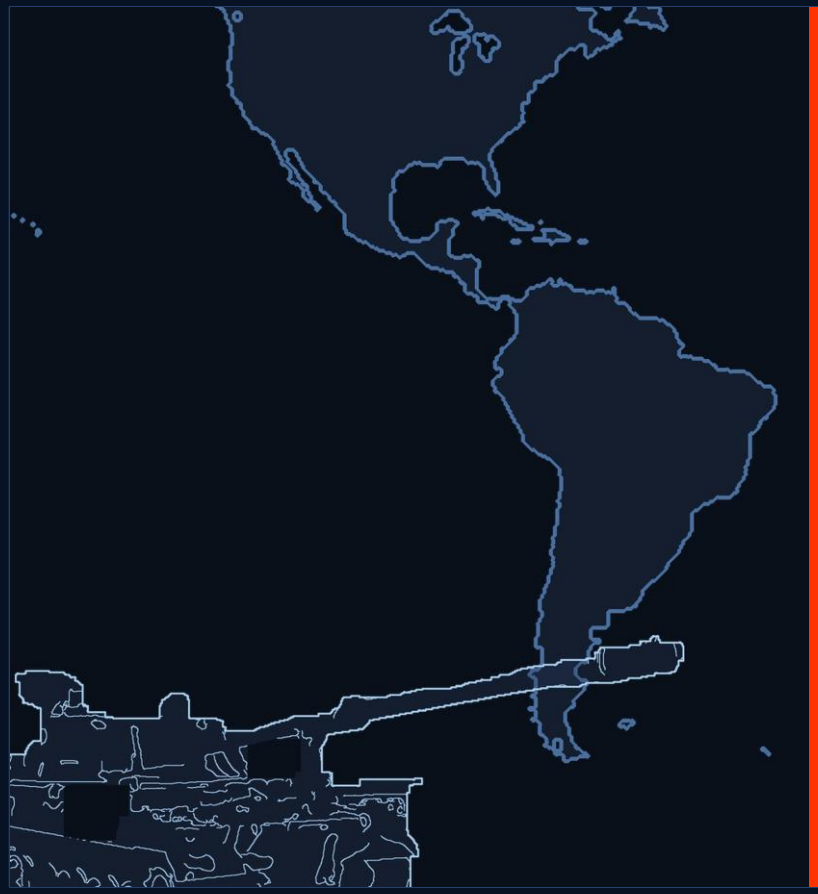
③ Defense Budget Expansion and Increased Capacity to Introduce New Weapons

- **New weapons introduction expected to increase with defense budget expansion** After the end of the Cold War, many Western countries cut defense budgets or restrained increases as military tension eased. As a large portion of the limited defense budget went to maintaining and operating existing forces and personnel costs, resources available for introducing new weapons were insufficient. In periods when the defense budget growth rate was low, extending the service life and upgrading the performance of existing equipment took priority over acquiring new forces. Even though the need to replace aging forces was steadily raised, actual procurement programs were often postponed or reduced. It was not that weapons demand was lacking, but that there was not enough budget to translate it into actual orders.
- In 2025, NATO agreed on a target of raising defense and security-related spending to 5% of GDP by 2035. Of this, core defense spending is at least 3.5% of GDP, a level well above the previous 2% target. If major countries implement this target going forward, it will be possible to allocate more budget to new weapons procurement while covering the maintenance and operating costs of existing forces. This means that force replacement and new force acquisition programs that had been postponed owing to budget shortages could translate into actual orders.
- The trend of defense budget expansion is not expected to be limited to NATO. Global military spending as a share of GDP in 2025 was 2.5%, still low compared with 5.6% in 1985 during the Cold War. Even though military spending has recently been rising, there remains room for further expansion compared with the past. If geopolitical uncertainty continues, moves to increase defense budgets are highly likely to spread not only in Europe but also in Asia, the Middle East and South America. Global defense budget expansion is expected to convert the weapons demand accumulated to date into actual orders and support medium- to long-term demand growth.

[Figure 72] Trend in global military spending as a share of GDP



Source: SIPRI, Hanwha Research Center



V

Weapons Supply-Demand Imbalance Expected to Persist Despite Supply Expansion

Expanding Defense Demand; Supply Capacity Is the Key · Artillery Systems · Air Defense Systems · Fighter Aircraft · Tanks

Defense Sector: Beyond Infinity

Supply Also Increased in Line With Increased Weapons Demand (1)

- **Need to check global weapons production increases** Since the 2022 Russia-Ukraine war, global defense companies have expanded production capacity in response to surging weapons demand. The expansion was not limited to specific weapon systems. Capacity expansion has proceeded in most weapon systems, from land weapons to air defense systems, missiles and fighters. New production facilities were built or existing lines were expanded, and investment continued to rebuild supply chains that had been scaled back for a long time. 2026 is the fifth year of expansion since the outbreak of the Russia-Ukraine war. It is necessary to check not only the increase in weapons demand but also how much global weapons supply capacity has actually expanded over the past five years. If supply capacity grows faster than demand, the current supplier-favorable environment could weaken.
- Production capacity is also an important variable for the medium- to long-term earnings of defense companies. When supply begins to exceed demand, competition among companies to secure orders inevitably intensifies. This could lead to price competition and lower profitability, and over the long term could weigh on revenue growth. To judge the future earnings of defense companies, it is necessary to compare how much weapons demand increases with how fast the production capacity to supply it is expanding.

Supply Also Increased in Line With Increased Weapons Demand (2)

[Table 35] Production Capacity and Expansion Status of Weapon Systems Competing with Korean Defense

Category	Weapon System	Annual CAPA Before the Russia-Ukraine War	Current Annual CAPA	Target CAPA	Remarks
Self-Propelled Howitzers	M109A7	60 guns (est.)	60 guns (est.)	Not disclosed	Low likelihood of expansion
	PzH2000	5 guns (est.)	10-15 guns (est.)	Not disclosed	Low likelihood of expansion
	RCH155	0 guns (before mass production)	25-35 guns (est.)	Expanding	Figures undisclosed
	CAESAR	24 guns	120 guns	144 guns	Low likelihood of further expansion
	ARCHER	10-15 guns (est.)	15 guns (est.)	Expanding	Low likelihood of further expansion
Multiple Launch Rocket Systems	HIMARS	48 units	96 units	Expansion completed	High likelihood of expansion
	PULS	Unknown	Unknown	Expanding (est.)	High likelihood of further expansion
Air Defense	Patriot	Unknown	12 batteries	Expanding	Figures undisclosed
	SAMP/T	1-3 batteries (est.)	4-6 batteries (est.)	Expanding	Figures undisclosed
	IRIS-T	Unknown	10 batteries	16 batteries	Long-term target of increasing production to 30 batteries
	NASAMS	1-3 batteries (est.)	3-5 batteries (est.)	Not disclosed	High likelihood of expansion
Fighters	F-16	Mass production started after plant relocation	20-30 units (est.)	48 units	Limited likelihood of further expansion
	F-15	10-15 units (est.)	12-17 units (est.)	24 units	Possibility of further expansion exists
	Rafale	12-20 units (est.)	20-30 units (est.)	48 units	Possibility of further expansion exists
	Eurofighter	10-12 units (est.)	12-14 units (est.)	20-30 units	Limited likelihood of further expansion
	Gripen	6-10 units (est.)	15 units (est.)	20-30 units	Limited likelihood of further expansion
Tanks	Leopard 2	Unknown	20-30 units (est.)	40-50 units (est.)	Low likelihood of further expansion
	M1A2	180 units (est.)	180 units (est.)	Expanding	Investing in facility modernization (est.)

Source: Companies, press reports, Hanwha Research Center

European Defense Weapons Supply Expansion: Whether It Is Absorbed Within the Region Is the Key (1)

- **European defense weapons production increases: whether they can be absorbed within the region is the key** European defense companies are rapidly expanding production capacity. Although part of the purpose of the expansion is to target global export markets, it is primarily aimed at responding to rearmament demand within Europe. The expansion of production capacity by European defense companies in itself is not a threat to Korean defense companies.
- What matters is whether the increase in European weapons supply exceeds intra-regional weapons demand. If the increasing weapons demand in the process of European rearmament sufficiently absorbs the expanded production capacity, the impact on the global arms market will be limited. Conversely, if the production capacity of European defense companies grows faster than intra-regional weapons demand, the situation changes. Weapons that cannot be absorbed within Europe will ultimately head to export markets outside the region. Competition for orders among defense companies is also likely to intensify.
- Korean defense companies could also be directly affected. This is because many of the weapon systems produced by European defense companies compete with Korea's main export products. Competing products exist not only in land weapons such as self-propelled howitzers, tanks and armored vehicles, but also in air defense systems and fighters. This is expected to become an important criterion for gauging the order visibility of Korean defense companies over the medium to long term.
- The US also needs to be examined from the same perspective. This is because if the production capacity of US defense companies exceeds demand from the US itself and existing customers, it could lead to an increase in global supply. However, we judge that US weapon systems have relatively limited areas of direct competition with Korea's main export products compared with European weapon systems.

European Defense Weapons Supply Expansion: Whether It Is Absorbed Within the Region Is the Key (2)

[Table 36] Key Competing Weapon Systems of Korea and Europe

Category	Korea	European Competing Products	Key Companies	Capacity Direction
Self-Propelled Howitzers	K9	CAESAR	KNDS France	Expanding
		RCH 155	KNDS Deutschland	Expanding
		Archer	BAE Systems	Expanding
		PzH 2000	KNDS Deutschland	Unknown
Multiple Launch Rocket Systems	Chunmoo	PULS	Elbit Systems	Expanding
		EuroPULS	KNDS / Elbit	Expansion planned
Tanks	K2	Leopard 2	KNDS / Rheinmetall	Expanding
Air Defense	Cheongung II	SAMP/T NG	MBDA	Expanding
		IRIS-T SLM	Diehl Defence	Expanding
Fighters	KF-21	Rafale	Dassault Aviation	Expanding
		Eurofighter	Airbus / BAE / Leonardo	Expanding
		Gripen	Saab	Expanding

Source: Companies, Hanwha Research Center

Existing Weapon Systems Are as Important as Future Growth Drivers (1)

- **From a short- to medium-term earnings perspective, exports of existing weapon systems are important** Domestic and overseas defense companies are currently preparing new sources of revenue for growth in the 2030s and 2040s. The combination of AI and defense, space and unmanned systems, and low-cost weapon systems are representative examples. Analysis of future growth drivers is important for gauging a company's long-term business direction and growth potential. This is because investors can also judge which businesses will lead company growth going forward.
- As much as future growth drivers, analysis of existing weapon systems that will be sold as main products through the late 2020s is also important. This is because considerable time is needed before new businesses translate into full-fledged revenue. Until then, it is the existing products that determine the earnings and order performance of defense companies. For Korean defense companies, we expect fighters, tanks, armored vehicles, self-propelled howitzers and air defense systems to drive earnings over the next several years. It is necessary to check the supply capacity of each weapon system to analyze the future supply-demand environment.

[Figure 73] Future growth drivers drawing attention in the defense industry



Source: Hanwha Research Center

Existing Weapon Systems Are as Important as Future Growth Drivers (2)

[Table 37] Existing Weapon Systems Drive Earnings Growth Until Future Businesses Generate Revenue

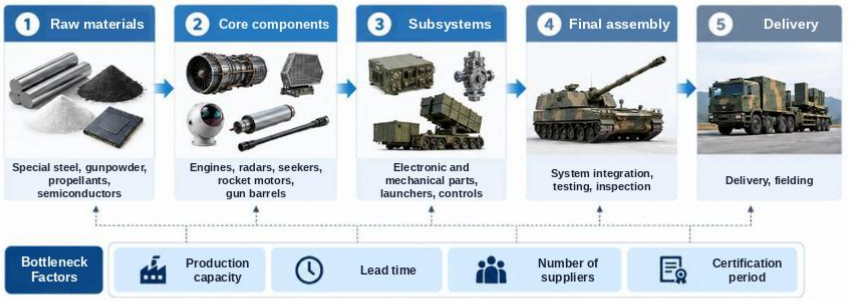
Category	Key Business	Current Stage	Full-Scale Revenue Timing	Company-Wide Earnings Contribution in the Late 2020s
Existing Weapon Systems	Tanks	Mass production, exports	Present	High
	Self-Propelled Howitzers	Mass production, exports	Present	High
	Multiple Launch Rocket Systems	Mass production, exports	Present	High
	Air Defense Systems	Mass production, exports	Present	High
	Fighters	Mass production	Present	High
Future Growth Drivers	Unmanned Systems	Development, initial mass production	2030s	Limited
	AI Application	Development	2030s	Limited
	Space	Development, market formation	2030s	Limited
	Low-Cost Systems	Development, initial mass production	Late 2020s	Gradual

Source: Hanwha Research Center

Gauging Supply Expansion Through the Capacity Expansion Plans of System Integrators

- **Weapons supply outlook through system integrators' expansion plans** It is difficult to judge overall production capacity from the final assembly capacity of a system-integrating defense company alone. This is because a single weapon system consists of hundreds to thousands of components, and a finished product cannot be produced if even one key component is not supplied. For actual weapons production capacity to expand, the production capacity of the lower-tier component supply chain must increase together with the final assembly line.
- There are limits to analyzing the lower-tier supply chain. This is because information disclosure in the defense industry is limited. While the output and target production capacity of system integrators are relatively easy to confirm, it is difficult to determine the production capacity, utilization rate, expansion scale and completion timing of component suppliers. The production capacity of key components is often not disclosed, which limits estimating when target production capacity will actually be achieved.
- Although it is difficult to pin down the exact completion timing of production increases, the direction and scale of capacity expansion can be gauged. This is because the current production capacity, expansion targets, investment scale and completion timing presented by system integrators can be relatively confirmed. It is also possible to gauge how far future weapons supply capacity can expand and from when supply shortages could ease. Below, we review the scale and timing of supply capacity expansion by weapon system through the production plans of major system integrators.

[Figure 74] Defense supply chain structure that determines weapon system production capacity



Source: Hanwha Research Center

Self-Propelled Howitzers and Multiple Launch Rocket Systems: We Judge Oversupply Concerns Are Very Limited

- **Tracked self-propelled howitzers: very limited capacity expansion burden** [Wheeled self-propelled howitzers: some burden, but demand is sufficient] [Multiple launch rocket systems: long lead times persist despite the burden] Although global supply capacity for self-propelled howitzers and multiple launch rocket systems is expanding, we judge it will be difficult for supply conditions to ease quickly. Since the Russia-Ukraine war, European force rebuilding and support demand for Ukraine have risen simultaneously, so defense companies have rapidly expanded production capacity. For self-propelled howitzers, production capacity has increased significantly centered on CAESAR and ARCHER, and mass production of RCH 155 is also expected to expand in line with rising new orders. For multiple launch rocket systems, HIMARS annual production capacity doubled from 48 to 96 units, and PULS is also pursuing expanded local production in Europe. For EuroPULS, which could become a variable going forward, securing large-scale mass production capability has not yet become visible. As supply constraints were so severe, long lead times will persist even if additional expansion proceeds.

- An environment in which artillery system supply constraints persist is positive for Korean defense companies, which have already secured high market shares. In self-propelled howitzers, the K9 accounts for about half of the global export market and has established a firm position, particularly in the tracked segment. An expansion of the product range into wheeled self-propelled howitzers, where its presence has been relatively limited, is expected going forward. For multiple launch rocket systems, suppliers are even more limited than for self-propelled howitzers. The systems capable of large-scale exports in the Western bloc are effectively narrowed down to the US HIMARS, Korea's Chunmoo and Israel's PULS. HIMARS has abundant potential demand despite increased production. EuroPULS is expected to need time to expand production capacity. The global multiple launch rocket system market is likely to evolve in the direction of Chunmoo and PULS gradually eroding the existing HIMARS-centered market. Considering the K9's high share secured in tracked self-propelled howitzers and the expansion of Chunmoo's global customer base, we expect the benefits to Korean defense to continue over the medium to long term.

[Table 38] Global Artillery System Expansion Status and CAPA Outlook

Category	Annual Production Capacity	CAPA Expansion Plan	Outlook
Self-Propelled Howitzers: M109A7	60 guns (est.)	Not disclosed	Very low likelihood of CAPA expansion
PzH 2000	10-15 guns (est.)	Not disclosed	Low likelihood of CAPA expansion
CAESAR	120 guns	144 guns	Low likelihood of additional CAPA expansion
RCH 155	25-35 guns (est.)	Expanding	Very high likelihood of additional CAPA expansion
ARCHER	15 guns (est.)	Expanding	Very low likelihood of additional CAPA expansion
Multiple Launch Rocket Systems: HIMARS	96 units	Not disclosed	High likelihood of CAPA expansion
PULS	Unknown	Expanding (est.)	Very high likelihood of additional CAPA expansion
THUNDART	In development	In development	Mass production targeted around 2030. Limited CAPA
X-FIRE	In development	In development	Whether development continues is key

Source: Hanwha Research Center

① Self-Propelled Howitzers: Supply Capacity Expansion Concentrated in Wheeled Types (1)

- **Self-propelled howitzer production increases are proceeding centered on wheeled types** Self-propelled howitzers are broadly divided into tracked and wheeled types by drive method. Tracked types mount a turret on a continuous-track hull similar to a tank. They have high mobility on soft ground and rough terrain and excellent armor protection. Because they are heavy, a separate means of transport is often needed for long-distance movement. The maintenance burden is also relatively large. Wheeled types mount the gun on a truck or armored vehicle wheeled platform. Their mobility on rough terrain and protection are lower than tracked types, but they can move quickly over long distances on roads. They also have the advantage of relatively low acquisition prices and maintenance costs.
- In Europe's new self-propelled howitzer procurement since the Russia-Ukraine war, the share of wheeled types is rapidly rising. This is because the widespread use of drones and counter-battery radar has changed how artillery survives. As the time between an artillery position being exposed and counterattack has shortened, the "shoot-and-scoot" ability to quickly leave the position immediately after firing has become important. Changes in the European operating environment are also affecting the expansion of demand for wheeled types. During the Cold War, Western European countries assumed large-scale armored warfare in their own or adjacent territory. The value of tracked self-propelled howitzers, which can maneuver across rough terrain together with tanks and armored vehicles, was high. Today NATO's defense line has extended to Eastern Europe, and units located in Western and Central Europe must move hundreds to thousands of kilometers to the eastern front in a contingency. This is why the strategic mobility of wheeled types has become more important than before. Countries close to Russia that must operate with armored units on soft ground and unpaved areas, such as in Eastern and Northern Europe, still regard the high rough-terrain mobility of tracked types as important.
- The expansion of European self-propelled howitzer supply capacity is concentrated in the wheeled CAESAR, RCH 155 and ARCHER. The production capacity of Germany's tracked PzH 2000 is still estimated at around 5-10 guns per year, and no large-scale expansion plan has been confirmed. For the US M109A7 Paladin as well, although the existing production line is maintained, the possibility of long-term large-scale CAPA expansion is declining. Although the self-propelled howitzer supply capacity of European defense companies is expanding since the Russia-Ukraine war, most of the increase is concentrated in wheeled types.

① Self-Propelled Howitzers: Supply Capacity Expansion Concentrated in Wheeled Types (2)

[Table 39] Comparison of Tracked and Wheeled Self-Propelled Howitzers

Category	Tracked Self-Propelled Howitzers	Wheeled Self-Propelled Howitzers
Hull	Continuous track	Wheeled truck, wheeled armored vehicle
Rough-Terrain Mobility	High	Relatively low
Road Mobility	Relatively low	High
Protection	Generally high	Relatively low
Weight	High	Relatively low
Operating Cost	High	Relatively low
Operational Concept	Accompanies mechanized units	Rapid, long-distance deployment
Ease of Transport	Relatively low	Relatively high
Maintenance Burden	High	Relatively low
Road Network Dependence	Low	High
Long-Distance Self-Propelled Movement	Limited	Excellent
Acquisition Price	Generally high	Generally low

Source: Hanwha Research Center

Tracked Self-Propelled Howitzers: Additional Capacity Expansion of K9 Competing Systems Expected to Be Limited (1)

▪ **Competing tracked self-propelled howitzers: M109, PzH 2000** The major tracked self-propelled howitzer systems are Hanwha Aerospace's K9, BAE Systems' M109A7 Paladin and KNDS Deutschland's PzH 2000. The system with the broadest customer base and mass production setup in the global export market is the K9. We judge that the possibility of a significant increase in the production capacity of the Paladin and PzH 2000 is currently limited.

[Table 40] Comparison of Major Global Tracked Self-Propelled Howitzers

Category	K9A1	M109A7	PzH 2000
Developer / Manufacturer	Korea / Hanwha Aerospace	US / BAE Systems (US)	Germany / KNDS Deutschland
Barrel	155mm / 52 calibers	155mm / 39 calibers	155mm / 52 calibers
Range	40km+ / extended-range projectile 50km+	22km / extended-range projectile 30km	40km / extended-range projectile 54km
Crew / Loading Method	4-5 persons / semi-automatic	4 persons / manual	3-5 persons / automatic projectile loading
Weight	47t	38-39t	57t
Market Position	Leader in the global tracked export market	Operated mainly in the US	Operated mainly in Europe

Note: Range varies depending on ammunition type
Source: Hanwha Research Center

Tracked Self-Propelled Howitzers: Additional Capacity Expansion of K9 Competing Systems Expected to Be Limited (2)

[Figure 75] New self-propelled howitzer procurement map by European country: tracked (red) vs wheeled (blue)



Source: Hanwha Research Center

BAE Systems (M109A7): Limited Possibility of Additional Expansion. Attention on the Direction of the M109-52 (1)

- **The possibility of increased M109A7 production is slim** The M109A7 Paladin is centered on US Army procurement. Production is not ending immediately. BAE Systems secured a multi-year contract of about USD 970 million in 2025 and also won an additional mass-production contract worth USD 540 million in 1H26. The existing US production line is expected to operate for the time being. Current annual production capacity is 60 units, but realistic output is estimated at 40-50 units per year.
- The possibility that M109A7 production capacity will expand significantly from now is low. This is because the US Army has begun to review a next-generation tracked self-propelled howitzer rather than procuring additional existing Paladins in large quantities. Last August, the US Army began reviewing candidates for a next-generation tracked self-propelled howitzer through the MTC-Tracked (MTC-T) program. The existing Paladin production line is likely to be maintained centered on the volume already secured, and we judge that the possibility of supply capacity expanding enough to threaten the K9 in the global export market is limited.
- The competing system to watch over the medium to long term is the M109-52 being developed by BAE Systems. The M109-52 integrates Rheinmetall's 155mm/L52 gun on the M109A7 hull. The key is to expand range and firepower by replacing the existing 39-caliber gun with a 52-caliber one while utilizing the proven M109 platform and the US production base. It has the advantage of lowering development risk and production conversion burden compared with developing an entirely new self-propelled howitzer.
- The M109-52 has not yet entered the mass production stage. BAE Systems disclosed that as of 2025 it plans to further mature the design over the next several years. The M109-52 is appropriately viewed not as a mass-production system currently competing with the K9 in the global market but as a candidate system likely to compete in the future US Army MTC-T program. Compared with the K9, its strength is that it can leverage the existing M109 production base, but the K9 has already secured experience in large-scale mass production and exports. While there is a possibility that the M109-52 becomes a strong competing system to the K9 in the MTC-T, we judge that the possibility of it significantly expanding global tracked self-propelled howitzer supply capacity in the short term is limited.

BAE Systems (M109A7): Limited Possibility of Additional Expansion. Attention on the Direction of the M109-52 (2)

[Table 41] M109A7 Procurement Plan by Year in the US President's Budget (Unit: guns)

Budget Year	FY20	FY21	FY22	FY23	FY24	FY25	FY26	FY27	FY28	FY29	Total
2020	53	37	35	43	45						213
2021		30	44	43	45	45					207
2022			25	0	0	0	0				25
2023				27	27	25	46	45			170
2024					24	22	37	40	39		162
2025						20	25	19	20	21	105
2026							10	0	0	0	10
2027								0	0	0	0

Note: Based on the procurement plan at the time each year's President's Budget was submitted. For 2027, the procurement quantity over the next five years is 0, but a planned procurement volume of 127 guns still remains
Source: U.S. Department of the Army, President's Budget Justification Books (FY2020~FY2027), Hanwha Research Center

[Table 42] Comparison of Key Specifications of the M109A7 and M109-52

Category	M109A7	M109-52	Key Changes
Manufacturer	BAE Systems	BAE Systems	Same
Caliber	155mm	155mm	Same
Barrel Length	39 calibers	52 calibers	33% increase
Main Gun	M284	Rheinmetall L52	Main gun replaced
Standard Projectile Range	22km	30km	36% increase
Rocket-Assisted Projectile Range	30km	60km	100% increase
Maximum Rate of Fire	4 rounds/min	Undisclosed	To be confirmed later
Crew	4 persons	Undisclosed	To be confirmed later
Hull	M109A7	M109A7	Existing hull utilized
Development Stage	In operational use	Development/testing	M109-52 is under development

Note: For 2022, the expected procurement quantities for future years were not reflected in the budget documents
Source: U.S. Department of the Army, President's Budget Justification Books (FY2020~FY2027), Hanwha Research Center

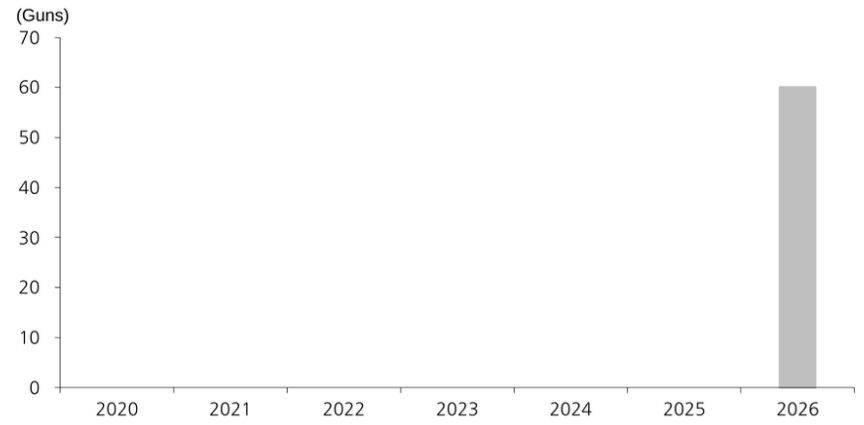
BAE Systems (M109A7): Limited Possibility of Additional Expansion. Attention on the Direction of the M109-52 (3)

[Table 43] US Army MTC-T Program Schedule

Category	Key Milestone	Details
August 2026	Request for Information (RFI) released	August 31, tracked self-propelled howitzer market research
September 2026	Industry briefing	September 8, program requirements and schedule disclosed
FY2027 1Q	Solicitation released	Request for solutions proposals from each company
FY2027 3Q	Target for prototype contract award	OTA-type contract pursued
FY2027 4Q	Target for initial prototype delivery	First prototype delivered within 90 days of contract
FY2027 ~ FY2029	Prototype testing/evaluation	Delivery of at least 5 units; ammunition/fire control network integration testing
FY2029 1Q	Army operational test demonstration	Operational testing using actual troops
FY2029 2Q	Target for production contract award	Review of transition to mass production based on test results

Source: U.S. Army, MTC-T Industry Engagement, SAM.gov, Hanwha Research Center

[Figure 76] Trend in new M109A7 orders excluding the US



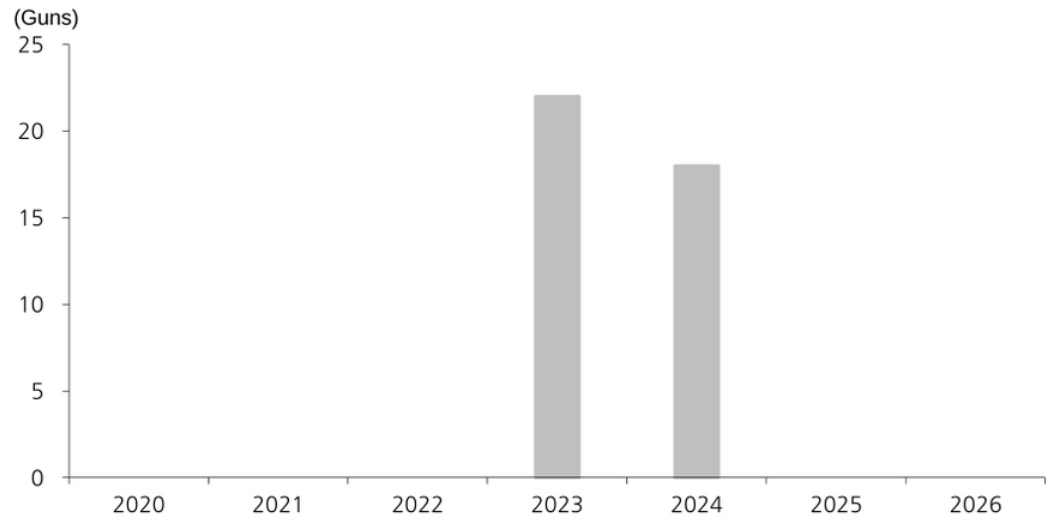
Source: BAE Systems, Hanwha Research Center

KNDS Deutschland (PzH 2000): Limited Possibility of Expansion. Expansion Focused on RCH 155 (1)

- **Possibility of PzH 2000 production increase also appears low** As with the M109A7, we judge that the possibility of expanding PzH 2000 supply capacity is not large. For the PzH 2000, KNDS Deutschland is responsible for system integration, and Rheinmetall supplies major components such as the L52 main gun and hull. KNDS does not disclose the official annual production capacity of the PzH 2000. Working backward from recent contracts and delivery schedules, current production is estimated at around 10 guns per year. Compared with past delivery pace as well, there is limited evidence to suggest that production capacity has expanded significantly.

- More important is that KNDS Deutschland's new capacity expansion investment is concentrated in the RCH 155 rather than the PzH 2000. The German Army has adopted the RCH 155 as the main self-propelled howitzer of the medium forces it is newly establishing. A considerable portion of future new self-propelled howitzer forces is judged likely to consist of the RCH 155. No large-scale new volume sufficient to significantly expand PzH 2000 production capacity has been confirmed at present. Even though European self-propelled howitzer demand has increased significantly since the Russia-Ukraine war, CAPA expansion of the PzH 2000, the representative tracked type, has been limited. We judge there are currently few competing systems that directly threaten the K9's supply capacity in the global tracked self-propelled howitzer market.

[Figure 77] Trend in new PzH 2000 orders



Source: KNDS Deutschland, Hanwha Research Center

KNDS Deutschland (PzH 2000): Limited Possibility of Expansion. Expansion Focused on RCH 155 (2)

[Table 44] Delivery Schedule of New PzH 2000 Orders Since 2020 and Estimated Annual Production Capacity

Category	Contract Date	Quantity	Delivery	Remarks
German Army main contract	2023.03	10 guns	2025~2026	
German Army option exercised	2023.05	12 guns	2027~2029	
Ukraine contract	2024.02	18 guns	2026~2027	
Total		**40 guns**		**Estimated annual CAPA of about 10-15 guns**

Source: KNDS Deutschland, Hanwha Research Center

Wheeled Self-Propelled Howitzers: The Center of CAPA Expansion Since the Russia-Ukraine War

- **Competing wheeled self-propelled howitzers: CAESAR, RCH155, ARCHER** Since the Russia-Ukraine war, the expansion of self-propelled howitzer production capacity by global defense companies has proceeded centered on wheeled types. The major systems are KNDS France's CAESAR, KNDS Deutschland's RCH 155 and BAE Systems' ARCHER.
- The European wheeled self-propelled howitzer market is being reorganized around the RCH 155 and CAESAR. The strengths of the two systems differ. The strengths of the RCH 155 are its high level of automation and commonality with the BOXER hull platform. Germany and the UK, which operate BOXER hulls on a large scale, can utilize existing maintenance systems, parts and training infrastructure, which favors adoption of the RCH 155. The strengths of the CAESAR are its relatively simple structure and low acquisition and operating costs. The structure is such that countries that value high automation and integration with existing BOXER forces choose the RCH 155, while countries seeking to expand wheeled artillery forces at relatively low cost choose the CAESAR.
- All three systems are expanding production capacity, but the stage of expansion differs. CAESAR has already completed a considerable portion of its large-scale expansion, and the RCH 155 has now entered the full-fledged mass-production expansion stage. For ARCHER as well, expansion of production facilities and the supply chain is under way.

[Table 45] Comparison of Major Global Wheeled Self-Propelled Howitzers

Category	K9MH	RCH 155	CAESAR
Developer / Manufacturer	Korea / Hanwha Aerospace	Germany / KNDS Deutschland	France / KNDS France
Maximum Road Speed	100km/h	100km/h	80km/h+
Weight	Approx. 40t	39t or less	27t or less
Barrel	155mm / 52 calibers	155mm / 52 calibers	155mm / 52 calibers
Range	40km / extended-range projectile 60km+	40km / extended-range projectile 54km	40km / extended-range projectile 55km+
Crew / Loading Method	3 persons / fully automatic	2 persons / fully automatic	3-5 persons / semi-automatic
Market Position	Stage of entering the global market	Capturing next-generation European demand	Leader in the European wheeled export market

Note: CAESAR specifications are based on the latest Mk2
Source: Companies, press reports, Hanwha Research Center

KNDS France (CAESAR): Expansion Largely Complete. Limited Possibility of Additional Expansion (1)

- **CAESAR production increase mostly complete; possibility of further increase appears low** The first wheeled self-propelled howitzer to expand CAPA was KNDS France's CAESAR. Before the Russia-Ukraine war, the production rate was about 2 guns per month. It then raised production capacity to 6 guns per month in 2023 and 10 guns per month in 2026. Converted to an annual basis, this is about 120 guns. Compared with before the Russia-Ukraine war, supply capacity has increased about fivefold. We estimate that it holds about 2.6 years' worth of workload.

- A characteristic is that KNDS France moved to expand capacity preemptively, departing from the previous approach of expanding production capacity in line with new orders. It even introduced a method of providing long-term orders to the supply chain and producing before customer orders are confirmed. The supply period, which previously required several years, has been greatly shortened. Since the Russia-Ukraine war, the CAESAR has had its production system expanded the fastest among European wheeled self-propelled howitzers, and it is understood that a considerable portion of the targeted CAPA has now been completed. Unless additional large-scale orders follow, we judge there is little need to expand significantly again from the current CAPA. Rather than additional CAPA expansion, the key is how high a utilization rate can be maintained on the production capacity already secured.

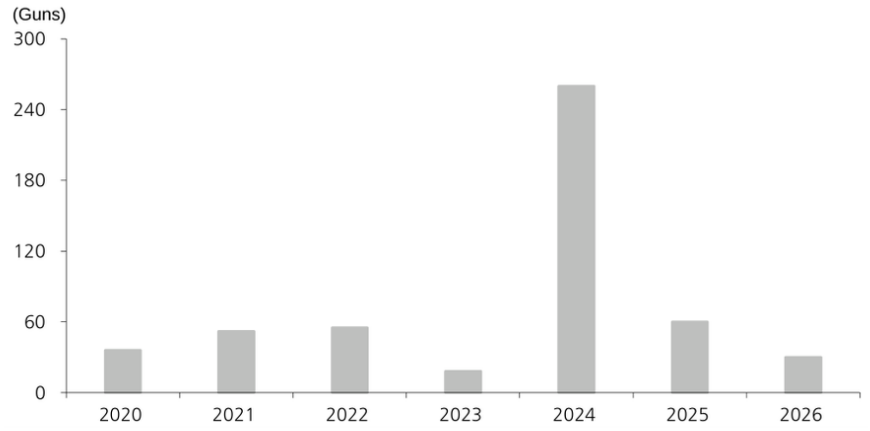
[Table 46] CAESAR Development and Mass Production History

Year	Key Details
1992	Development of CAESAR begins
1994	First prototype system unveiled
2000	France places advance order for 5 guns
2004	France places mass-production order for 72 guns
2006	Saudi Arabia orders 76 guns, Thailand orders 6 guns. Exports begin
2008	Deployment to the French military begins
2012	Indonesia orders 37 guns
2017	Indonesia orders 18 guns. Denmark orders 15 guns
2019	Denmark orders 4 guns
2020	Morocco orders 36 guns
2021	Czech Republic orders 52 guns. Development of CAESAR MKII begins
2022	France orders 18 guns, Czech Republic 10 guns, Belgium 9 guns, Lithuania 18 guns
2023	Ukraine orders 18 guns
2024	France orders 109 CAESAR MKII guns; Armenia orders 36 guns, Estonia 12 guns, Belgium 19 guns
2025	Slovenia orders 12 guns, Croatia 18 guns, Lithuania 30 guns
2026	Estonia orders 12 guns, Malaysia 18 guns

Source: KNDS France, Hanwha Research Center

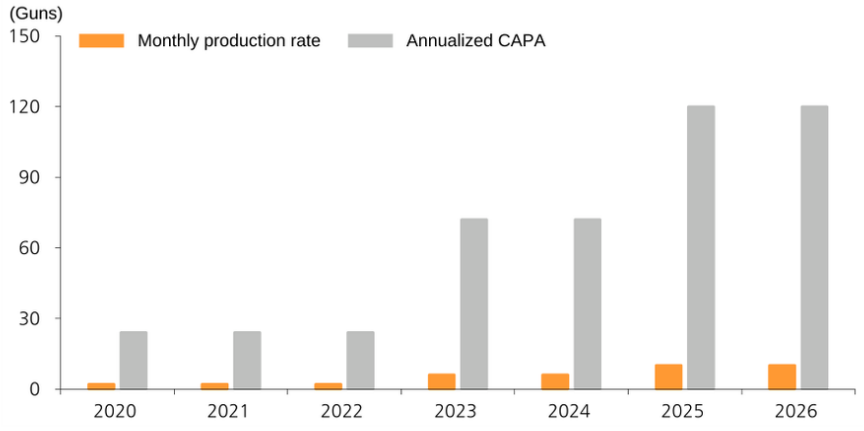
KNDS France (CAESAR): Expansion Largely Complete. Limited Possibility of Additional Expansion (2)

[Figure 78] Trend in new CAESAR orders



Source: KNDS France, Hanwha Research Center

[Figure 79] Trend in CAESAR annual production rate



Source: KNDS France, Hanwha Research Center

KNDS Deutschland (RCH 155): Full-Scale Mass Production Starting Now. High Likelihood of Expansion (1)

- **RCH155 production increase to get under way; potential demand is very large** The RCH 155 is about to undergo full-fledged production capacity expansion. When the Russia-Ukraine war broke out in 2022, the RCH 155 was not a weapon system with a large-scale mass production setup. Beginning with Ukraine's first order of 18 guns in 2022, additional orders followed and the volume for Ukraine expanded to a total of 54 guns. In 2025, 84 guns from Germany and 72 from the UK were added, securing volume sufficient to build a large-scale mass production base. Confirmed volume from the three countries currently totals 210 guns.
- KNDS does not officially disclose the annual production capacity of the RCH 155. Considering the current order volume and delivery periods, it is estimated that securing production capacity of at least 40 guns per year will be necessary going forward. Considering that the German and UK production lines are being expanded at the same time and that BOXER hull production capacity is also increasing, there is ample possibility that RCH 155 production capacity will rise further over the medium to long term. As the first confirmed volume under Germany's 500-gun framework is only 84 guns, the possibility of additional orders also remains open.
- We judge it is unlikely that an expansion of RCH 155 production capacity will lead to oversupply in the wheeled self-propelled howitzer market in the short term. Supply of the 72 guns for the UK will continue from 2028 to 2032. The German and Ukrainian volumes must be produced at the same time. Although the RCH 155 is expanding production capacity, its existing order volume is large, so the supply shortage is expected to persist for the time being. Even if additional expansion proceeds, it must be confirmed whether actual lead times shorten quickly.

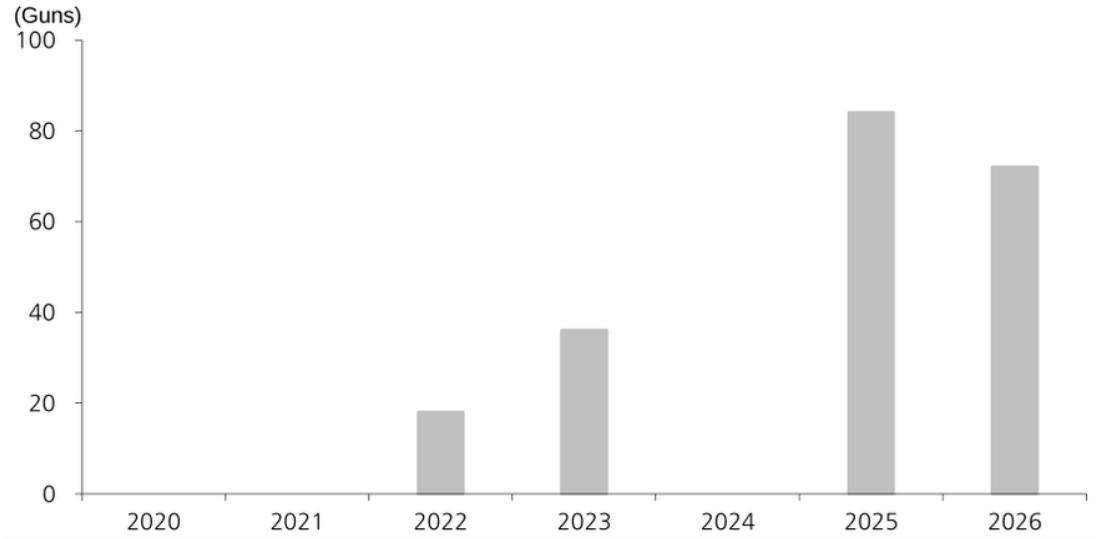
[Table 47] RCH 155 Development and Mass Production History

Year	Key Details
2014	Development of BOXER-based RCH 155 begins, prototype system unveiled
2014~2021	Firing and driving tests and system improvements carried out
2021	Improved version unveiled, firing-on-the-move demonstration
2022	Ukraine's first mass-production contract for 18 guns
2023	Additional contract with Ukraine for 36 guns. Cumulative 54 guns ordered
2024	UK selects RCH 155 as its next-generation self-propelled howitzer
2025	Germany orders 84 guns. Framework agreement for up to 500 guns signed
2026	UK main contract for 72 guns

Source: KNDS Deutschland, Hanwha Research Center

KNDS Deutschland (RCH 155): Full-Scale Mass Production Starting Now. High Likelihood of Expansion (2)

[Figure 80] Trend in new RCH 155 orders



Source: KNDS Deutschland, Hanwha Research Center

[Table 48] Delivery Schedule of New RCH 155 Orders Since 2020 and Estimated Annual Production Capacity

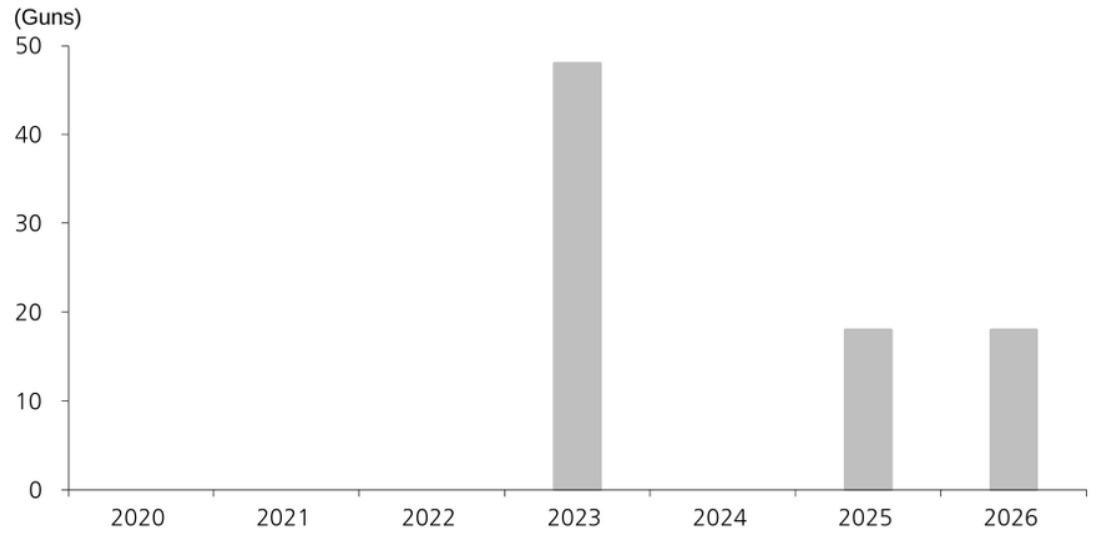
Category	Contract Date	Quantity	Delivery	Remarks
Ukraine 1st	2022	18 guns	2025~	First delivery in 2026
Ukraine additional	2023	36 guns	Undisclosed	-
German Army	2025.12	84 guns	2027~2029	Framework contract for up to 500 guns
UK Army	2026.04	72 guns	2028~2032	-
Total		**210 guns**		**Estimated current annual CAPA of about 25-35 guns**

Source: KNDS Deutschland, Hanwha Research Center

BAE Systems (ARCHER): We Judge the Possibility of Expansion Is Very Low (1)

▪ **The possibility of ARCHER production increase is slim** BAE Systems has also not disclosed a specific target CAPA for ARCHER. Its customer base is much more limited than that of the CAESAR self-propelled howitzer. It should also be considered that although the UK urgently introduced 14 used ARCHER guns, it selected the RCH 155 as its long-term main system. BAE Systems is gradually expanding ARCHER production capacity to handle existing orders centered on Sweden. As it is not a large-scale expansion, we judge the possibility of oversupply to be low.

[Figure 81] Trend in new ARCHER orders



Source: BAE Systems, Hanwha Research Center

BAE Systems (ARCHER): We Judge the Possibility of Expansion Is Very Low (2)

[Table 49] Delivery Schedule of New ARCHER Orders Since 2020 and Estimated Annual Production Capacity

Category	Contract Date	Quantity	Delivery	Remarks
Swedish Army	2023.09	48 guns	2026~2030	First delivery in 2026
Ukraine 1st	2025.03	18 guns	2026~	Additional 18 guns to be ordered
Total		**66 guns**		**Estimated current annual CAPA of about 10-15 guns**

Source: KNDS Deutschland, Hanwha Research Center

Self-Propelled Howitzer Conclusion: Oversupply Concerns Very Limited. Export Competitiveness of Korean Defense Expected to Be Maintained

- **Self-propelled howitzer supply conditions: all clear** Global self-propelled howitzer supply capacity is expanding, but no expansion significantly ahead of demand is appearing. Production capacity is expanding centered on wheeled types, but most of it is investment to produce already-secured order volume and to shorten lead times. Expansion of tracked types is even more limited. Based on the expansion plans confirmed so far, we judge it will be difficult for self-propelled howitzer output to significantly exceed demand in the short term.
- It is a favorable supply environment for Hanwha Aerospace. In the tracked self-propelled howitzer market, where the K9 has strengths, we judge its competitive advantage will continue. In the wheeled self-propelled howitzer market, new export opportunities are expected to expand, starting with entry into the US market. If it expands its product range to wheeled types while maintaining competitiveness in the existing tracked market, its business area within the self-propelled howitzer market could widen further. This is why the results of the US MTC program are important going forward.

② Multiple Launch Rocket Systems: A Market Where Alternatives to HIMARS Are Expanding

▪ **Competing multiple launch rocket systems: HIMARS, PULS** The major weapon systems in the global multiple launch rocket system market can be divided into Lockheed Martin's HIMARS, Hanwha Aerospace's Chunmoo and Israel's Elbit Systems' PULS. HIMARS is the weapon system that has secured the broadest overseas customer base and mass production setup. The multiple launch rocket system market is characterized by a few companies leading the global export market while participation by Western European defense companies is limited. This contrasts with the self-propelled howitzer market, where Western European defense companies such as those in Germany, France and the UK have established themselves as major suppliers.

▪ As the way Western European countries introduce and maintain weapons changed after the end of the Cold War, the supply structure of the multiple launch rocket system market also changed. During the Cold War, the US and Western European countries co-developed the M270 multiple launch rocket system, and it was also produced locally in Europe. After the end of the Cold War, as the possibility of large-scale land warfare declined, Western European countries chose upgrading and extending the service life of the existing M270 rather than developing new multiple launch rocket systems. In fact, before the Russia-Ukraine war, Western European multiple launch rocket system forces were mostly of the M270 family. By contrast, for self-propelled howitzers, although attempts by Western European countries to co-develop self-propelled howitzers during the Cold War did not succeed, independent development and production by each country continued. Even after the end of the Cold War, new development continued for overseas deployment, improved mobility and reduced operating personnel, allowing Western European defense companies to maintain diverse self-propelled howitzer production bases.

[Table 50] Comparison of Major Global Multiple Launch Rocket Systems

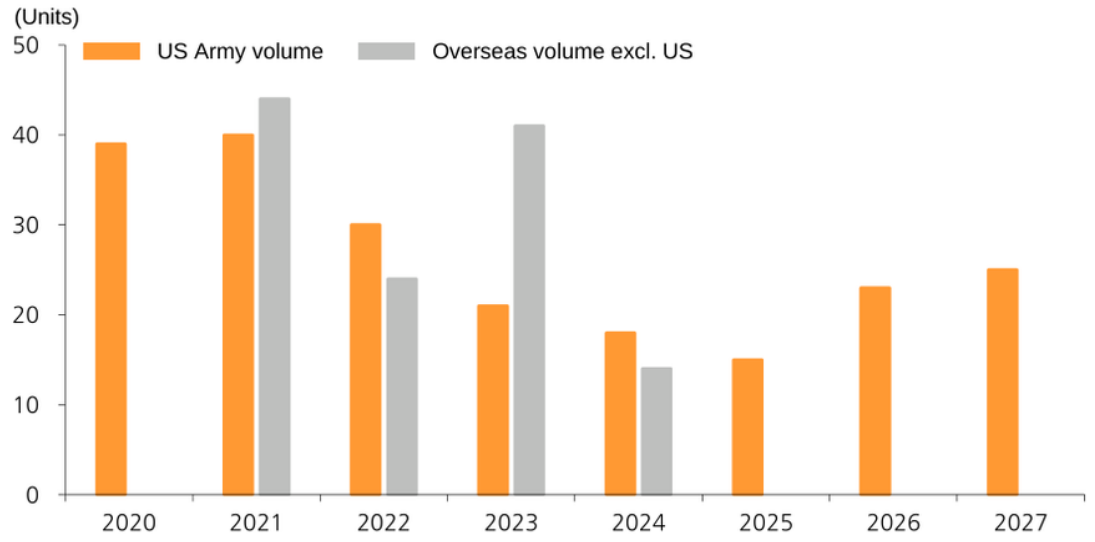
Category	Chunmoo (K239)	HIMARS	PULS
Developer / Manufacturer	Korea / Hanwha Aerospace	US / Lockheed Martin	Israel / Elbit Systems
Vehicle	8x8 high-mobility truck	6x6 FMTV	Selectable platform, such as 6x6
Launch Pods	2	1	2
Reload Time	Within 7 minutes	5-8 minutes	Within 10 minutes
Range by Major Munition Type	CGR-080: 80km; CTM-MR: 160km; CTM-290: 290km	GMLRS: 70km+; ER GMLRS: 150km; ATACMS: 300km; PrSM: 499km+	Accular 122: 35km; Accular 160: 40km; EXTRA: 150km; Predator Hawk: 300km
Rounds per Pod	CGR-080: 6 rounds; CTM-MR: 4 rounds; CTM-290: 1 round	GMLRS: 6 rounds; ER GMLRS: 6 rounds; ATACMS: 1 round; PrSM: 2 rounds	Accular 122: 18 rounds; Accular 160: 10 rounds; EXTRA: 4 rounds; Predator Hawk: 2 rounds
Total Rounds per Launcher	**CGR-080: 12 rounds; CTM-MR: 8 rounds; CTM-290: 2 rounds**	**GMLRS: 6 rounds; ER GMLRS: 6 rounds; ATACMS: 1 round; PrSM: 2 rounds**	**Accular 122: 36 rounds; Accular 160: 20 rounds; EXTRA: 8 rounds; Predator Hawk: 4 rounds**
Market Position	Large order from Poland, expanding in Europe	Global standard, market-leading system	Adopted by multiple European countries, localization expanding

Source: Companies, press reports, Hanwha Research Center

Lockheed Martin (HIMARS): Limited Delivery Time Reduction Expected Despite Completed Expansion (1)

▪ **HIMARS production increase already complete** Lockheed Martin has expanded production capacity centered on HIMARS since the Russia-Ukraine war. For the M270, the business is proceeding centered on upgrading existing equipment and extending service life, and production of new launchers is concentrated on HIMARS. HIMARS annual production capacity increased from 48 units before the Russia-Ukraine war to 96 units in 2024. Production capacity for guided missiles is also expanding. The representative guided rocket, GMLRS, secured annual production capacity of 10,000 rounds in 2023, and expansion to 14,000 rounds was carried out in 2025. For the next-generation precision strike missile, PrSM, expansion is under way targeting annual production capacity of 400 rounds. HIMARS has built a large-scale mass production system by expanding supply capacity not only for launchers but also for guided missiles.

[Figure 82] Annual M270A2 multiple launch rocket system upgrade volume



Note: US Army quantities are based on the number requested for M270A2 upgrades in each fiscal year (FY) President's budget
Source: U.S. Department of the Army, UK Ministry of Defence, Italian Ministry of Defence, Finnish Ministry of Defence, Hanwha Research Center

Lockheed Martin (HIMARS): Limited Delivery Time Reduction Expected Despite Completed Expansion (2)

[Table 51] Comparison of CAPA for the HIMARS System and Key Munitions Before and After the Russia-Ukraine War

Category	CAPA Before 2022	CAPA After 2022	Remarks
HIMARS system	48 units/year	96 units/year	100% increase
GMLRS (70km-range missile)	10,000 rounds/year	14,000 rounds/year	40% increase; includes ER GMLRS (150km)
ATACMS (300km-range missile)	320 rounds/year	500 rounds/year	US: transition from ATACMS to PrSM; overseas: demand continues
PrSM (499km-range missile)	Early mass-production stage	400 rounds/year	Maximum of 664 rounds/year in FY2027; pursuing a fourfold expansion of long-term production capacity

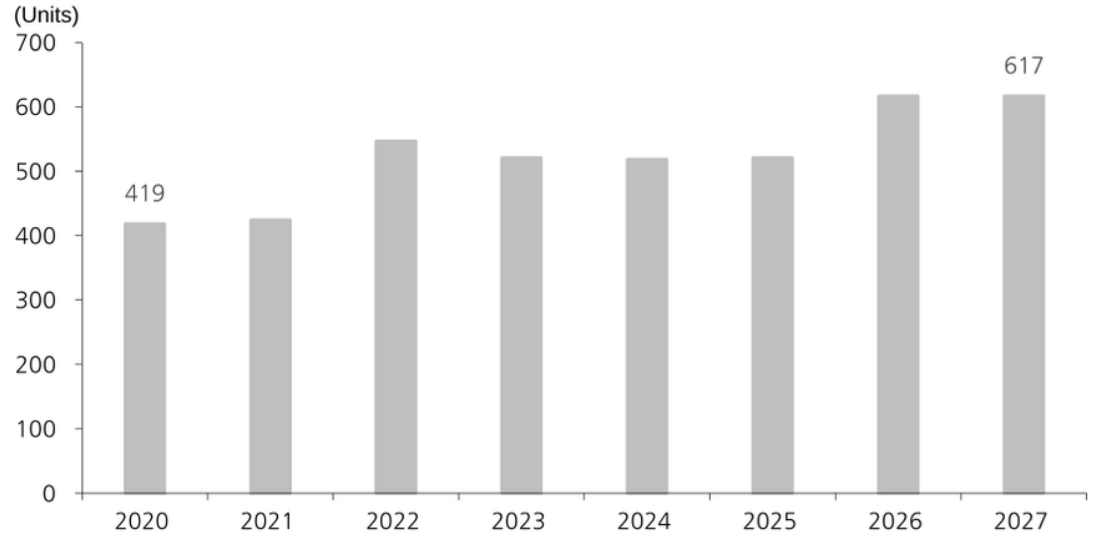
Source: U.S. Department of Defense, U.S. Department of the Army, Lockheed Martin, Hanwha Research Center

- **Considering demand, there is room for further expansion** Considering the expansion of HIMARS production capacity, we judge supply constraints have eased considerably compared with the past. It is difficult to determine precisely how many years' worth of volume the current order backlog represents on the basis of the annual production capacity of 96 units. Depending on the size of additional orders from the US and Poland, production volume could again increase rapidly and long delivery times could persist.
- US HIMARS procurement targets are continuing to be raised. The US Army's HIMARS Army Acquisition Objective (AAO) was 521 units in the FY2025 President's Budget but rose to 617 units in the FY2027 budget. The cumulative planned procurement quantity reflecting the FY2027-FY2031 procurement plan is 715 units, exceeding the current acquisition objective by 98 units. The US is strengthening mobile long-range precision strike forces in preparation for a potential conflict with China in the Indo-Pacific. We judge there is a possibility that the official acquisition objective will be raised further along with the expansion of HIMARS operating units.
- Following the US, the size of Poland's additional orders is also a major variable that will determine future HIMARS production volume. In 2023, Poland signed a framework agreement with Lockheed Martin for the introduction of 486 HIMARS-based Homar-A launchers. This volume has not yet been converted into a firm implementation contract and thus is difficult to include in the current order backlog. If the entire volume is converted into an implementation contract in the future, it would correspond to about 5.1 years of volume based on annual production capacity of 96 units. As local production in Poland is included, not all 486 units will be produced in the US. The volume that must be produced in the US has not yet been determined, but once the contract gets under way in earnest, we judge Lockheed Martin's production burden will increase significantly.

Lockheed Martin (HIMARS): Limited Delivery Time Reduction Expected Despite Completed Expansion (3)

- We judge that long delivery times are highly likely to persist even though HIMARS production capacity has expanded significantly. This is because the upward revision of US procurement targets, the possibility of conversion of Poland's implementation contract and additional demand from countries other than the US remain. There is also room for additional expansion of both launchers and guided missiles going forward. If expansion materializes, delivery times for new customers will shorten. Considering the time required to expand production capacity and potential demand together, we judge it is unlikely that HIMARS' long delivery times will shorten sharply through the mid-2030s.

[Figure 83] Trend in US Army HIMARS target inventory (AAO) by year



Note: Target inventory refers to the US Army's Army Acquisition Objective (AAO). 2022~2023 are based on the targets to pursue increases in the budget documents

Source: U.S. Department of the Army, President's Budget Justification Books (FY2020~FY2027), Hanwha Research Center

Lockheed Martin (HIMARS): Limited Delivery Time Reduction Expected Despite Completed Expansion (4)

[Table 52] HIMARS Procurement Plan by Year in the US President's Budget (Unit: units)

Budget Year	FY20	FY21	FY22	FY23	FY24	FY25	FY26	FY27	FY28	FY29	Total
2020	0	0	11	4	0						15
2021		5	11	5	0	0					21
2022			19	0	0	0	0				19
2023				23	27	19	18	18			105
2024					28	26	20	20	20		114
2025						10	10	12	14	14	60
2026							6	0	0	0	6
2027								96	12	4	112

Note: Based on the procurement plan at the time each year's President's Budget was submitted
Source: U.S. Department of the Army, President's Budget Justification Books (FY2020~FY2027), Hanwha Research Center

[Table 53] Key Milestones of Poland's Homar-A Program

Year	Key Details
2012	Concept for the Homar long-range multiple launch rocket system begins
2015	Official Homar acquisition process launched
2017	PGZ and Lockheed Martin/HIMARS selected; US approves first HIMARS FMS sale
2018	Existing industrial cooperation approach discontinued; switch to direct introduction through the US government FMS; LoR for one HIMARS battalion submitted to the US
2019	US approves FMS of 20 HIMARS; main contract for 20 HIMARS
2023	US approves large additional HIMARS FMS; first HIMARS arrives in Poland; Homar-A prototype vehicle unveiled; Homar-A framework contract
2024	Delivery of the 2019 contract volume completed
2024~2025	Negotiations on first implementation contract and local production conditions
Current	Implementation contract for the 486-unit framework not concluded

Source: Polish Ministry of National Defence, Armament Agency, U.S. DSCA, Lockheed Martin, Hanwha Research Center

Lockheed Martin (HIMARS): Limited Delivery Time Reduction Expected Despite Completed Expansion (5)

[Table 54] HIMARS New Order Trend (Excluding the US)

Category	Contract Date	Quantity	First Delivery	Final Delivery
Taiwan	2021.06	11 units	2024	Completed 2024.10
Australia 1st	2022.11	20 units	2025.03	Targeted 2026
Estonia 1st	2022.12	6 units	2025.04	Completed 2025.04
Lithuania 1st	2022.12	8 units	Scheduled 2026	Undisclosed
Taiwan 2nd	2023.04	18 units	Scheduled 2026	Targeted 2026
Australia 2nd	2023.12	22 units	Undisclosed	Targeted 2028
Latvia	2023.12	6 units	Scheduled 2027	Targeted 2029
Italy 1st	2024.02	7 units	Undisclosed	Undisclosed
Croatia	2024.12	8 units	Scheduled 2027	Undisclosed
Italy 2nd	2025.01	14 units	Undisclosed	Undisclosed
Canada	2026.01	26 units	Scheduled 2029	Undisclosed
Estonia 2nd	2026.04	3 units	Scheduled 2027	Undisclosed
Lithuania 2nd	2026.04	*8 units	Undisclosed	Undisclosed
Taiwan 3rd	2026.04	82 units	Scheduled 2030	Targeted 2032
Morocco	Undisclosed	10 units	Undisclosed	Undisclosed

Source: Ministries of defense of each country, Lockheed Martin, press reports, Hanwha Research Center

Elbit Systems (PULS): Delivery Periods Estimated at 3-5 Years. Scale of EuroPULS Facilities Is Important Going Forward (1)

- **PULS production capacity not disclosed; contract performance period around 3-5 years** Elbit Systems expanded its multiple launch rocket system business by acquiring Israel's IMI Systems in 2018. IMI Systems developed and produced the Lynx multiple launch rocket system and various guided missiles. Based on the acquisition, Elbit Systems unveiled the new multiple launch rocket system PULS in 2020. Beginning with adoption by the Israeli military, PULS exports took off in earnest after the Russia-Ukraine war. It entered the European market by winning orders from Denmark and the Netherlands in 2023. We judge that its relatively lower price and faster delivery compared with HIMARS worked as export competitiveness. At that time, HIMARS delivery times were lengthening owing to rising global demand, and Chunmoo also had to work through Poland's large order volume. PULS is presumed to have emerged as an alternative capable of relatively fast supply.
- PULS's production capacity and order backlog are difficult to confirm specifically. Elbit Systems has been expanding capital expenditure in response to the weapons demand that has increased since the Russia-Ukraine war, but it does not separately disclose the investment amount allocated to PULS or its target production capacity. As it is difficult to directly compare current production capacity with the scale of future expansion, the level of production capacity expansion can be viewed indirectly through the performance periods of recent overseas export contracts. The contract performance period of export projects is generally around 3-5 years. Supply conditions differ by contract, so actual launcher delivery times differ. This is because overall contract periods can lengthen when local production and guided missile supply are included. For cases where PULS launchers are introduced without separate localization conditions, there are examples where delivery of about one year was offered.
- We judge that PULS was able to offer relatively fast delivery in some contracts. From the publicly available information alone, it is difficult to assert that there is ample room in current production capacity.

[Table 55] PULS New Order Trend (Excluding Israel)

Category	Contract Date	Launcher Quantity	Contract Performance Period	Remarks
Denmark	2023.02	8 units	3 years (~2026)	-
Netherlands	2023.05	20 units	5 years (~2028)	-
Undisclosed	2023.07	Undisclosed	3 years (~2026)	USD 150 million
Undisclosed	2024.08	Undisclosed	4 years (~2028)	USD 270 million
Undisclosed	2024.11	Undisclosed	4 years (~2028)	USD 340 million, European country
Germany	2025.02	5 units	Undisclosed	-
Undisclosed	2025.08	Undisclosed	5 years (~2030)	USD 1.64 billion, Serbia
Greece	2026.04	*36 units	4 years (~2030)	Launcher quantity per press reports

Source: Elbit Systems, press reports, Hanwha Research Center

Elbit Systems (PULS): Delivery Periods Estimated at 3-5 Years. Scale of EuroPULS Facilities Is Important Going Forward (2)

- **The scale of future EuroPULS production increases is key** The most important variable in the future expansion of PULS supply capacity is the EuroPULS program. EuroPULS is a European multiple launch rocket system program pursued by Elbit Systems and Germany's KNDS Deutschland. The key is to reflect the requirements of European countries on the basis of the existing PULS and to build a local production base. After signing a cooperation memorandum of understanding in 2022, the two companies firmed up joint development cooperation. A development cooperation contract was signed in 2023, the German parliament approved the introduction of 5 launchers in 2024, a purchase contract was concluded in 2025, and plans to establish a joint venture were even announced in 2026. In other words, the program is expanding from initial development cooperation to the stage of actual launcher introduction and building a local production base.
- The timing at which EuroPULS supply capacity expands in earnest depends on the schedule for building launcher and ammunition production facilities in Germany. What matters is whether not only launcher assembly but also guided missiles can be produced locally in Europe. This is because even if launcher production capacity increases, the actual pace of force fielding may be limited if the supply of rockets fails to keep up. It will be necessary to watch the joint venture's capital expenditure scale, production capacity by launcher and guided missile, and the start of local production.
- Over the medium to long term, there is ample possibility of large-scale EuroPULS expansion. This is because considerable production capacity would be needed if the German Army's demand for about 200 units and the roughly 300-unit joint procurement concept of Germany and its allies materialize. These volumes are procurement plans for which actual purchase contracts have not yet been concluded, and the scale of expansion could change depending on whether they convert to contracts. If large-scale procurement materializes, new production volume is likely to be supplied first to Germany and European allies.
- EuroPULS could become a factor expanding supply in the global multiple launch rocket system market over the medium to long term. Building a European local production base requires considerable time and investment, and the large potential demand of Germany and its allies is also ample. Even if expansion gets under way in earnest, we expect a considerable portion of new production volume to be absorbed within Europe. We judge that the possibility of EuroPULS causing oversupply in the global multiple launch rocket system market through the mid-2030s is low.

Elbit Systems (PULS): Delivery Periods Estimated at 3-5 Years. Scale of EuroPULS Facilities Is Important Going Forward (3)

[Table 56] Key Milestones of the EuroPULS Program

Category	Key Details
2022.06	KMW and Elbit Systems sign cooperation MOU
2022.12	EuroPULS development plan announced
2023.09	KNDS Deutschland and Elbit Systems sign joint business agreement
2024.09	Diehl Defence and Elbit Systems sign rocket co-development agreement
2024.12	German parliament approves budget for initial EuroPULS introduction
2025.02	German Army announces initial purchase contract (5 launchers, EUR 57 million)
2025	Preliminary testing of EuroPULS training rockets
2026.02	Pre-test launch of 122mm training rocket
2026.03	KNDS Deutschland and Elbit Systems agree to establish joint venture (EuroPULS GmbH, 50:50 equity stake)
2026.03	Successful test launch of EuroPULS training rocket
2027 (target)	Delivery and certification of German Army's initial launchers

Source: KNDS, Elbit Systems, Diehl Defence, Janes, Hanwha Research Center

[Table 57] Key Details of the EuroPULS Development Plan (December 2022)

Category	Key Details
Development Purpose	Modernization of and successor system for the MLRS/MARS II operated in Europe
Base System	Utilizes Elbit's PULS launcher and precision-guided missiles already in operation
Role of KMW	Improves and integrates the system to European standards using its experience integrating existing artillery systems
Munition Integration	Secures a structure capable of integrating various rockets to suit customer requirements
Local Production in Europe	Concept of local rocket production to secure European supply independence
Commercialization	At the time, the EuroPULS concept had already been proposed to initial European customers

Source: KNDS Deutschland, Elbit Systems, Hanwha Research Center

Elbit Systems (PULS): Delivery Periods Estimated at 3-5 Years. Scale of EuroPULS Facilities Is Important Going Forward (4)

[Table 58] Key Details of the EuroPULS Joint Business Agreement (September 2023)

Category	Key Details
System Integration	KNDS Deutschland leads European-type system integration
Launcher	Utilizes Elbit's existing PULS modular launcher
Fire Control System	KNDS Deutschland's modular fire control system applied
Rockets, Guided Missiles	Integrates Elbit precision-guided missiles and training rounds; third-party missile companies may participate
Vehicle Platform	System can be integrated onto various mobility platforms
Target Market	Europe and NATO countries

Source: KNDS Deutschland, Elbit Systems, Hanwha Research Center

[Table 59] Key Details of the EuroPULS Rocket Co-Development Agreement (September 2024)

Category	Key Details
Cooperation Target	Rockets/high-performance training rounds for PULS and EuroPULS
Development Direction	Developing ammunition to suit the rocket launcher operating requirements of the German Army and European countries
Local Production	Pursuing rocket development/production capabilities in Europe
Logistics Support	Expanding local support and follow-on logistics support capabilities for European operating countries
Technical Cooperation	Combining Diehl Defence's 40+ years of MLRS rocket production experience with Elbit Systems Land's rocket launcher technology

Source: Diehl Defence, Hanwha Research Center

France's Multiple Launch Rocket System Development: Supply Expansion After the 2030s

- **France developing a multiple launch rocket system; first delivery estimated around the 2030s** Over the medium to long term, France's independent development of a multiple launch rocket system warrants attention in terms of global supply expansion. The French military has operated the LRU, an upgraded version of the US M270. For its next multiple launch rocket system, it decided to develop its own without depending on the US. Thales' X-FIRE and MBDA/Safran's THUNDART competed for the next-generation multiple launch rocket system program. Neither weapon system is currently in mass production; both are at the development stage.

- In June 2026, France selected THUNDART as its next-generation multiple launch rocket system. This program is part of a multiple launch rocket system modernization program to replace the aging LRU. First delivery of THUNDART is expected around 2030, and the specific quantity to be introduced by the French military has not yet been officially disclosed. THUNDART's output may begin to rise in earnest from the early-to-mid 2030s. Initial production volume is expected to be supplied first to the French military. Even if THUNDART mass production begins, we judge the impact on supply expansion in the global export market through the mid-2030s will be limited.

[Table 60] Comparison of France's Next-Generation Multiple Launch Rockets THUNDART and X-FIRE

Category	THUNDART	X-FIRE
Developer	MBDA, Safran Electronics & Defense	Thales, ArianeGroup, Soframe
Selected for Next-Generation Program	Selected 2026.09	Not selected
Development Status	Development/testing stage	Development/testing stage
Maximum Range	150km	150km+
Launcher Payload	8 rounds (4 rounds x 2 pods)	Undisclosed
Munition Compatibility	Centered on THUNDART's own guided missiles	Pursuing integration of French/foreign munitions
First Rocket Test Launch	Successful 2026.04.14	Successful 2026.05.05
Fielding Target	Target of initial operational capability in 2029; target of securing operational sovereignty in 2030	Undisclosed because the program was not selected
Annual Production Capacity	Undisclosed	Undisclosed
Local Production Plan	Pursuing development and production in France	Conceptualizing development and production in France

Source: Companies, press reports, Hanwha Research Center

Thales (X-FIRE): Possibility of Cooperation With Hanwha Aerospace Stands Out

- **Hanwha-X-FIRE guided missile cooperation** The moves of X-FIRE, which was eliminated from the French military's multiple launch rocket system development program, also warrant attention. Although Thales failed to supply the French military, it signed a cooperation memorandum of understanding with Hanwha Aerospace at the 2026 Eurosatory defense exhibition. The key is to integrate Hanwha Aerospace's guided missiles into the X-FIRE launcher. It is interpreted as a move to pursue overseas exports despite the failure to win the French military program. As X-FIRE is also at the development stage, it will be necessary to check its future development schedule and whether it actually goes into mass production.
- If X-FIRE's development is completed and overseas exports get under way in earnest, it could also be an opportunity for Korean defense companies. European countries place importance on local production and the participation of domestic companies in the weapons introduction process. Combining a French-made launcher with Hanwha Aerospace's guided missiles could expand exports of Korean-made guided missiles while responding to Europe's localization requirements. We judge X-FIRE to be both a new competing system in the global multiple launch rocket system market and a cooperative project that could expand Hanwha Aerospace's entry into the European market.

Multiple Launch Rocket System Conclusion: Supply Shortage Expected to Be Prolonged. Chunmoo's Export Competitiveness Expected to Expand

- **Production increases of multiple launch rocket systems expected to continue; demand is that strong; excess demand expected to be prolonged** Global multiple launch rocket system production capacity is expanding, but we judge there is limited possibility that lead times will shorten significantly through at least 2030. Even though HIMARS doubled its production capacity, considering the upward revision of US procurement targets and Poland's large potential demand, it is hard to say there is ample supply capacity. Even if large-scale expansion of EuroPULS is decided in the future, given the considerable potential demand of Germany and European allies, new production capacity is likely to be used first to absorb intra-regional demand. France's THUNDART is still at the development stage and needs considerable time before full-scale mass production. While the oligopoly of a few companies such as HIMARS, Chunmoo and PULS in the multiple launch rocket system market is maintained for the time being, supply shortages are expected to persist owing to high demand despite production capacity expansion. We judge that the possibility of oversupply in global multiple launch rocket systems will be limited.
- The market environment is expected to remain favorable for Hanwha Aerospace's multiple launch rocket system business. As Chunmoo had to prioritize producing Poland's large order volume after the Russia-Ukraine war, we estimate it was difficult to offer fast delivery times like Poland's until 2024. As deliveries of Chunmoo systems to Poland have since progressed considerably, supply capacity and delivery competitiveness for new customers are likely to have recovered considerably. In fact, following Poland, Chunmoo's export markets are expanding to Norway, Croatia and others. With global multiple launch rocket system supply shortages continuing, this export expansion trend is expected to continue for a considerable period.
- When Poland's joint venture for guided missile production is fully operational, local supply capacity for guided missiles as well as launchers is expected to expand. This could act as a factor that raises supply stability for existing customers while improving the speed of execution of new export contracts. As the prolonged long lead times in the global multiple launch rocket system market, the recovery of Chunmoo's delivery competitiveness and the expansion of local production bases in Europe come together, we judge that Hanwha Aerospace's multiple launch rocket system export competitiveness will be further strengthened.

Long Delivery Times Expected to Persist Despite Production Capacity Expansion

- **Air defense systems: excess demand expected to persist despite the burden of capacity expansion** We judge that the export competitiveness of Korean air defense systems will be maintained for a considerable period despite the expansion of global air defense system supply capacity. Through the Russia-Ukraine war and the US-Iran war, drone and ballistic missile threats have expanded simultaneously, and the importance of medium- and long-range air defense networks, not just short-range ones, has risen. Expansion of interceptor missile and air defense system production capacity is getting under way in earnest, centered on US and European defense companies. Even so, the pace of supply expansion of competing systems is limited. In Europe, MBDA's SAMP/T is the most direct competing system, but expanding production capacity of the next-generation Aster 30 B1NT interceptor missile is expected to take time. Diehl Defense's IRIS-T and Kongsberg's NASAMS are also strengthening ballistic missile response capability, but considerable time is needed before full-scale mass production. Although expansion of local production in Europe is being pursued for Patriot as well, local production of the PAC-3 MSE is highly uncertain owing to technology transfer issues, and it is also difficult for GEM-T production capacity to expand significantly in a short period. We judge that even if global air defense capacity expansion continues, the long lead times of major medium- and long-range air defense systems are likely to be maintained until at least the early-to-mid 2030s.
- Recent US Patriot supply chain expansion is weighted more toward expanding interceptor missile production capacity than supplying new batteries. This is because, with interceptor missile inventories having fallen through the US-Iran war, the appropriate inventory level required going forward is also likely to be higher than in the past. We judge that expansion of Patriot production capacity is unlikely to lead to an increase in overall system export supply capacity in the short term. Europe is also rapidly expanding air defense system and interceptor missile production capacity, but a considerable portion of the volume is likely to be allocated first to expanding its own forces and rebuilding inventories of existing operating countries. We judge that, despite large-scale expansion by competitors, the possibility of intensified export competition for Korean air defense systems in the short term is low. Korea is expanding both engagement altitude and systems simultaneously through the development of Cheongung-III and mass production of L-SAM, based on its experience in mass production, actual interceptions and exports of Cheongung-II. If development through Cheongung-III is completed, the export competitiveness of the Cheongung family is highly likely to be maintained for a considerable period. We judge that expanding demand for building multi-layered defense networks could also lead to additional export opportunities for L-SAM.

[Table 61] Global Air Defense System Expansion Status and CAPA Outlook

Category	Annual Production Capacity	CAPA Expansion Plan	Outlook
Patriot	12 batteries	Expanding	Specific battery CAPA not disclosed; missile CAPA expanding
SAMP/T	4-6 batteries (est.)	Expanding	SAMP/T NG to enter full-scale mass production; missile CAPA expanding
NASAMS	3-5 batteries (est.)	Not disclosed	High likelihood of CAPA expansion
IRIS-T	10 batteries	Expanding	16 batteries in 2028, 30 batteries over the medium to long term

Source: Hanwha Research Center

Review of the European Air Defense System Supply Chain (1)

- **European competing air defense systems: SAMP/T, IRIS-T, NASAMS** Supply capacity expansion of European air defense systems is proceeding centered on MBDA (unlisted), Diehl Defence (unlisted) and Kongsberg (Norway). As of 2025, annual revenues are about EUR 5.8 billion, EUR 2.3 billion and EUR 2.2 billion, respectively. None of the three companies discloses the share of revenue from missiles, including air defense systems. Considering product portfolios, MBDA, with a missile-centered lineup, is estimated to have a relatively higher missile revenue share than the other companies.
- MBDA is effectively the core supplier of Europe's missile industry. It was established in 2001 by integrating the main missile businesses dispersed across France, the UK and Italy. Airbus currently holds 37.5%, BAE Systems 37.5% and Leonardo 25.0%. It is distinctive in being the only European defense company able to produce missile systems for all land, sea and air domains.
- Diehl Defence is Germany's leading missile and air defense company. It mainly produces guided missiles and ammunition, and based on its representative product IRIS-T, it has built a product family ranging from air-to-air missiles to ground-based air defense systems such as the IRIS-T SLS and SLM.
- Kongsberg Defence & Aerospace is Norway's leading defense company. It mainly produces air defense systems, anti-ship and air-to-ground missiles and remote weapon systems, and its representative products are the NASAMS air defense system, NSM missile and JSM missile. NASAMS is a medium-range air defense system co-developed with the US's RTX, and it is distinctive in that Kongsberg is responsible for system integration centered on launchers and command and control systems.

[Table 62] Summary of Europe's Key Air Defense System Companies

Category	Missile Business Dependence	Remarks
MBDA	Very high	Unlisted; revenue EUR 5.8 billion (2025); representative air defense system is the SAMP/T family
Diehl Defence	High	Unlisted; parent company Diehl Group; revenue EUR 2.3 billion (2025, Defence division); representative air defense system is the IRIS-T family
Kongsberg Gruppen	Medium	Listed in Norway; revenue EUR 2.2 billion (2025, missiles/space business); representative air defense system is NASAMS

Source: MBDA, Diehl, Kongsberg Gruppen, Hanwha Research Center

Review of the European Air Defense System Supply Chain (2)

[Table 63] Performance Comparison of Major European Air Defense Systems

Category	SAMP/T	IRIS-T	NASAMS
Manufacturer	Eurosam	Diehl Defence	Kongsberg, RTX
Developing Country	France, Italy	Germany	Norway, US
Existing Operational Model	SAMP/T	IRIS-SLM	NASAMS
Main Interceptor	Aster 30 Block 1	IRIS-T SL	AIM-120 (B, C-5, C-7)
Maximum Interception Range	120km	40km	20-30km
Maximum Interception Altitude	20km	20km	10-16km
Main Interception Targets	Aircraft/cruise missiles/ballistic missiles	Aircraft/cruise missiles/ballistic missiles	Aircraft/cruise missiles/drones
Ballistic Missile Response	Short-range ballistic missiles	Short-range ballistic missiles	Official capability not confirmed
Basis for Response Capability	Verified through testing	Manufacturer's claim. Verification unconfirmed	-
Interceptor Guidance Method	Active radar	Infrared imaging seeker	Active radar
Target Destruction Method	Direct impact and proximity detonation/fragments	Combined direct impact/proximity detonation	Proximity detonation/fragmentation warhead
Rounds per Launcher	8	8	6

Source: Companies, press reports, Hanwha Research Center

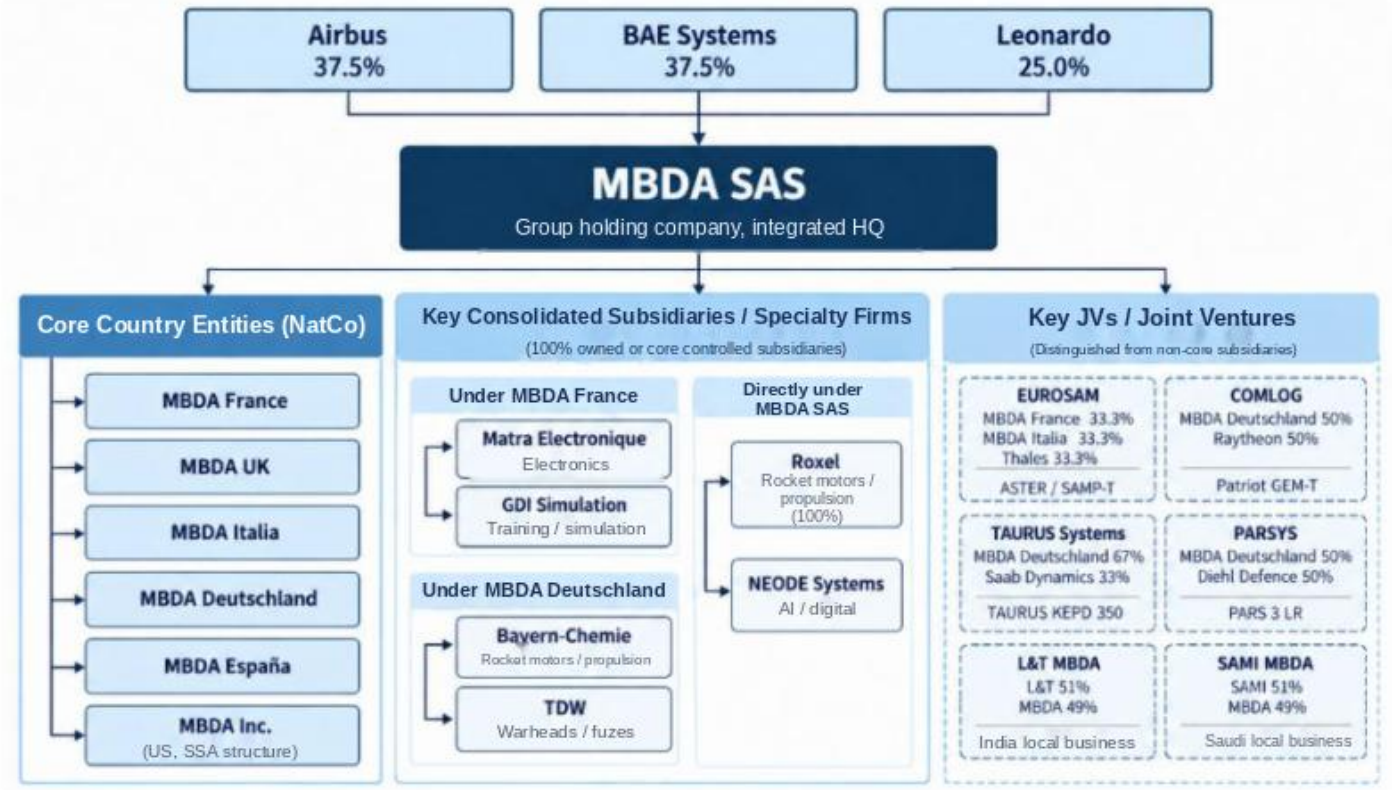
[Table 64] Comparison of Performance and Development Status of Major European Next-Generation Air Defense Systems

Category	SAMP/T NG	IRIS-T SLX	NASAMS (AMRAAM-ER)
Upgrade Target	Entire air defense system	Centered on new interceptor	Centered on new interceptor
Radar Upgrade	New radar introduced	Existing radar linked/utilized	Existing radar linked/utilized
Command and Control System	New system introduced	Existing system linked/utilized	Existing system linked/utilized
Nature of Upgrade	Performance upgrade of the entire system	Interceptor-centered performance upgrade	Interceptor-centered performance upgrade
Next-Generation Interceptor	Aster 30 B1NT	IRIS-T SLX	AMRAAM-ER
Maximum Interception Range	150km+	80km	+50% vs. existing
Maximum Interception Altitude	Expanded vs. existing	25km+	+70% vs. existing
Main Interception Targets	Aircraft/cruise/ballistic missiles	Aircraft/cruise missiles, etc.	Aircraft/cruise missiles/drones
Ballistic Missile Response	Short-/medium-range ballistic missiles	Not confirmed	Not confirmed
Basis for Response Capability	Presented by manufacturer. Verification unconfirmed	-	-
Development/Testing Status	Testing, pursuing fielding	Test launch in 2027	Conversion to mass production under way
First Mass Production Schedule	End-2027	2029	-

Source: Companies, press reports, Hanwha Research Center

Review of the European Air Defense System Supply Chain (3)

[Figure 84] MBDA governance structure and status of key consolidated subsidiaries and JVs



Source: MBDA, Hanwha Research Center

Review of the European Air Defense System Supply Chain (4)

[Table 65] MBDA Status by Key Country Entity

Category	Key Products
MBDA France	Air defense/air-to-air/anti-ship/cruise missiles: Aster, SCALP, Exocet, MICA, Akeron
MBDA UK	Air-to-air/air defense/air-to-ground missiles: ASRAAM, CAMM, Storm Shadow, SPEAR
MBDA Italia	Air defense/anti-ship missiles and guidance devices: Aster, CAMM-ER, Marte, Teseo
MBDA Deutschland	Air defense/air-to-air/air-to-ground/anti-tank missiles: GEM-T, Meteor, Taurus, RAM, Enforcer
MBDA España	Short-range air defense systems, system integration: Mistral 3, SIMBAD-RC
MBDA Inc.	Anti-tank/air-to-ground precision-guided weapons: Dual Mode Brimstone, Viper-E, Diamond Back

Source: MBDA, Hanwha Research Center

[Table 66] Status of Key MBDA Subsidiaries

Category	Key Products
Matra Electronique	Electronic circuit boards, wiring and electronic equipment
GDI Simulation	Military training simulators
Bayern-Chemie	Solid-fuel propellants, gas generators
TDW	Warheads, fuzes, safe-and-arm devices
Roxel	Solid-propellant rocket motors, gas generators
Neode Systems	Defense artificial intelligence, digital technology

Source: MBDA, Hanwha Research Center

[Table 67] Status of Key MBDA Joint Ventures

Category	Ownership Structure	Key Business
EUROSAM	MBDA France/Italy 66.7%, Thales 33.3%	Development and integration of Aster-based air defense systems - SAMP/T, PAAMS
TAURUS Systems	MBDA Deutschland 67%, Saab Dynamics 33%	Development and production of the Taurus KEPD 350 family long-range air-to-ground cruise missile
PARSYS	MBDA Deutschland 50%, Diehl Defence 50%	PARS 3 LR anti-tank missile program
COMLOG	MBDA Deutschland 50%, RTX 50%	Patriot missile logistics support and GEM-T production and integration in Germany
L&T MBDA Missile Systems	L&T 51%, MBDA 49%	Localization and production of missiles in India; Make in India program
SAMI-MBDA Missile Systems	SAMI 51%, MBDA 49%	Localization of the Saudi missile industry; development and building of production base

Source: MBDA, Hanwha Research Center

MBDA (SAMP/T NG): A Strong Competitor to Cheongung-II in the Patriot Replacement Market (1)

- **Strong competing air defense system: SAMP/T** We judge MBDA's SAMP/T NG, along with Cheongung-II, to be a strong air defense system able to complement the Patriot amid a continuing global shortage of interceptor missiles. SAMP/T is an air defense system developed/produced by Eurosam, a joint venture established by MBDA France/Italy and Thales. It is divided into the existing SAMP/T and the next-generation SAMP/T NG. The existing SAMP/T has the ability to intercept aircraft, cruise missiles and short-range ballistic missiles. SAMP/T NG applies a new radar and engagement control center and has expanded its ballistic missile response range. It is differentiated from the existing system in that it can respond not only to short-range ballistic missiles but also to some medium-range ballistic missiles.

- To understand SAMP/T's performance, it is necessary to look at the Aster interceptor missile. The Aster family is divided into Aster 15, Aster 30, Aster 30 B1 and Aster 30 B1NT. The Aster 15 and Aster 30 were developed to counter general airborne threats such as aircraft, cruise missiles and anti-ship missiles. From Aster 30 B1, short-range ballistic missile interception capability was added. The next-generation Aster 30 B1NT further expanded the ballistic missile response range. According to MBDA, the B1NT can respond to ballistic missiles of the 1,500km class. This means it can defend against some medium-range ballistic missile threats. First delivery of the B1NT is scheduled to begin in 2027.

[Table 68] Comparison of Key Specifications of SAMP/T and SAMP/T NG

Category	SAMP/T	SAMP/T NG
Development Entity	Eurosam	Eurosam
Program Launch	1990s	March 2021
Operational Status	In operation by French and Italian militaries	Initial delivery stage
First Fielding	Early 2010s	Target 2026
Main Interceptor	Aster 30 Block 1	Aster 30 B1NT
Interceptor Seeker	Ku-band active radar	Ka-band active radar
Radar	Arabel	France: Ground Fire 300; Italy: Kronos GM HP
Radar Detection Range	100km	350km+
Aircraft Engagement Range	100km	150km+
Ballistic Missile Response	Short-range tactical ballistic missiles (SRBM)	Short- and medium-range tactical ballistic missiles (SLBM, MRBM)
Launcher Configuration	8 rounds per launcher, up to 6 launchers	8 rounds per launcher, up to 6 launchers
Engagement Control System	Existing engagement control module	New engagement control module

Source: Eurosam, MBDA, Thales, Leonardo, Hanwha Research Center

MBDA (SAMP/T NG): A Strong Competitor to Cheongung-II in the Patriot Replacement Market (2)

[Table 69] Comparison of Key Specifications of Aster Interceptor Versions

Category	Aster 30	Aster 30 Block 1	Aster 30 Block 1NT
Developer	Eurosam	Eurosam	Eurosam
Main Applied System	Naval air defense systems	SAMP/T	SAMP/T NG
Main Interception Targets	Aircraft/cruise missiles	Aircraft/cruise missiles/SRBM	Aircraft/cruise missiles/SRBM/MRBM
Ballistic Missile Response Level	Limited	600km-class SRBM	Targeting 1,500km-class MRBM
Interception Range for Air Targets	120km+	120km+	150km
Interception Altitude	20km	20km	Undisclosed
Maximum Flight Speed	Mach 4.5 class	Mach 4.5 class	Undisclosed
Radar Seeker	Ku-band	Ku-band	Ka-band
Guidance/Control	PIF-PAF	Improved ballistic missile response performance	Improvements to seeker, electronics and guidance performance
Development/Operational Status	In operation	In operation	Mass production preparation stage

Source: MBDA, Eurosam, Hanwha Research Center

[Table 70] Key Development and Mass Production History of Aster 30 Block 1NT

Category	Key Details	Remarks
2015.12	First B1NT development contract signed	French defense procurement agency-Eurosam contract
2016.06	France and Italy agree on B1NT development cooperation	Joint development, targeting land/naval application
2016.12	Italy's participation in joint development confirmed	-
2023.01	First B1NT mass-production contract signed	700 rounds combined for Aster 15, B1 and B1NT
2024.10	First B1NT test launch succeeds	Test of identifying and intercepting enemy/friendly aircraft
2025.07	Second B1NT test launch succeeds	Long-range, high-altitude flight performance test
2025.12	SAMP/T NG system integration test launch succeeds	Radar/engagement control/launcher integration
2026.02	Italy B1NT production base expansion contract	Diversifying production facilities concentrated in France
2027	Target for initial delivery	-

Source: MBDA, Eurosam, Hanwha Research Center

Expansion of SAMP/T and Aster Interceptor Missile Production Under Way in Line With Rising Demand

- **SAMP/T interceptor missile production being expanded** In the past, demand for Aster expanded first at sea rather than on land. Aster was developed in the process of France and Italy building an independent fleet air defense system that does not depend on the US. At sea, it was applied to air defense systems such as SAAM and PAAMS and was mounted on Horizon-class destroyers, FREMM frigates and the UK's Type 45 destroyers. As countries that introduced European-built ships also adopted Aster, the number of naval operating countries grew. Operating countries of the land-based SAMP/T were relatively limited. Before the 2022 Russia-Ukraine war, the only countries in Europe operating the SAMP/T were the developers France and Italy.

- Demand for the land-based SAMP/T is likely to expand going forward. This is because, with the global shortage of interceptor missiles continuing, the ballistic missile response capability of the SAMP/T NG is being strengthened. Whereas the existing SAMP/T mainly responded to short-range ballistic missiles, the key point is that the SAMP/T NG can defend against some medium-range ballistic missile threats. This means that the competitiveness of the SAMP/T NG could rise in the ballistic missile defense market that the Patriot has mainly covered. Ukraine received SAMP/T and Aster interceptor missiles from France and Italy and operated them in actual combat. Denmark also selected the SAMP/T NG in 2025 as its long-range ground-based air defense system, with deliveries scheduled to begin in 2028. Given that land-based operating countries were previously limited to France and Italy, this is a meaningful change.

- MBDA is responding by expanding Aster production capacity. In March 2026, MBDA's CEO said it plans to double Aster production in 2026 compared with the previous year. It did not disclose a specific annual production capacity. Based on what the director general of the French defense procurement agency said at a defense committee hearing in December 2024, 2025 Aster output is estimated at about 80-100 rounds. In 2026, output is expected to increase to about 160-200 rounds. Through additional capital expenditure, it plans to expand annual production capacity to 300 rounds or more in 2028.

[Table 71] Status of Land and Naval Aster Interceptor Operating Countries

Naval Aster Operating Countries	Land Aster Operating Countries (SAMP/T Family)
Total of 8 countries operate	**Total of 5 countries operate**

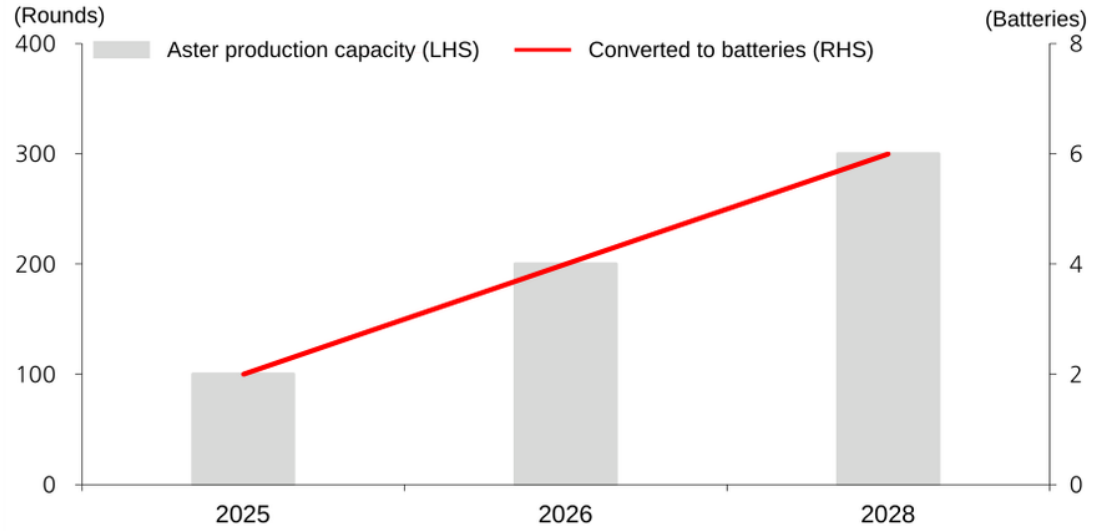
Note: Denmark decided in 2026 to introduce the SAMP/T NG. Not yet in a fielded state
Source: Each country, Hanwha Research Center

Difficult for SAMP/T NG System Lead Times to Shorten Quickly (1)

- **SAMP/T NG: possibility of a sharp production increase in the short term is limited** The point to watch going forward is the production capacity of the Aster 30 B1NT. The B1NT is the key interceptor missile raising the ballistic missile response capability of the SAMP/T NG. With first delivery scheduled from 2027, its share of total Aster output is expected to be limited for the time being. Because Aster is used not only for the land-based SAMP/T but also for naval air defense systems, not all of the increased output is supplied to the SAMP/T NG. Over the medium to long term, if mass production of the B1NT gets under way in earnest, the number of new SAMP/T NG customers is likely to expand even more quickly.
- Aster interceptor missile output can be roughly estimated, but the annual production capacity of SAMP/T NG batteries is difficult to confirm. Theoretical supply scale can be calculated on the basis of Aster output. One SAMP/T NG battery consists of up to 6 launchers and 48 interceptor missiles. If Aster production capacity reaches 300 rounds per year in 2028, it could supply the missiles needed for about 6 batteries. If it expands to 500 rounds per year, that corresponds to about 10 batteries. These figures assume that all Aster produced is supplied to new SAMP/T NG batteries, with 48 rounds allocated per battery. Considering inventory replenishment for existing operating countries, naval demand and additional reserve rounds, the volume actually available for new batteries is less than this. The production capacity of radars, engagement control centers and launchers must also be considered separately.
- When judging SAMP/T NG production capacity, B1NT interceptor missile output is more important than total Aster output. Even if Aster output increases, if B1NT production is not sufficiently expanded, supply of the SAMP/T NG, with its strengthened ballistic missile defense capability, could be limited. As deliveries of the B1NT begin from 2027, how much output will expand by around 2030 is highly uncertain. From a medium- to long-term perspective, the possibility of expansion in the SAMP/T NG's competitiveness is high, but we judge it will be difficult for supply capacity to increase quickly until at least around 2030.

Difficult for SAMP/T NG System Lead Times to Shorten Quickly (2)

[Figure 85] Comparison of Aster interceptor missile production capacity expansion and supply volume converted into SAMP/T batteries



Note: Aster production capacity for 2025 is the forecast of the French defense procurement agency. Battery conversion assumes 6 launchers per battery and 8 interceptor missiles each

Source: French National Assembly Defense Committee hearing, Eurosam, press reports, Hanwha Research Center

Kongsberg (NASAMS): Time Needed Before Securing Ballistic Missile Defense Capability (1)

- **NASAMS ballistic missile interception performance is limited** Norwegian defense company Kongsberg is also expanding production capacity centered on missiles, air defense systems and remote weapon systems (RWS) since the Russia-Ukraine war. Its representative air defense system is NASAMS. NASAMS is a short- and medium-range ground-based air defense system co-developed by Kongsberg and the US's RTX. Kongsberg produces the fire distribution center (FDC), an engagement control device, and launchers, while RTX supplies radars and interceptor missiles. It uses the AMRAAM as the basic interceptor missile, and the AMRAAM-ER and AIM-9X can also be integrated into the same system.

- The biggest difference between NASAMS and SAMP/T NG is that NASAMS currently has no ballistic missile interception capability. The AMRAAM, AMRAAM-ER and AIM-9X fired from NASAMS are centered on intercepting aircraft, cruise missiles and drones. In 2023, development began with RTX and the Norwegian government to expand NASAMS' ability to respond to future threats, and this was concretized in 2025 as the development of Full Spectrum Air Defence (FSAD). FSAD aims to respond even to tactical ballistic missiles by integrating long-range radars and a new interception system into the existing NASAMS. Kongsberg expects first supply around 2030 if prospective purchasers confirm FSAD purchases early. It is uncertain whether supply can actually begin from 2030. FSAD is at a stage where development is still in progress, and the specific interceptor missile and sensor configuration to be used for ballistic missile interception, purchasing countries and mass production schedule have not been determined. Even in its home country Norway, where there is a gap in ballistic missile air defense, a ballistic missile defense system has not yet been selected. Despite the 2030 supply target, we judge that considerable time is likely to be needed before it is actually fielded as a ballistic missile defense system. The impact on the export competitiveness of Korean air defense systems, which have already secured ballistic missile interception capability, is also expected to be limited.

[Figure 86] NASAMS operating countries and new contracting countries since 2022 (existing operating countries: red, new contracting countries: blue)



Note: Taiwan, Belgium and Kuwait have not yet been fielded after signing contracts
Source: Kongsberg Defence & Aerospace, RTX (Raytheon), ministries of defense of each country, Hanwha Research Center

Kongsberg (NASAMS): Time Needed Before Securing Ballistic Missile Defense Capability (2)

[Table 72] NASAMS New Orders and Delivery Status Since 2020

Category	Contract Date	Contract Amount	Scheduled Contract End Date	Remarks
Hungary	2020.11	USD 470 million	Undisclosed	-
Ukraine 1st	2022.08	USD 180 million	2027.12	-
Ukraine 2nd	2022.11	USD 1.2 billion	2028.06	-
Canada	2023.01	USD 300 million	Undisclosed	Donation to Ukraine
Lithuania 1st additional	2024.12	USD 220 million	Undisclosed	New introduction + upgrade of existing equipment
Norway 1st	2024.01	USD 140 million	Undisclosed	-
Spain	2024.05	USD 440 million	Undisclosed	-
Norway 2nd	2024.06	USD 220 million	Undisclosed	-
Lithuania 2nd additional	2024.10	USD 210 million	Undisclosed	New introduction + upgrade of existing equipment
Netherlands	2024.11	USD 1.09 billion	Undisclosed	-
Norway 3rd	2024.12	USD 240 million	Undisclosed	-
Taiwan	2025.01	USD 800 million	Undisclosed	-
Denmark	2025.11	USD 1.02 billion	Undisclosed	-
Kuwait	2026.05	USD 1.02 billion	2031.09	-
Belgium	2026.09	USD 1.04 billion	Undisclosed	-
Netherlands additional	2026.09	USD 160 million	Undisclosed	-

Note: Contract amount currency converted to US dollars
Source: Kongsberg, RTX, governments of each country, Hanwha Research Center

[Table 73] Key Milestones of the Full Spectrum Air Defence (FSAD) Program

Category	Key Details	Remarks
2024.04	Norwegian defense plan announced	Pursuing introduction of long-range air defense and SLBM defense capability
2024	Kongsberg formalizes FSAD development strategy	NASAMS: responding to a wider range and diverse threats
2025.05	Kongsberg details FSAD development strategy	Extends NASAMS performance to long-range, ballistic missile defense; ready to supply by 2030 if customers invest; describes Patriot and SAMP/T as competitors
2026.06	Kongsberg specifies FSAD target market	Integrates interception capability into existing systems rather than developing new systems

Source: Norwegian Ministry of Defence, Kongsberg, Hanwha Research Center

Diehl Defence (IRIS-T): Likewise Limited Ballistic Missile Response Capability (1)

- **IRIS-T ballistic missile interception performance is limited** Germany's Diehl Defence is rapidly expanding production capacity of IRIS-T air defense systems and interceptor missiles. The IRIS-T SLM and IRIS-T SLS systems can be produced at up to 10 batteries per year in 2026, with plans to expand this to 16 batteries in 2028. Production of IRIS-T family missiles has also increased significantly since the Russia-Ukraine war. To this end, Diehl is continuously expanding its production sites in Germany, and in 2026 it began operating a new missile integration production facility in Nonnweiler.
- IRIS-T was originally developed as a short-range air-to-air missile. The system applying the IRIS-T missile for ground launch is the short-range air defense IRIS-T SLS. The SLS can use the existing air-to-air IRIS-T missile without hardware changes, and its maximum range is about 12km. It is responsible for defending maneuver units and key facilities against threats such as aircraft, helicopters and drones. The IRIS-T SLM, developed afterward, uses the IRIS-T SL interceptor missile, which significantly upgraded the rocket motor of the existing air-to-air type to raise range and interception performance. Maximum range and interception altitude have been expanded to 40km and 20km, respectively. Main targets are aircraft, helicopters, cruise missiles and drones, and ballistic missile interception is not presented as a core mission.
- The next stage of the IRIS-T SLM is the IRIS-T SLX under development. The SLX expands interception range to a maximum of 80km and interception altitude to 25-30km or more. It is being developed in the direction of raising compatibility with the existing SLM system. Radars and engagement control systems are designed to be usable with only partial upgrades. The next-generation launcher is being developed to carry both SLM and SLX interceptor missiles. Diehl is pursuing a strategy of expanding the range and operating scope of interceptor missiles while utilizing the main equipment of existing air defense systems.

[Figure 87] IRIS-T air defense system operating countries and new contracting countries since 2022 (existing operating countries: red, new contracting countries: blue)



Note: Estonia, Latvia, Slovenia, Bulgaria, Switzerland and Denmark are not yet confirmed as fielded
Source: Ministries of defense of each country, Diehl Defence, Hanwha Research Center

Diehl Defence (IRIS-T): Likewise Limited Ballistic Missile Response Capability (2)

▪ **IRIS-SLX is still at the development stage** It still needs to be confirmed whether the IRIS-SLX can directly enter the same ballistic missile defense market as Patriot or Cheongung-II. Diehl explains that the SLX can reduce dependence on Patriot interceptor missiles, but the publicly disclosed SLX performance does not specify ballistic missile interception capability. Reducing dependence on Patriot could mean partially replacing air and cruise missile defense missions rather than ballistic missile interception. The SLX is still at the development stage, and Diehl is targeting the start of production around 2029. Although IRIS-T's long-range air defense capability is rapidly strengthening, we judge that in the short to medium term it is difficult to view it as a system that competes directly for supply with the Patriot or Cheongung-II, which are designed with ballistic missile interception as a core mission.

[Table 74] Production Capacity of Major Weapon Systems, Mostly More Than Doubled Since 2022

Category	IRIS-T SLS	IRIS-T SLM	IRIS-T SLX
System Classification	Short-range air defense	Medium-range air defense	Long-range air defense
Maximum Interception Range	12km	40km	80km
Maximum Interception Altitude	6km	20km	25km+
Main Interception Targets	Aircraft/helicopters/cruise missiles/drones	Aircraft/helicopters/cruise missiles/drones	Aircraft/helicopters/cruise missiles/long-range platforms
Ballistic Missile Interception	Capability not secured	Capability not secured	Capability not secured (under development)
Main Interceptor	IRIS-T air-to-air missile	IRIS-T SL	IRIS-T SLX
Development/Mass Production Status	In mass production/operation	In mass production/operation	Under development
First Fielding	2019	2022	Target 2029

Source: Diehl Defence, Hanwha Research Center

[Table 75] Key Milestones of IRIS-T SLX

Category	Key Details	Remarks
2023.05	Diehl officially announces development of IRIS-T SLX	Target range of 80km, interception altitude of 30km
2024.06	Physical mockup of SLX unveiled at the ILA Berlin exhibition	-
2026.01	Diehl announces SLX production target schedule	Production target in 2029
2026.03	Diehl and Indra sign agreement for joint development/production of air defense systems	Priority cooperation target is medium-range air defense, including SLX
2026.07	Diehl and AEL sign MOU on air defense cooperation	Plans to build a local industrial base for SLX in Brazil going forward

Source: Diehl Defence, press reports, Hanwha Research Center

Patriot Production Increase Strategy: (1) Local European Production of Interceptor Missiles, (2) Expansion of the European Parts Network

- **Patriot production increasing globally, but local PAC-3 production not smooth** Localization of Patriot interceptor missile production in Europe could ease the global interceptor missile supply shortage, which is a negative factor for Korean air defense exports. We judge that the impact will be limited until at least the early 2030s. Patriot's overseas production increase is currently proceeding along two main axes: local European production of the GEM-T and expansion of the European supply chain for the PAC-3 MSE. Local production plans for the GEM-T have already been concretized, but by its nature the product has limits in ballistic missile response capability. For the PAC-3 MSE, which specializes in ballistic missile interception, expansion of the European supply chain is being pursued, but whether finished products will be produced locally has not yet been determined. Even as the Patriot is localized in Europe, it will be difficult to resolve the global interceptor missile supply shortage in a short period.

- Local European production of the GEM-T is already gathering pace. The GEM-T is an interceptor missile of the Patriot PAC-2 family produced by RTX. In Europe, COMLOG, a joint venture in which MBDA Deutschland and RTX each hold 50%, is responsible for supply. COMLOG has mainly handled maintenance, repair and performance upgrades of Patriot missiles operated in Europe. Since the Russia-Ukraine war, the need to rebuild Patriot interceptor missile inventories in Europe has grown, and its role is expanding to new production. COMLOG is building a new GEM-T production facility in Germany, scheduled to start production at the end of 2026 and deliver the first volumes from 2027. Subcomponents are produced in the US and Europe, with final assembly and testing handled in Germany.

[Table 76] Comparison of Key Specifications of Patriot Interceptor Missiles

Category	PAC-2 GEM-T	PAC-3 CRI	PAC-3 MSE
Manufacturer	RTX	Lockheed Martin	Lockheed Martin
First Fielding	2002	Around 2007	2016
Main Interception Targets	Aircraft/cruise and ballistic missiles	Aircraft/cruise and ballistic missiles	Aircraft/cruise and ballistic missiles
Ballistic Missile Interception	Capable	Capable	Capable
Interception Method	Proximity burst/fragmentation	Direct impact (hit-to-kill)	Direct impact (hit-to-kill)
Guidance Method	Ground radar-based track-via-missile	Active radar terminal guidance	Active radar terminal guidance
Propulsion	Solid rocket	Solid rocket	Dual-pulse solid rocket
Loadout per Launcher	4 rounds	16 rounds	12 rounds
Features	Fragmentation warhead-based; multipurpose air defense	Direct-impact method introduced; production-cost-reduced type	Dual-pulse propulsion-based; expanded engagement envelope versus CRI
Production/Operating Status	In production/operation	Existing inventory in operation (estimated)	Main product in production/operation

Source: Lockheed Martin, RTX, Hanwha Research Center

GEM-T Local Production Alone Unlikely to Resolve Interceptor Missile Supply Shortage

- **Even with large-scale GEM-T production increase, Europe can fully absorb it internally** In 2024, the NATO Support and Procurement Agency (NSPA) signed a EUR 5.1 billion contract with COMLOG to supply up to 1,000 GEM-T rounds. Germany, the Netherlands, Romania and Spain jointly participated in the GEM-T purchase, with Germany the largest source of demand. Delivery is scheduled for 2027-2033. Assuming all 1,000 optional rounds are exercised, about 143 rounds a year would be supplied on average from 2027. In 2026, an additional GEM-T contract for several hundred rounds for Ukraine was also signed. This means the new European production line is likely to be used not only to rebuild inventories of existing NATO operating countries but also to replenish Ukraine's depleted stocks.

- Despite the expansion of GEM-T production in Europe, we judge that the impact on Korean air defense exports will be limited. The GEM-T is an interceptor missile that strengthened ballistic missile response capability on the basis of the PAC-2, and it clearly possesses ballistic missile interception capability in itself. The GEM-T intercepts by using a fragmentation warhead that detonates near the target. The PAC-3 MSE adopts the hit-to-kill method of colliding directly with the target and specializes in ballistic missile interception. It is hard to see GEM-T production expansion alone resolving the shortage of Patriot's high-performance ballistic missile defense interceptors. New GEM-T output is expected to go first to rebuilding inventories of existing operating countries and to supporting Ukraine. We judge the volume available to supply new export markets to be limited.

[Table 77] Structure of the NSPA-COMLOG Joint Procurement Contract for GEM-T

Category	Details
Contract Name	NSPA-COMLOG GEM-T Joint Procurement Contract
Ordering Entity	NSPA (NATO joint procurement platform)
Supplier	COMLOG (MBDA Deutschland-Raytheon joint venture)
Contract Date	2024.01
Participating Countries	Germany, the Netherlands, Romania, Spain
Contract Size	Up to EUR 5.1 billion
Maximum Quantity	1,000 rounds
Average Annual Supply	Delivery period 2027-2033; if all 1,000 optional rounds are exercised, average annual supply of about 143 rounds
Production Base	Expansion of GEM-T production capacity in Europe
Expected Effects	Rebuilding inventories of existing NATO operating countries; expanded capacity to replenish Ukraine

Source: MBDA Deutschland, COMLOG, Hanwha Research Center

Local PAC-3 MSE Production Is the Key Variable, but Not Yet Determined

- **PAC-3 MSE supply chain expanding, but local production appears difficult** Whether the PAC-3 MSE will be localized in Europe is the more important variable. In October 2025, Lockheed Martin signed an MOU with Germany's Diehl Defence and began reviewing cooperation to expand the PAC-3 MSE global supply chain. The background is that, following the Russia-Ukraine war and combat use in the Middle East, demand for the PAC-3 MSE has surged, while it has become difficult to meet US and allied demand simultaneously with US production capacity alone. European participation in the supply chain is gradually expanding. However, it is too early to view local European production of the PAC-3 MSE as confirmed. The Diehl-Lockheed Martin MOU is at the level of reviewing the possibility of cooperation such as PAC-3 MSE supply chain expansion and R&D. In Spain and Poland, only some components are being produced locally, and no plan to build a finished-product PAC-3 MSE production line, including final assembly, has been disclosed. There is a possibility that Europe's role will expand from component supply and maintenance to finished-product production, but production location, investment size and mass production timing have not yet been determined.
- Even if local European production of the PAC-3 MSE is decided in the future, considerable time will be needed for building production facilities, certifying the supply chain and ramping up mass production. Even assuming a positive scenario for European localization, we judge that the point at which European-made PAC-3 MSE contributes meaningfully to easing the global interceptor missile shortage is likely to come no earlier than the early-to-mid 2030s.

[Table 78] Key Status of PAC-3 MSE European Supply Chain Expansion

Category	Partner	Announcement Date	Key Details	Progress
Poland	WZL-1	2023	Local production of PAC-3 MSE launch tubes	Production started; first volumes delivered in April 2025
Poland	WZE	2023	Production of missile attitude control system (ACS) and attitude control motors (ACM)	Production line verified in 2025; first ACM delivery to the US planned for 2027
Poland	MESKO	2024.09	Training and equipment support for solid rocket motor production	Production capability building stage
Poland	PGZ, PCO	2024.09	Expansion of PAC-3 MSE missile system test and verification capability (HIL)	Project implementation agreement signed in 2024; HIL demonstration with PGZ succeeded in 2026
Spain	Sener	2024.12	Production of PAC-3 MSE actuators	Construction started in 2026; targeting operation in 2027
Spain	Grupo Oesía-Tecnobit	2026.01	Development of missile cable and harness production line	Order completed; production system build-out under way
Germany	Diehl Defence	2025.10	Cooperation on PAC-3 MSE global supply chain expansion	MOU signed; reviewing supply chain expansion and joint R&D cooperation
Europe joint	Germany, the Netherlands, Poland, Sweden, US	2026.07	Review of European PAC-3 missile maintenance and repair facilities	Government-to-government joint letter of intent signed

Source: Lockheed Martin, Diehl Defence, company data, Hanwha Research Center

Air Defense Conclusion: Supply Shortage Expected to Persist; Export Competitiveness of Korean Air Defense Systems

- **Air defense system supply shortage expected to persist** Although global air defense production capacity is expanding, we judge that the supply shortage is likely to persist until at least the early-to-mid 2030s. Through the Russia-Ukraine war and the US-Iran war, drone and ballistic missile threats have grown, and the need to rebuild inventories has increased owing to large-scale interceptor missile expenditure. The key point is that supply of air defense systems with ballistic missile interception capability remains limited. At present, only a handful of major medium- and long-range air defense systems in the global market have secured ballistic missile response capability, including the Patriot, Cheongung-II and SAMP/T NG. For the SAMP/T NG, first delivery of the next-generation interceptor missile Aster 30 B1NT is scheduled from 2027 and battery production has also begun, but whether there will be a large-scale production increase is hard to confirm yet. Patriot interceptor missile production capacity is expanding, but demand from the US and existing operating countries for force build-up and interceptor missile inventory accumulation is substantial. The IRIS-T and NASAMS need time to secure ballistic missile response capability. Even if global air defense production continues to increase, we judge it unlikely that the long lead times of ballistic missile defense systems will shorten greatly in a short period.
- The export environment is expected to remain favorable for Korean air defense systems. Demand for air defense systems that can complement the Patriot is likely to remain high going forward. In particular, as ballistic missile response capability emerges as an important criterion in air defense system selection, we judge that the export competitiveness of the Cheongung-II will also be maintained. The Cheongung-II has already secured mass production, exports and combat operation experience, and Korea is continuously strengthening its ballistic missile defense capability through development of the Cheongung-III and mass production of the L-SAM. It is also positive that, even as competing systems expand production capacity, a considerable share of the volume is likely to go first to the countries' own force build-up and to rebuilding inventories of existing operating countries. In conclusion, with the supply shortage of global ballistic missile defense systems continuing, we judge that Korea will maintain its medium- to long-term export competitiveness not only by expanding exports of the existing Cheongung-II but also by broadening its product line through the Cheongung-III and L-SAM.

We Judge That the Possibility of Global Fighter Aircraft Lead Times Shortening Quickly Is Limited

- **Fighter makers are aggressively expanding, but bottlenecks persist** Global fighter aircraft production capacity is expanding, but we judge that the possibility of lead times shortening quickly is limited. Since the Russia-Ukraine war, demand has concentrated on ground weapons such as artillery and air defense systems, but fighter aircraft remain a core force in modern warfare. Whereas deliveries of major Western fighters have not changed much since 2022, new orders have increased rapidly. Because it usually takes several years from contract to delivery, the recent increase in orders is expected to lead to higher deliveries going forward.
- Fighter makers have been expanding CAPA since 2021, and if targets are achieved, annual supply capacity is expected to increase to about 200 aircraft. The combined order backlog of major Western fighter makers is estimated at about 700 aircraft. Applying current production capacity, about 6-7 years would be needed to deliver all of them, and even if target CAPA is achieved, about 3.5 years would be needed. The makers' capacity expansions are being delayed from the original targets owing to supply chain bottlenecks and production process stabilization issues. Long delivery times are expected to persist despite capacity expansion, and we judge that this situation will continue until at least the early-to-mid 2030s.
- The KF-21's faster delivery time relative to competing aircraft is likely to act as an important export competitive strength. For fighter aircraft, not only performance and delivery time but also the operational sovereignty a buyer can secure, such as technology transfer, local production and weapons integration, is important. Since there are as yet no export cases for the KF-21, it is difficult to judge the scope of operational sovereignty it can offer. However, considering that the competing Gripen has used active technology transfer and localization strategies as an export strength, we judge that the KF-21 could also secure medium- to long-term order competitiveness if it combines fast supply capability with technology transfer and flexible weapons integration.

[Table 79] Global Major Fighter Aircraft Capacity Expansion Status and CAPA Outlook

Category	Annual Production Capacity	CAPA Expansion Plan	Outlook
F-16	20-30 aircraft (estimated)	48 aircraft	Target achievement delayed; additional expansion unlikely
F-15	12-17 aircraft (estimated)	24 aircraft	Target achievement delayed; additional expansion possible
Rafale	20-30 aircraft (estimated)	48 aircraft	Target achievement delayed; additional expansion possible after achievement
Eurofighter	12-14 aircraft (estimated)	20-30 aircraft	Additional expansion unlikely after target achievement
Gripen	15 aircraft (estimated)	20-30 aircraft	Additional expansion unlikely after target achievement

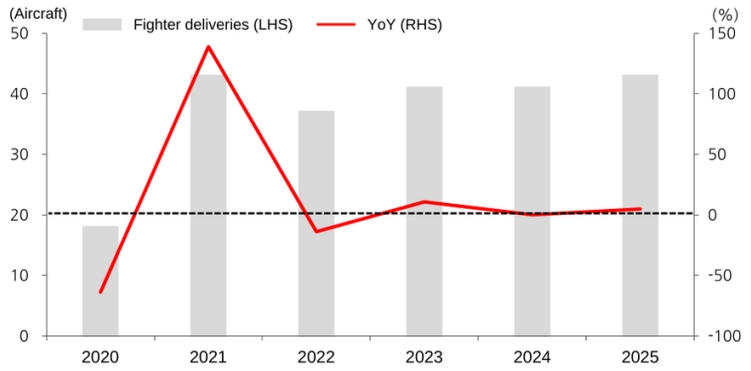
Source: Hanwha Research Center

Fighter Aircraft Supply Expansion Has Only Just Begun (1)

- **With new fighter orders rising, defense companies are responding with capacity expansion** Since the Russia-Ukraine war, global defense market attention has concentrated on ground weapons such as artillery and air defense systems. Demand growth for fighter aircraft has been relatively less pronounced, but there is no doubt that they are a core force in modern warfare, for securing air superiority, suppressing enemy air defenses and precision strikes. As global rearmament expands beyond short-term urgent force replenishment to overall force rebuilding, replacement and force expansion of fighter aircraft are also likely to be unavoidable.
- Annual deliveries of major Western 4.5-generation fighters have not increased significantly since 2022. Excluding the Gripen, which does not disclose deliveries, and the F-16, which has a base effect from the relocation of its production site, annual deliveries of Western 4.5-generation fighters were 38 aircraft in 2022, 40 in 2023, 41 in 2024 and 43 in 2025. Over the same period, new orders accumulated steadily. Replacement of existing fighters and force expansion proceeded simultaneously, centered on Europe, and new introductions continued in the Middle East and Asia as well. Although deliveries have stagnated, the volume to be produced going forward has actually increased.
- Even when new demand arises for fighter aircraft, it takes a long time before actual delivery. After contract signing, parts procurement, airframe manufacturing and test evaluation usually take several years before delivery. Production capacity is also hard to expand in a short period. Not only securing skilled personnel and expanding production facilities but also the supply capacity of major components such as engines, radars and electronic equipment must increase together. We judge that the fact that new orders increased since 2022 have not yet translated sufficiently into higher deliveries is due more to the long production time and limited supply capacity of the fighter aircraft industry than to insufficient demand.
- Fighter makers are expanding production capacity in response to their increased order backlogs. As production capacity rises going forward, annual deliveries are also expected to increase gradually. Because the order backlog currently secured is substantial and introductions of new fighters continue, we judge the possibility that higher output leads to oversupply to be low. The fighter market, where new orders increased first from 2022, is entering a stage in which production and deliveries are beginning to rise in earnest.

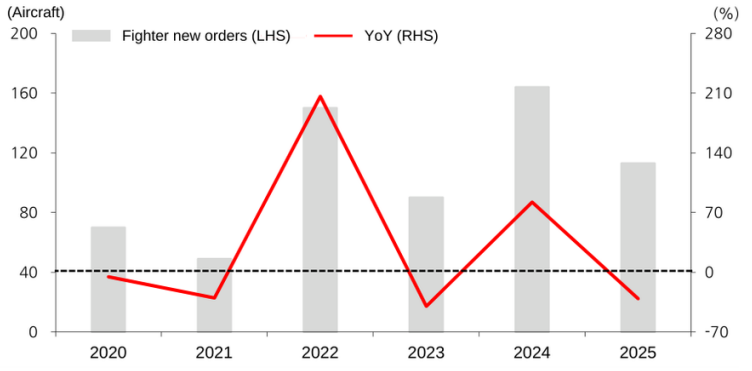
Fighter Aircraft Supply Expansion Has Only Just Begun (2)

[Figure 88] Trend in Annual Deliveries of Western 4.5-Generation Fighters



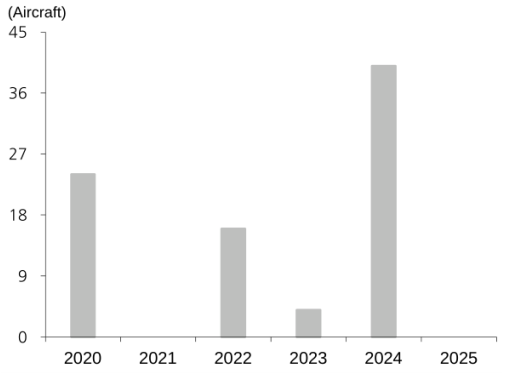
Note: F-15, Rafale and Eurofighter. Excludes the F-16 family, which had a base effect from the relocation of its production site, and the Gripen family, which does not disclose deliveries
Source: Boeing, Dassault Aviation, Airbus, Hanwha Research Center

[Figure 89] Trend in Annual New Orders for Western 4.5-Generation Fighters



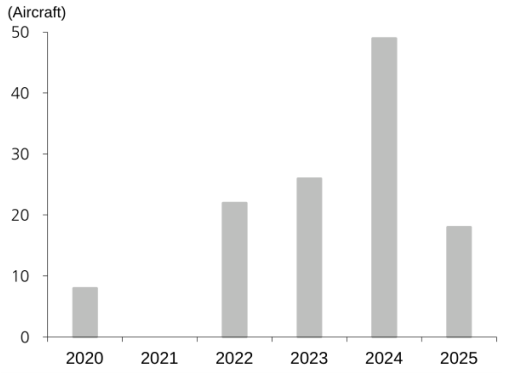
Source: Hanwha Research Center

[Figure 90] Trend in F-16 Fighter New Orders



Source: Lockheed Martin, Hanwha Research Center

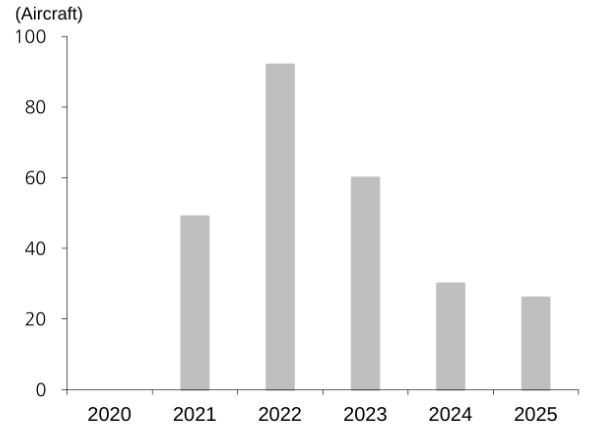
[Figure 91] Trend in F-15 Fighter New Orders



Source: Boeing, Hanwha Research Center

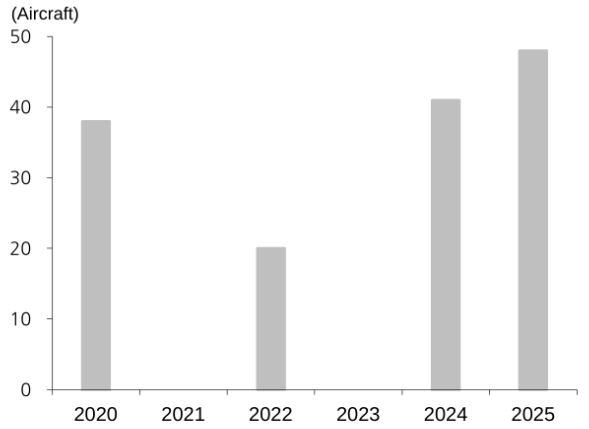
Fighter Aircraft Supply Expansion Has Only Just Begun (3)

[Figure 92] Trend in Rafale Fighter New Orders



Source: Dassault Aviation, Hanwha Research Center

[Figure 93] Trend in Eurofighter Fighter New Orders



Source: Airbus, Eurofighter, Hanwha Research Center

Lockheed Martin (F-16): Deliveries Continue to Fall Short of Target CAPA (1)

- **F-16: production increase delayed; estimated 7-8 years of backlog** Lockheed Martin's F-16 was formerly produced at the Fort Worth plant. As that plant was reorganized around F-35 production, the production line was moved to the Greenville plant in 2017. After building new production facilities and preparing for mass production, Block 70/72 production got under way in earnest from 2020, and actual customer deliveries began in 2024. Lockheed Martin has presented the Greenville line's target production capacity as 4 aircraft per month, or 48 per year. Actual F-16 deliveries were only 16 in 2024 and 16 in 2025. In 2026 as well, software problems and test delays on the F-16s for Taiwan have continued, leaving deliveries far short of the target CAPA.
- The order backlog is still large. As of February 2025, Lockheed Martin stated that the F-16 Block 70/72 backlog to be produced at Greenville was 117 aircraft. Combining aircraft delivered since then with new orders, we estimate that it currently holds an order backlog of about 120 aircraft.
- What matters going forward is whether Lockheed Martin can achieve its annual production target of 48 aircraft. The current backlog equals about 2-3 years on a target CAPA basis. Calculated on the basis of actual annual deliveries of 16 aircraft, it amounts to about 7-8 years. Even if new customers order the F-16, we expect initial delivery to be possible only around 2030. Lockheed Martin has pursued production increases since relocating the production line but has not yet secured a stable production system of 48 aircraft a year, so we judge the impact on supply to the global fighter market in the short term to be limited.

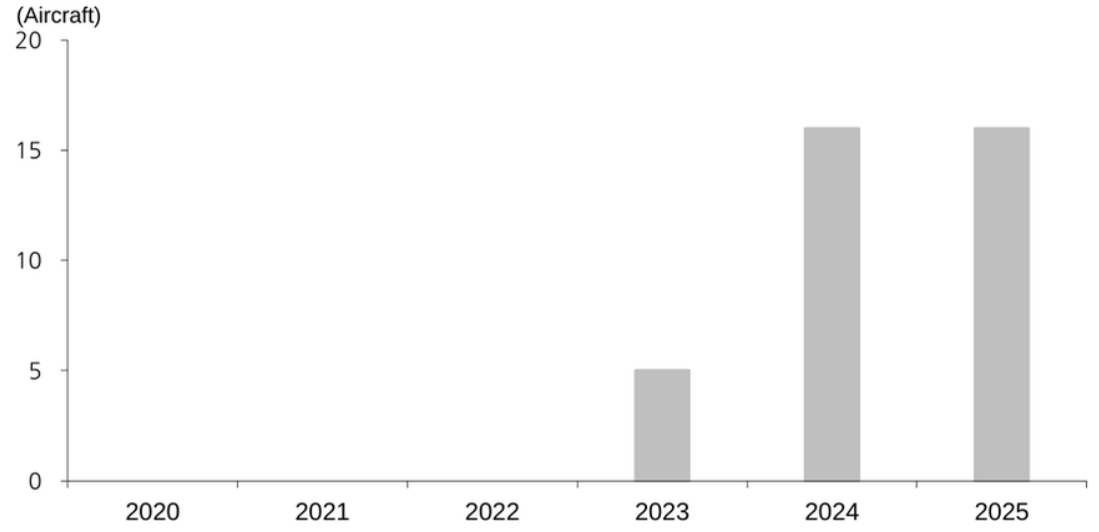
[Table 80] Key History of the F-16 Production Line Relocation and Mass Production

Category	Key Details	Remarks
2017	Fort Worth, Texas, US production ended	F-16 line moved to Greenville to expand F-35 production
2019.04	Greenville, South Carolina production facility opened	-
2019.11	F-16 Block 70 production started	-
2022.11	First F-16 Block 70 airframe rolled out	-
2023.01	Initial rollout	Additional flight testing and certification after first flight of F-16 Block 70
2024.03	First F-16 Block 70 overseas delivery	3 F-16 Block 70 for Bahrain
2024.07	Deliveries to European customers began	2 F-16 Block 70 for Slovakia
2025.02	Deliveries to Bulgaria began	First F-16 Block 70 for Bulgaria
2025.11	Taiwan delivery delay made official	Delivery completion delayed by production line relocation and supply chain problems
2026.10	First 2 aircraft for Taiwan arrived	2 of 66 aircraft; 3 years behind the initial delivery plan

Source: Lockheed Martin, Hanwha Research Center

Lockheed Martin (F-16): Deliveries Continue to Fall Short of Target CAPA (2)

[Figure 94] Trend in F-16 Fighter Deliveries



Source: Lockheed Martin, Hanwha Research Center

[Table 81] Trend in F-16 Fighter Annual Deliveries and Order Backlog

Category	2020	2021	2022	2023	2024	2025
Deliveries	0	0	0	5	16	16
Target Annual Production Capacity	48	48	48	48	48	48
Order Backlog	128	128	136	125	121	104

Note: Excludes F-16 upgrade/maintenance projects. Production capacity is Lockheed Martin's target
Source: Lockheed Martin, Hanwha Research Center

Boeing (F-15): Likewise Difficult to Expand Supply Conditions Significantly in a Short Period (1)

- **F-15: production increase delayed; about 5 years of backlog if target achieved** Since the US Air Force's last F-15E order in 2001, the F-15 has kept its production line going on the basis of overseas export volumes to Saudi Arabia, Qatar, Korea and others. Over the past roughly 10 years, annual deliveries mostly stayed at 20 aircraft or fewer. When the US Air Force began introducing the F-15EX, a substantially upgraded version of the existing F-15, in 2020, an opportunity for production expansion arose. In response to increased US and overseas demand, Boeing has since 2021 presented a target of raising the production rate of the St. Louis line to 2 aircraft per month, or 24 per year. Actual F-15 deliveries were only 14 in 2024 and 9 in 2025, and remained at 4 in the first half of 2026. Compared with the target CAPA of 24 per year, actual deliveries remain low.

- Boeing's F-15 family order backlog is estimated to exceed at least 120 aircraft. Even applying Boeing's target annual CAPA of 24, this corresponds to about 5 years of production volume. Long delivery times are unavoidable even if the target CAPA is achieved. Considering the delivery pace, we judge that the time needed to deliver the entire current backlog will be longer than 5 years. The F-15 order backlog is also likely to grow further. Owing to the aging of the existing F-15E model, the US expanded its final F-15EX procurement plan from 129 to 267 aircraft, of which 120 are currently under contract. In addition, Israel firmly contracted for 25 F-15IA at the end of 2025 and secured an option for another 25. We judge that long delivery times will persist even if F-15 production capacity reaches the target level.

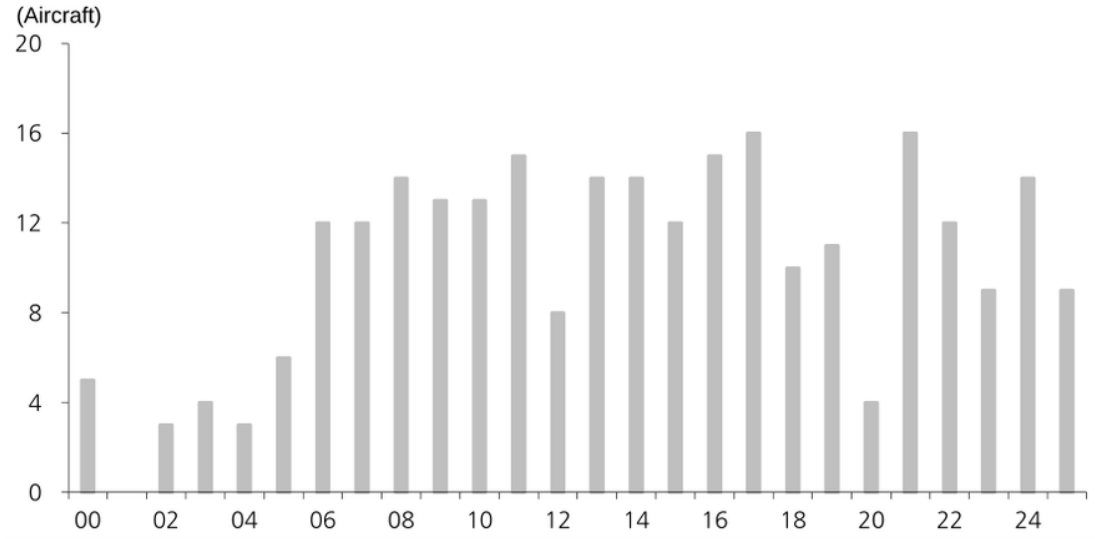
[Table 82] F-15 Fighter New Orders and Delivery Schedule by Country Since 2000

Category	Ordering Country	Model	Contracted Aircraft	Delivery Period	Remarks
2001.04	US	F-15E	10	2002-04	Delivery completed
2002.06	Korea	F-15K	40	2005-08	Delivery completed
2005.12	Singapore	F-15SG	12	2008-09	Delivery completed
2007.01	Singapore	F-15SG	12	Original delivery date unconfirmed	Delivery completed
2008.04	Korea	F-15K	21	2010-12	Delivery completed
2011.12	Saudi Arabia	F-15SA	84	2015-19	Delivery delayed. Completed December 2020
2017.12	Qatar	F-15QA	36	2021-early 2023	Delivery completed
2020.07	US	F-15EX Lot 1	8	FY21-FY23	Delivery delayed. Completed in FY24
2021.11	US	F-15EX Lot 2	12	FY24-	Delivery delayed. Completion in FY26
2022.12	US	F-15EX Lot 3	12	FY25-	Delivery delayed. First delivery in FY26
2023.09	US	F-15EX Lot 4	24	Original delivery date unconfirmed	-
2024.07	US	F-15EX Lot 5	24	Original delivery date unconfirmed	-
2024.11	Israel	F-15IA	25	2031-	-
2025.01	US	F-15EX Lot 6	18	Original delivery date unconfirmed	-
2026.09	US	F-15EX Lot 7	22	Original delivery date unconfirmed	-

Source: Boeing, US Air Force, US Department of Defense, Hanwha Research Center

Boeing (F-15): Likewise Difficult to Expand Supply Conditions Significantly in a Short Period (2)

[Figure 95] Trend in F-15 Fighter Annual Deliveries



Source: Boeing, Hanwha Research Center

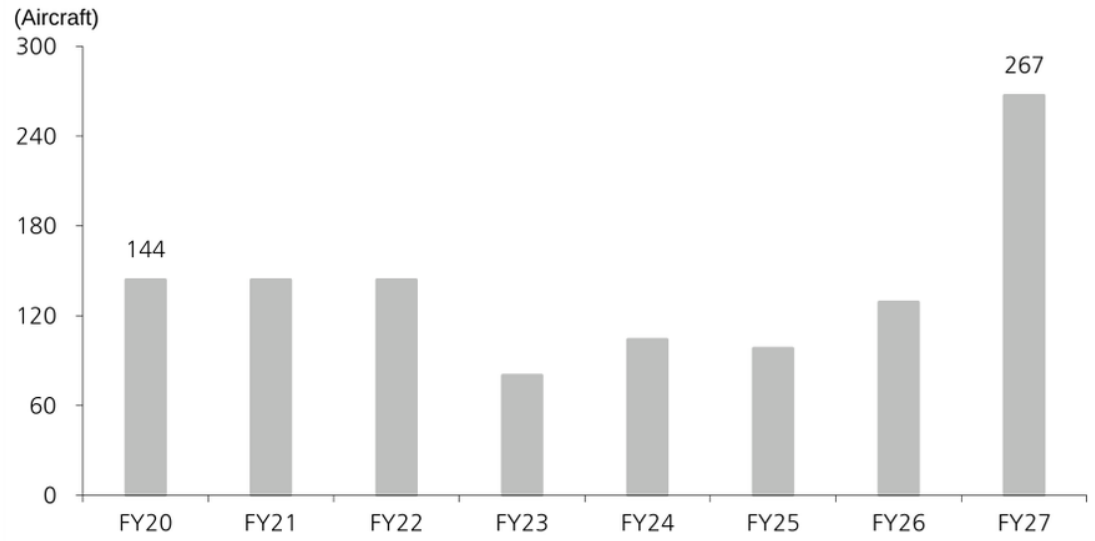
[Table 83] Trend in F-15 Fighter Annual Deliveries and Order Backlog

Category	2020	2021	2022	2023	2024	2025
Deliveries	4	16	12	9	14	9
Target Annual Production Capacity	-	24	24	24	24	24
Order Backlog	44	40	40	55	90	99

Note: Order backlog is a Hanwha Research Center estimate back-calculated from deliveries
Source: Boeing, Hanwha Research Center

Boeing (F-15): Likewise Difficult to Expand Supply Conditions Significantly in a Short Period (3)

[Figure 96] Changes in the US Air Force's Final F-15EX Procurement Target by Fiscal Year

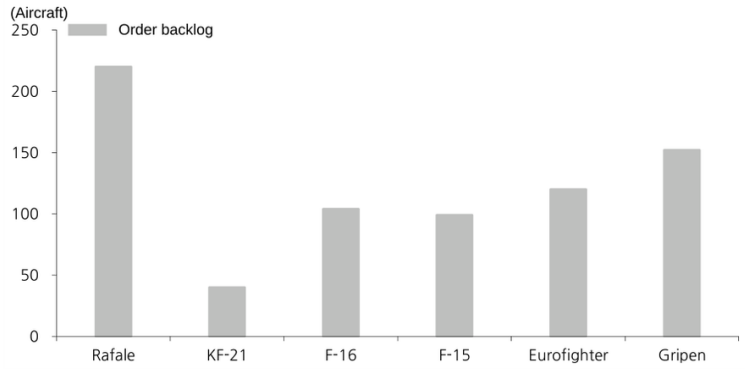


Source: U.S. Air Force, U.S. Department of Defense, Congressional Research Service, Government Accountability Office, Hanwha Research Center

Dassault Aviation (Rafale): Capacity Expansion Focused on Shortening the Backlog Delivery Schedule (1)

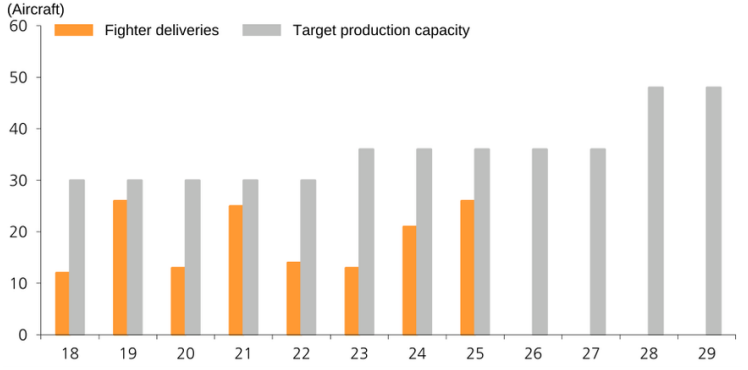
- **Rafale: target of 48 per year; 5-6 years of backlog held even if achieved** Dassault Aviation's Rafale is the fighter with the largest order backlog relative to current production capacity. At the end of 2025, Dassault's Rafale order backlog was about 220 aircraft. The large orders of 80 aircraft from the UAE in 2021 and 42 from France in 2023 had a big impact. Of the order backlog of about 220 aircraft, we estimate that export volumes are at about 170 and French forces' volumes at about 50. Annual Rafale deliveries were only 21 in 2024 and 26 in 2025. In 2025, with 26 new export Rafale orders won, the year-end order backlog stayed at about 220, the same as the prior year. A high order backlog is being maintained as the delivery pace has stagnated relative to increased demand.
- Dassault is continuously expanding CAPA to respond to the increased demand. The Rafale's previous production capacity was about 1 aircraft per month, or 11 per year. From 2015, as overseas orders expanded in earnest, Rafale production capacity expanded to about 2 per month, or 24 per year. In 2023, in response to the large accumulated order backlog, it targeted raising production capacity to about 3 per month, or 36 per year, by the end of 2024. Considering cumulative deliveries in 2024, 2025 and 2026, it appears that achieving the production capacity target is still difficult. Over the medium to long term it targets a CAPA of about 4 per month, or 48 per year, in 2028-29, and assuming it is achieved, we estimate it holds about 4.5 years of work. Simply applying the current annual delivery pace of 26-28 aircraft, about 8 years would be needed just to work through the existing order backlog.
- We judge that the Rafale's CAPA expansion is more likely to be allocated first to reducing the accumulated order backlog than to leading to intensified competition for new orders in the global fighter market.

[Figure 97] Comparison of Order Backlogs of Major Western 4.5-Generation Fighters



Note: Rafale, F-15 and Eurofighter are estimates
Source: Companies, press reports, Hanwha Research Center

[Figure 98] Trend in Rafale Fighter Annual Deliveries and Target Production Capacity



Note: Target production capacity is the monthly production rate annualized, based on the annual production increase plans
Source: Dassault Aviation, AFP, Hanwha Research Center

Dassault Aviation (Rafale): Capacity Expansion Focused on Shortening the Backlog Delivery Schedule (2)

[Table 84] Rafale Fighter New Orders and Delivery Schedule by Country Since 2015

Contract Date	Ordering Country	Model	Contracted Aircraft	Delivery Period	Remarks
2015.02	Egypt	Rafale B/C	24	2015-19	Delivery completed
2015.05	Qatar	Rafale B/C	24	2019-20	Delivery completed
2016.09	India	Rafale B/C	36	2019-22	Delivery completed
2017.12	Qatar	Rafale B/C	12	2021-22	Delivery completed
2021.01	Greece	Rafale B/C	6	2021-23	Delivery completed
2021.01	France	Rafale B/C	12	2025 → end of 2026	Schedule postponed
2021.05	Egypt	Rafale B/C	31	2024-2026 → end of 2027 expected	Delivery delayed
2021.12	UAE	Rafale F4	80	2026-31	-
2022.02	Indonesia	Rafale B/C	42	From 2026, end unconfirmed	-
2022.03	Greece	Rafale B/C	6	From summer 2024	Delivery completed
2023.12	France	Rafale F4	42	From 2027, end unconfirmed	-
2024.08	Serbia	Rafale	12	2029-30	-
2025.04	India	Rafale M	26	2028-30	-

Source: Dassault Aviation, Janes, Hanwha Research Center

Eurofighter GmbH (Eurofighter): CAPA Expansion Proceeding at a Level Matching Demand (1)

- **Eurofighter: annual production capacity target of 20-30 aircraft; 5-7 years of backlog held even if achieved** The Eurofighter consortium is pursuing production capacity expansion in response to demand. Since the end of 2024, a total of 89 new orders have been added from Spain, Italy, Germany and Türkiye alone. This is a steep increase from 38 new orders in 2020, 0 in 2021 and 20 in 2022. The Eurofighter's past annual production capacity is estimated at 12-14 aircraft. In 2025 it presented a target of expanding production capacity over the next 3 years to secure a CAPA of about 20 aircraft a year by 2028. If additional export contracts are secured, it is also reviewing expanding production capacity to 30 or more aircraft a year. Compared with existing production capacity, the 2028 target CAPA is an expansion of about 40-70%, and with additional expansion it is more than double.

- Eurofighters not yet delivered are estimated at about 140. Assuming the 2028 target CAPA is achieved, this corresponds to about 7 years of work. Germany's 20-aircraft contract signed in 2025 is scheduled to begin initial delivery in 2031 and end final delivery in 2034. We judge that the order backlog alone has secured production volume through the early-to-mid 2030s. Even if production capacity is expanded, working through the current order backlog is expected to take until the early-to-mid 2030s. Considering that in the 2010s the Eurofighter consortium voluntarily adjusted production capacity out of concern about a gap in its order backlog, we judge the impact on global fighter supply to be limited.

[Table 85] Eurofighter Production Share Structure by Country

Category	UK	Germany	Italy	Spain
Main Producer	BAE Systems	Airbus	Leonardo	Airbus
Dedicated Production Parts	Front fuselage; canard; vertical tail fin; part of rear fuselage	Central fuselage	Left main wing; part of rear fuselage	Right main wing; leading-edge high-lift devices

Source: Eurofighter GmbH, Airbus, Fondation pour la Recherche Stratégique (FRS), Hanwha Research Center

Eurofighter GmbH (Eurofighter): CAPA Expansion Proceeding at a Level Matching Demand (2)

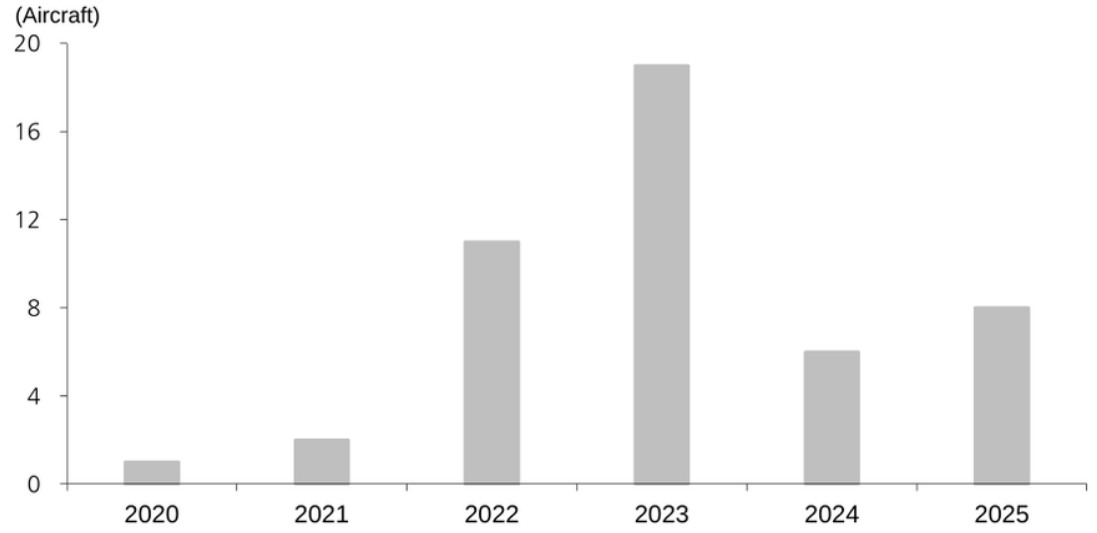
[Table 86] Eurofighter Production Capacity Adjustments and Production Increase Plan

Category	Annual Production Capacity	Key Details
2010.07	60 aircraft	Combined annual output of 60 aircraft across the four countries' production lines
2011.09	Reduction	Decided on production cut in response to decline in new orders; 2011 deliveries of 53 aircraft, reduced to 43 and 39 in 2012 and 2013
2015.11	Further reduction	BAE Systems further reduced production rate; pursuing about 50% cut in in-house assembly and parts production rate
2017.01	UK reduction	Announced plan to cut UK output from 20 aircraft in 2017 to 11 in 2018
2020.11	Maintained	Germany signed 38-aircraft contract
2022.06	Maintained	Spain signed 20-aircraft contract
2024.12	Maintained	Spain signed 25-aircraft and Italy up to 24-aircraft contracts
2025.06	Production increase decided	Announced expansion of annual production capacity from 12-14 to 20 aircraft within 36 months, targeting mid-2028; mentioned possible expansion to 30 or more if additional export contracts are secured, timing undetermined

Source: Eurofighter GmbH, Hanwha Research Center

Eurofighter GmbH (Eurofighter): CAPA Expansion Proceeding at a Level Matching Demand (3)

[Figure 99] Trend in Eurofighter Fighter Deliveries



Source: Eurofighter GmbH, Hanwha Research Center

[Table 87] Trend in Eurofighter Fighter Annual Deliveries and Order Backlog

Category	2020	2021	2022	2023	2024	2025
Deliveries	1	2	11	19	6	8
Target Annual Production Capacity	-	-	-	-	-	20
Order Backlog	90	88	96	77	112	152

Note: Order backlog is an estimate, calculated as cumulative orders less cumulative deliveries
Source: Airbus, Eurofighter GmbH, Janes, Hanwha Research Center

Saab (Gripen): Additional Expansion Unlikely After Target CAPA Achieved (1)

- **Gripen: production increase delayed; annual production target of 20-30 aircraft; 3-5 years of backlog held even if achieved** Saab is aggressively targeting a Gripen production increase. Whether mass production of the latest version, the Gripen E, has entered a stabilization phase needs to be confirmed. Gripen E mass production began around 2019. In 2020, Saab stated that it could later raise the Gripen production rate to about 24 aircraft a year. However, even in April 2026, six years later, the actual production rate remains at about 15 aircraft a year. This is because additional functionality development continued during Gripen E mass production, and the plan to reuse existing Gripen C/D equipment was changed to applying new equipment, disrupting the production schedule.
- Saab plans to expand Gripen production capacity to about 20-30 aircraft a year. According to Saab, production at about 20 aircraft a year is expected to be possible in 2027, with the production rate likely to rise further in 2028. The current Gripen order backlog is estimated at about 90-100 aircraft. Applying the current production rate of 15 a year, this corresponds to more than 6 years of production volume. Even applying the 2027 mass production target of 20 a year, it is about 4-5 years of volume.
- The order backlog is also likely to increase further going forward. Potential demand includes the possible introduction of the Gripen by Ukraine, follow-on introductions by Thailand and additional orders from Sweden. Even if production capacity expands according to Saab's plan, considerable time is expected to be needed to work through the existing order backlog. If new orders also increase, we judge that the Gripen's long delivery times could persist into the 2030s.

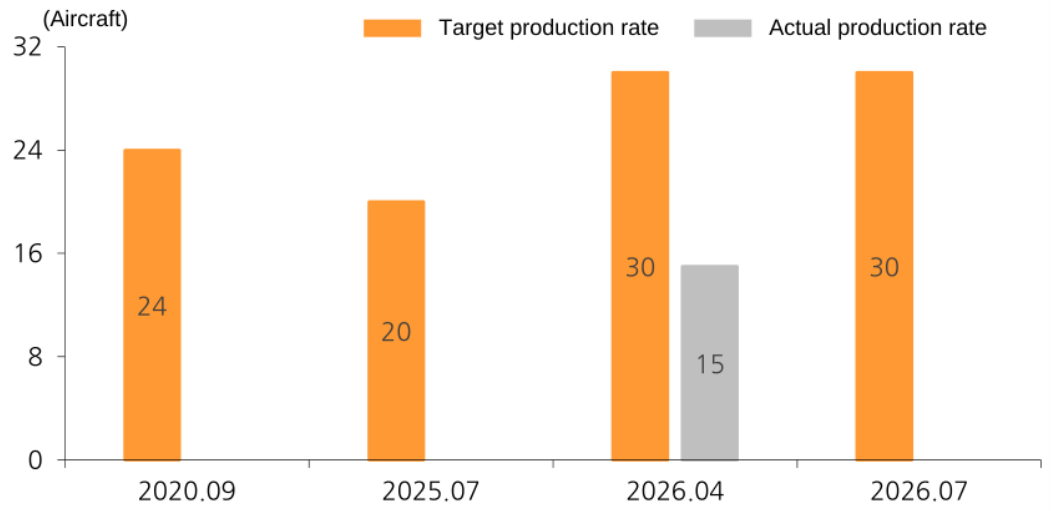
[Table 88] Comparison of Key Performance and Specifications of Gripen Generations

Category	Gripen A/B	Gripen C/D	Gripen E/F
First Flight	1988	2002	2017
Engine	RM12	RM12	F414G
Radar	PS-05/A	PS-05/A (improved)	Raven ES-05 AESA
IRST	Not fitted on baseline version	Not fitted on baseline version	Fitted
Electronic Warfare Capability	Basic threat detection and defense	Improved electronic warfare and self-defense	New system, strengthened active jamming
Weapon Stations	8	8	10
Main Features	Early multirole fighter	NATO compatibility, expanded weaponry	Strengthened AESA, IRST, electronic warfare

Source: Saab, JetPhotos, Aviation Week, Wikimedia Commons, Hanwha Research Center

Saab (Gripen): Additional Expansion Unlikely After Target CAPA Achieved (2)

[Figure 100] Comparison of Gripen Fighter Target Production Rate and Actual Production Rate



Source: Saab, Hanwha Research Center

[Table 89] Gripen Fighter Overseas New Orders and Delivery Schedule Since 2010

Contract Date	Ordering Country	Model	Contracted Aircraft	Delivery Period	Remarks
2010.11	Thailand	Gripen C	6	2010-2013	Delivery completed
2014.10	Brazil	Gripen E/F	36	2019-2024	Delivery delayed
2024.02	Hungary	Gripen C	4	Original delivery date undisclosed	Delivery completed in June 2026
2025.08	Thailand	Gripen E/F	4	2025-2030	-
2025.11	Colombia	Gripen E/F	17	2026-2032	-
Total			**67**		

Source: Saab, Hanwha Research Center

Operational Sovereignty, Another Competitive Factor in Fighter Exports

- **Operational sovereignty, as important as delivery time** Production capacity and delivery time are key factors determining competitiveness in winning fighter orders, but from the buyer's standpoint, how autonomously it can operate the fighter after introduction is also important. The value of operational sovereignty rises further when new weapons and sensors must be introduced, or new threats responded to, as the security environment changes.
- Operational sovereignty is a broader concept than simple technology transfer. Typical examples are whether the buyer can directly reflect in the fighter the information needed to identify and respond to a new enemy radar when one appears, and whether it can use missiles or electronic warfare equipment developed domestically without the manufacturing country's help. Whether maintenance and upgrades can be performed domestically is also important. The more of these authorities a buyer secures, the wider the scope in which it can respond independently to new threats and changes in the operating environment. Local production and transfer of core technology are also means of expanding operational sovereignty. Depending on the scope of operational sovereignty the buyer secures after introducing the fighter, its dependence on the supplying country can differ.
- Operational sovereignty has long been a major competitive factor in the fighter export market. The Eurofighter emphasizes that when a new threat appears, the buyer can directly reflect the information it needs in the fighter and respond. In selling the Gripen, Saab promotes as a major strength that buyers can apply weapons and equipment they developed domestically to the fighter and participate directly in future upgrades. In the sale to Brazil, technology transfer was carried out across design and development, flight testing, avionics and weapons integration, and a local production line was even built.

[Table 90] Key Evaluation Items for Fighter Aircraft Operational Sovereignty

Category	Authority the Buyer Can Secure
Mission Data, Electronic Warfare	Own modification/updating of enemy radar information and electronic warfare data
Software	Software modification/upgrades without manufacturing country approval
Weapons Integration	Independent integration of domestically developed missiles/weapons
Technology Transfer	Securing fighter design/development/test technology
Local Production	Domestic assembly of fighters and parts production
Maintenance/Upgrades	Independent maintenance and upgrades of major equipment

Source: Hanwha Research Center

The US and Europe: Different Operational Sovereignty Strategies

- **US and European fighter makers actively marketing operational sovereignty** The biggest strength of the US-made F-16 and F-15 is the vast operating base built around the world. Many allies operate the same fighters and weapons, making joint operations easy, and a variety of already proven weapons such as US-made missiles and bombs can be used without separate development. However, the scope within which a buyer can directly change the fighter's core software or independently apply new weapons is affected by the range of technology and information the US government allows.
- Unlike US-made fighters, the Rafale promotes as a strength its relatively low dependence on US technology approval and export policy. This is because France can supply the fighter, radar, electronic warfare equipment and major weapons on its own. The Eurofighter emphasizes operational sovereignty more directly. It presents as a competitive strength that the buyer can manage mission data itself and operate the fighter to its own requirements. The Eurofighter's joint-development structure also has the limitation that prior coordination among the Eurofighter development countries is needed for new upgrades and weapons integration.
- The system showing the most active operational sovereignty strategy is the Gripen. A representative order is the Brazil case. In the Brazil program, Saab included in the scope of technology transfer not just simple local assembly but design and development, flight testing, avionics and weapons integration. It provided the related technology and authority so that Brazil could directly select the weapons and equipment it needs and apply them to the Gripen. We judge that the Gripen has chosen a strategy of differentiating on the buyer's ability to secure its own fighter operation and upgrade capability, rather than competing with the huge ecosystem of US-made fighters.

[Table 91] Comparison of Operational Sovereignty Strategies of Major Western 4.5-Generation Fighters

Category	F-16 / F-15	Rafale	Eurofighter	Gripen
Manufacturer	Lockheed Martin/Boeing	Dassault	European consortium	Saab
Main Strength	Broad weapon systems	French technological self-reliance	Mission data autonomy	Technology transfer/localization
Weapons Integration	Wide variety of US weapons	French/European weapons	US/European weapons	Weapons integration technology transfer
Software/Electronic Warfare	Dependent on US approval	French proprietary technology	Autonomous modification of mission data	Participation in software development/testing
Technology Transfer	Limited	Provided depending on negotiation	Provided depending on negotiation	Broad technology transfer
Local Production	Local assembly permitted	Local production cooperation	Local production cooperation	Local production support
Main Constraints	US government approval	Reliance on French technology	Coordination among developing countries	Reliance on foreign parts

Source: Lockheed Martin, Boeing, Dassault Aviation, Eurofighter, BAE Systems, Saab, Hanwha Research Center

KF-21: If Operational Sovereignty Is Added to Fast Delivery

- **KF-21: conditions for competitive differentiation** Since the KF-21 has no export contract yet, it is difficult to predict what level of operational sovereignty it will offer overseas customers. At this point it cannot be asserted that the KF-21 will offer the same level of technology transfer or weapons integration authority as the Gripen. Because engines and some components introduced from overseas are also applied to the KF-21, Korea cannot independently decide transfer authority for all technologies and weapons. The actual scope of operational sovereignty must be confirmed in the first export contract.
- We judge that the KF-21, like the Gripen, may pursue an export strategy that puts technology transfer and local production first. From the KF-21's development stage, Korea brought Indonesia in as a co-development country and included technology transfer and local industry participation in the program structure. In future exports as well, there is room to present local production and technology cooperation as a package rather than simple sales of finished aircraft. This could be an important competitive strength in targeting countries that want to foster their own defense industries.
- If price and delivery-time competitiveness are added to technology transfer and local production, we judge the KF-21's export competitiveness will rise further. 4.5-generation fighter makers hold large order backlogs, so even if a new customer signs a contract, considerable time is needed before actual delivery. The KF-21's production capacity is planned to expand gradually, starting with deliveries to the Korean Air Force. If it can offer a lower acquisition price and faster delivery than competing aircraft, it can turn the existing fighter suppliers' long delivery times into an export opportunity. It could be differentiated from existing competing aircraft in that it can support not only price and delivery time but also the buyer's development of its defense industry.

[Table 92] KF-21 Key Export Competitiveness and Future Tasks

Category	Direction for Securing Competitiveness	Current Status
Price	Lower acquisition price than competing aircraft	Overseas sale price undisclosed
Delivery Time	Faster supply than competing aircraft	Domestic mass production under way
Technology Transfer	Buyer participation in development/testing	History of Indonesia's participation in co-development
Local Production	Building production base in buyer's country	No overseas production contract
Weapons Integration	Support for buyer-developed weapons integration	Authority to be offered overseas undisclosed
Software	Support for buyer's independent modification/upgrade	Authority to be offered overseas undisclosed

Note: Authority related to technology transfer, local production, weapons integration and software may vary depending on the terms of future export contracts
Source: Hanwha Research Center

Limited Capacity Expansion by Competitors; K2 Export Competitiveness Expected to Be Maintained

- **Transition period to next-generation tanks; production increases expected to be very limited** Global tank production capacity is expanding, but we judge the possibility of oversupply to be limited. In Europe, KNDS Deutschland is expanding Leopard 2 production capacity, but with a large existing order backlog, it is unlikely that delivery times will shorten significantly in a short period. After reaching annual production capacity of about 40-50 tanks in the late 2020s, whether to expand further is likely to be decided by the size of new orders. In the US, General Dynamics is pursuing modernization of existing Abrams production facilities, but its focus is on the transition to mass production of the next-generation M1E3 rather than a large production increase. We judge that global tank production capacity is unlikely to greatly exceed demand until at least around 2030. The supply environment is favorable for Hyundai Rotem. The K2 has secured stable production capacity and export results in Poland, so limited supply expansion by competitors could lead to new order opportunities. Building a local production base in Poland is expected to raise supply stability and export competitiveness in the European market going forward.

- The Russia-Ukraine war has put renewed emphasis on the importance of ground warfare and has also re-illuminated the strategic value of tanks. The US-Iran war highlighted the importance of air defense systems, and with the spread of drones, concerns about tank survivability have also grown. However, tanks are an essential force for ground troop maneuver, fire support and area seizure. Rather than the role of tanks shrinking, they are expected to evolve toward strengthened protection against new threats. Hyundai Rotem plans to apply active protection systems and drone jammers to the K2PL and is also developing AI-based counter-drone protection systems and manned-unmanned teaming technology. With technology development and the expansion of the local production base in Poland interlocking, we expect the K2's medium- to long-term export competitiveness to be maintained as well.

[Table 93] Global Major Tank Capacity Expansion Status and CAPA Outlook

Category	Annual Production Capacity	CAPA Expansion Plan	Outlook
Leopard 2	20-30 tanks (estimated)	40-50 tanks (estimated)	Additional CAPA expansion unlikely
M1A2	180 tanks (estimated)	Capital investment under way	Plant modernization presumed to be for next-generation tank production
Leopard 2 A-RC 3.0	-	Under development	Mass production schedule undecided
KF51	-	Under development	Mass production schedule undecided
MGCS	-	Under development	Mass production targeted around 2040
M1E3	-	Under development	Mass production expected to begin in earnest before 2030

Source: Hanwha Research Center

KNDS Deutschland (Leopard 2): Capacity Expansion Expected to Peak in the Late 2020s

- **Leopard 2 production increase presumed under way; annual 40-50 tanks possible; additional expansion unlikely** KNDS Deutschland does not disclose annual production capacity of the Leopard 2 tank. Some overseas media report that production of about 50 a year is possible, but there is a lack of specific production results or official data to support this. Production facility expansion and supply chain expansion are under way in Germany and overseas. Judging from the delivery times of recent new order contracts, production capacity is presumed to be improving gradually. Looking at new-build tank contracts signed since 2020, the time from contract to first delivery has shortened from about 3 years in the past to about 2 years recently. Considering the pace of capacity expansion, we estimate that from 2028 the Leopard 2's annual production capacity will expand to about 40-50 tanks.

- The Leopard 2 order backlog is estimated at 300 or more tanks, and many contracts are scheduled to end deliveries around 2030. We judge that KNDS's production capacity expansion is focused on working through existing order volumes and shortening delivery times rather than preparing for new orders. Even if annual production capacity expands to about 40-50 tanks going forward, whether to expand further is likely to be decided by the size of new orders. Since a considerable portion of the current order backlog is expected to be exhausted around 2030, continuous new orders will be needed to maintain high utilization thereafter. If large additional orders are not secured, we judge that additional expansion after reaching the target production capacity in the late 2020s will be limited.

[Table 94] Leopard 2 Tank Ordering Countries and Delivery Schedule Since 2020

Category	Ordered Quantity	Delivery Period	Remarks
Norway	54	2026-2028	First 2 delivered in March 2026
Germany	18	2025-2026	Delivery delay presumed
Germany	105	2027-2030	-
Sweden	24	2028-2031	-
Sweden	20	2028-2031	-
Lithuania	44	2028-2030	Originally scheduled for 2029-2034, but shortened
Netherlands	46	2028-2031	-
Czech Republic	44	2028-2031	-
Croatia	44	2028-2030	-

Source: Hanwha Research Center

Europe's Next-Generation Tank Development: Full-Scale Mass Production of New Tanks After the 2030s (1)

- **Europe developing next-generation tanks; mass production in the 2030s or later** In Europe, along with expansion of production capacity of existing tanks, next-generation tank development is also under way. The main programs are KNDS Deutschland's Leopard successor, Rheinmetall's KF51 Panther, and the MGCS co-developed by France and Germany. Most are still at the development stage, so their impact on global tank supply capacity in the late 2020s is expected to be limited.
- Germany is pursuing development of a new intermediate-stage tank to be introduced before the MGCS is fielded. KNDS Deutschland is developing the Leopard 2 A-RC 3.0, which applies an unmanned turret and autoloader to the existing Leopard 2, and development of a successor tank referred to as the Leopard 2AX or Leopard 3 is also being pursued separately. The specific development completion date and mass production schedule have not been determined, and there are no volumes for which actual purchase contracts have been signed.
- Rheinmetall is pursuing entry into the next-generation tank market through the KF51 Panther. The KF51 is a new tank applying a 130mm main gun and autoloader, and in 2023 it signed a development contract of about EUR 290 million with Hungary. It is also pursuing local development and production in cooperation with Italy's Leonardo. Italy plans to introduce up to 132 tanks and 140 support vehicles. Neither Hungary nor Italy has yet signed a large-scale mass production contract.
- Over the long term, there is the MGCS program co-developed by France and Germany. The MGCS is a next-generation ground combat system to replace the Leopard 2 and Leclerc, and is being pursued with the aim of fielding around 2040. KNDS, Rheinmetall and Thales are participating in the co-development, but the specific mass production schedule and purchase volume have not yet been determined.
- Europe's next-generation tank development is a factor that could expand competition in the global tank market over the medium to long term. However, programs for which large purchase volumes are confirmed are limited, and considerable time is needed before full-scale mass production. We judge that the effect of new tank development in expanding global supply capacity will be limited until at least around 2030.

Europe's Next-Generation Tank Development: Full-Scale Mass Production of New Tanks After the 2030s (2)

[Table 95] Status of Major European Next-Generation Tank Development and Potential Demand

Category	Leopard 2 A-RC 3.0	KF51 Panther	MGCS
Developer	KNDS Deutschland	Rheinmetall	KNDS, Rheinmetall, Thales
Developing Country	Germany	Germany	Germany, France
Key Progress	Technology demonstrator unveiled in 2024	Prototype unveiled in 2022; Hungary development contract in 2023	Program launched in 2017; joint venture established in 2025
Features	Unmanned turret; autoloader; 120-140mm main gun	130mm main gun; autoloader	Integration of manned/unmanned combat systems
Development Status	Technology demonstration stage	Pursuing co-development with Italy	Pursuing co-development
Mass Production Target	Undecided	Undecided	Around 2040
Potential Buyers	Germany	Hungary, Italy	Germany, France
Potential Purchase Volume	Undisclosed	Italy up to 132 tanks	Undisclosed
Confirmed Mass Production Contract	None	None	None

Source: KNDS, Rheinmetall, Hanwha Research Center

General Dynamics (M1 Abrams): Focus on Production Facility Modernization and Transition to Next-Generation Tank

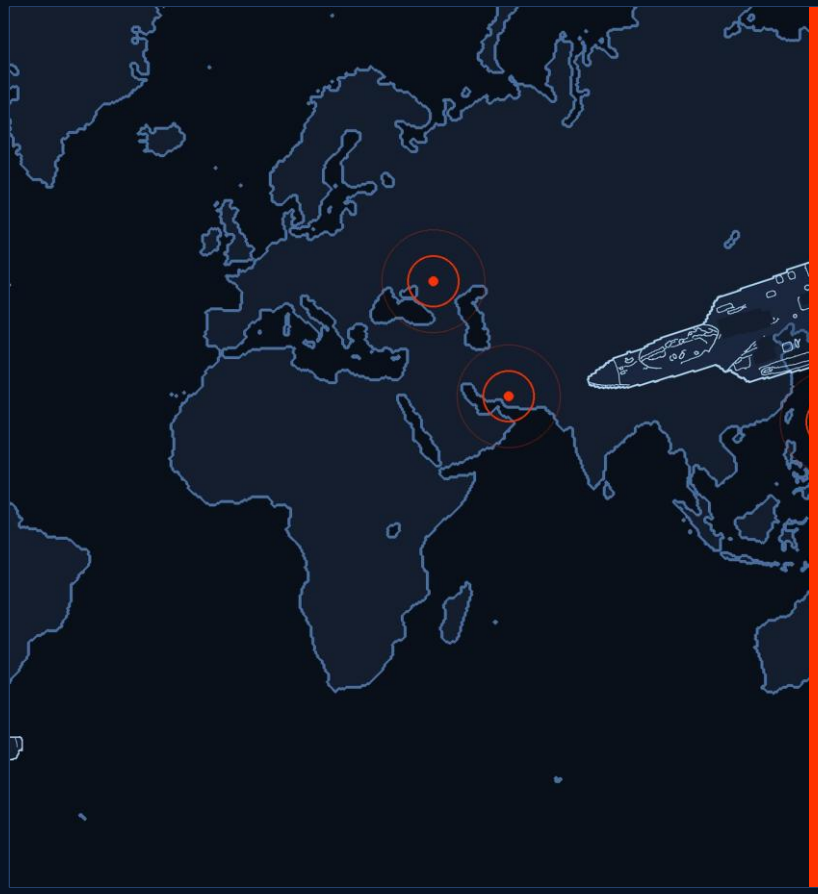
- **Abrams: capacity expansion unlikely; focus on next-generation tank** The Abrams discontinued new production in the 1990s and has since produced upgraded models by remanufacturing existing tanks. In the production process, the hulls and major parts of existing M1A1 and M1A2 tanks are disassembled and refurbished at the US Anniston depot, then finally assembled into the latest M1A2 SEPv3 at the Lima plant. The Abrams production rate mentioned in the US Congress in 2022 was about 15 tanks a month, or about 180 a year. Since the Russia-Ukraine war, it has been pursuing production facility modernization. In 2023 it announced investment of USD 558 million in the Lima plant over the next 15 years, and in 2026 it completed a new welding facility worth USD 44 million at the Anniston depot. Automation and expansion of turret fabrication facilities are also being pursued at Lima. How much annual production capacity will expand has not been disclosed.

- We judge that the expansion of Abrams production capacity going forward will focus on the transition to mass production of the next-generation M1E3 tank rather than a production increase of the existing model. In 2023, the US canceled development of the existing M1A2 SEPv4 and switched to M1E3 development. This is because the weight of the M1A2 rose as protective armor and electronic equipment were continuously added, which increased the mobility and logistics support burden. Since the M1E3 redesigns the hull and turret structure and the protection system, changes are also expected in the existing remanufacturing-centered production structure. The US plans to validate new manufacturing processes through initial production vehicles of the M1E3 from FY2027. In the same year, the procurement request for the existing Abrams upgrade program is only 22 tanks. Since General Dynamics' specific production capacity expansion target has also not been disclosed, we judge that the possibility of the Abrams causing global tank oversupply is limited.

[Table 96] Key Development and Mass Production Schedule of the Next-Generation Abrams M1E3

Schedule	Key Details
2023.09	M1A2 SEPv4 development ended; decision to switch to next-generation M1E3 development
2025.12	Initial M1E3 prototype vehicle secured
2026.01	M1E3 prototype vehicle unveiled for the first time
2026	Pursuing M1E3 prototype performance testing and operational verification
FY2027	Pursuing initial trial production and validation of new manufacturing processes
FY2028-29	Plan to deliver initial production vehicles based on the FY2027 budget
Early 2030s	Target to secure initial operational capability (IOC). Future schedule subject to change

Source: U.S. Army (2023.09), CRS (2026.04), U.S. Army FY2027 Budget, Hanwha Research Center



VI

Company Analysis

Hanwha Aerospace · Hanwha Systems · LIG Defense & Aerospace · Korea Aerospace Industries · Hyundai Rotem

Defense Sector: Beyond Infinity

Hanwha Aerospace (012450): Still Many Opportunities Bigger Than Expected



Buy (Initiation)

Target Price (New)

1,300,000 KRW

Current Price (10/23)

792,000 KRW

Upside

▲64.1%

Market Cap	**40,838.2 bn KRW**
Shares Outstanding	**51,563 thousand**
52-Week High / Low	**1,537,000 / 792,000**
90-Day Avg. Daily Value	**174.04 bn KRW**
Foreign Ownership	**44.9%**
Hanwha (+7)	**35.8%**
Natl. Pension Service (+1)	**7.9%**
Employee Stock Own. Assoc. (+1)	**1.3%**

Share Price Returns (%)

Return	1M	3M	6M	12M
Absolute	-22.4	-17.3	-44.4	-24.5
Rel. (KOSPI)	-0.8	4.5	-30.1	-68.8

▪ Despite the achievement of entering the US defense market, the share price has been sluggish, but earnings growth potential through 2029 remains valid on the back of the large order backlog. Despite production increases by European defense companies, the global artillery system supply shortage is expected to persist, and we judge that the company's opportunity to expand market share is actually growing.

Hanwha Aerospace Financial Summary (Unit: KRW billions, KRW, %, x)

Category	2024	2025	2026E	2027E
Revenue	11,240	26,703	31,721	33,251
Operating Profit	1,732	3,089	4,619	5,502
EBITDA	2,063	3,962	5,587	6,593
Net Income (Ctrl)	2,299	1,405	2,366	3,249
EPS	49,621	27,247	45,986	63,156
Net Debt	7,536	4,717	6,913	7,649
PER	16.0	29.1	17.2	12.5
PBR	7.2	4.2	3.1	2.6
EV/EBITDA	23.5	11.5	8.5	7.4
Dividend Yield	0.4	0.9	1.0	1.3
ROE	53.9	19.1	20.8	22.4

Entry Into the US Defense Market: An Achievement the Share Price Has Yet to Reflect

- In August 2026, the company was selected as a prototype program contractor for the US wheeled self-propelled howitzer. It is significant in that, after competition with major global defense companies, it secured an opportunity to enter the US defense market, which accounts for about one-third of global military spending. It is currently at the prototype contract stage, but we note the possibility of expansion into a mass production program in the future as the US Army's need for self-propelled howitzer modernization rises. We also rate highly the company's likelihood of entry into the tracked self-propelled howitzer modernization program. The share price has been sluggish owing to weak share prices of global defense peer groups and a shift of investor interest to other sectors.

Artillery Systems: Market Share Expected to Expand Amid Supply Constraints

- The global supply constraint on self-propelled howitzers and multiple launch rocket systems, which are core to the company's earnings, is expected to continue for a considerable period. European defense companies are pursuing production capacity expansion, but we judge that a considerable portion of the increasing supply will be absorbed by intra-European demand alone. It is unlikely that global order competition will intensify sharply over the medium to long term. While maintaining a stable market position in tracked self-propelled howitzers, the company is expected to expand its share in the wheeled self-propelled howitzer market as well. We judge that Chunmoo will rapidly erode HIMARS' global market share on the basis of relatively faster delivery times and price competitiveness.

Buy Rating, Coverage Initiated With a Target Price of KRW 1,300,000 (1)

- Despite various macro uncertainties, the company's large defense order backlog is the key basis for medium- to long-term earnings growth. As deliveries of existing orders expand, profit growth is expected to continue not only in 2027 but also in 2028-2029. Short-term share price momentum is expected to focus on the progress of export projects in the Middle East. Considering stable earnings growth and strengthening competitiveness in the global artillery system market, we initiate coverage of Hanwha Aerospace with a Buy rating and a target price of KRW 1,300,000. The target price was calculated by applying a target PER to expected EPS.

[Table 97] Hanwha Aerospace Valuation

Category	Value	Remarks
Expected EPS (KRW)	63,156	2027 expected EPS
Target P/E (x)	21.0	Applied average 2027 PER of 20.7x for two overseas defense companies (Rheinmetall, General Dynamics)
Fair Price (KRW)	1,326,266	Expected EPS * Target P/E
Target Price (KRW)	1,300,000	
Current Price (KRW)	792,000	
Upside (%)	64.1	

Source: Hanwha Research Center

[Table 98] Hanwha Aerospace 3Q26 Earnings Estimates (Unit: KRW billions)

Category	3Q25	4Q25	1Q26	2Q26	3Q26E Estimate	3Q26E Consensus	YoY	QoQ	Difference vs. Consensus
Revenue	6,486	8,421	5,751	9,293	7,128	7,507	9.9%	-23.3%	-5.0%
Operating Profit	890	808	639	1,365	1,127	1,167	26.6%	-17.5%	-3.4%
Net Income (Ctrl)	533	632	261	636	612	544	14.8%	-3.9%	12.4%
Operating Margin	13.7	9.6	11.1	14.7	15.8	15.5	2.1%p	1.1%p	0.3%p
Net Margin	8.2	7.5	4.5	6.8	8.6	7.2	0.4%p	1.7%p	1.3%p

Source: FnGuide, Hanwha Research Center

Buy Rating, Coverage Initiated With a Target Price of KRW 1,300,000 (2)

[Table 99] Hanwha Aerospace Quarterly and Annual Earnings Trend (Unit: KRW billions, %)

Category	2025	2026(E)	2027(E)	2028(E)	2026 YoY	2027 YoY
Revenue	26,703	31,721	33,251	36,443	18.8	4.8
Ground Defense	8,133	9,116	11,153	12,277	12.1	22.4
Aerospace	2,513	2,882	3,491	3,996	14.7	21.1
Hanwha Systems	3,664	4,107	4,520	5,308	12.1	10.1
Hanwha Ocean	12,688	15,806	14,562	15,290	24.6	-7.9
Other	-391	-485	-475	-428	-	-
Operating Profit	3,120	4,619	5,502	6,043	48.0	19.1
Pretax Income	2,241	4,075	5,030	5,493	81.9	23.4
Net Income	1,432	2,366	3,249	3,548	65.2	37.3
Operating Margin	11.7	14.6	16.5	16.6	-	-
Pretax Margin	8.4	12.8	15.1	15.1	-	-
Net Margin	5.4	7.5	9.8	9.7	-	-

Source: Hanwha Aerospace, Hanwha Research Center

Buy Rating, Coverage Initiated With a Target Price of KRW 1,300,000 (3)

Hanwha Aerospace Income Statement (Unit: KRW billions)

Dec Year-End	2023	2024	2025	2026E	2027E
Revenue	7,890	11,240	26,703	31,721	33,251
Gross Profit	1,428	2,870	5,450	6,538	8,156
Operating Profit	594	1,732	3,089	4,619	5,502
EBITDA	933	2,063	3,962	5,587	6,593
Net Interest	-69	-172	-238	-294	-225
FX Gains/Losses	5	92	-66	384	91
Equity Method	-28	1,240	-75	-164	-165
Pretax Income	1,131	2,647	2,210	4,075	5,030
Net Income	977	2,540	2,202	3,335	3,823
Net Income (Ctrl)	818	2,299	1,405	2,366	3,249
Growth (%)					
Revenue	11.7	42.5	137.6	18.8	4.8
Operating Profit	48.4	191.4	78.4	49.5	19.1
EBITDA	40.8	121.0	92.1	41.0	18.0
Net Income	566.6	160.0	-13.3	51.4	14.6
Margins (%)					
Gross Margin	18.1	25.5	20.4	20.6	24.5
Operating Margin	7.5	15.4	11.6	14.6	16.5
EBITDA Margin	11.8	18.4	14.8	17.6	19.8
Pretax Margin	14.3	23.6	8.3	12.8	15.1
Net Margin	12.4	22.6	8.2	10.5	11.5

Hanwha Aerospace Balance Sheet (Unit: KRW billions)

Dec Year-End	2023	2024	2025	2026E	2027E
Current Assets	9,192	22,982	30,705	33,721	36,982
Cash & Equiv.	1,909	3,654	8,294	9,438	10,414
Accounts Receivable	2,119	3,365	3,448	3,637	4,083
Inventory	2,869	6,404	7,742	8,285	9,623
Non-current Assets	10,351	20,580	23,248	26,526	29,449
Investment Assets	4,738	4,438	6,226	8,159	8,259
Tangible Assets	3,410	8,386	9,524	10,983	13,458
Intangible Assets	2,203	7,756	7,498	7,384	7,733
Total Assets	**19,543**	**43,562**	**53,954**	**60,247**	**66,431**
Current Liabilities	12,077	25,516	29,976	30,890	32,163
Accounts Payable	1,879	4,270	6,041	5,140	5,771
Current Debt	2,616	6,566	7,178	9,927	9,927
Non-current Liab.	2,782	6,554	7,189	8,032	10,105
Long-term Debt	1,644	4,623	5,832	6,424	8,456
Total Liabilities	**14,859**	**32,070**	**37,165**	**38,922**	**42,268**
Capital Stock	266	240	270	270	270
Capital Surplus	192	207	3,604	5,020	5,020
Retained Earnings	2,627	4,747	5,983	7,969	10,806
Capital Adjustments	443	-198	-173	-173	-173
Treasury Stock	-2	-21	-21	-21	-21
Total Equity	**4,684**	**11,492**	**16,788**	**21,325**	**24,163**

Buy Rating, Coverage Initiated With a Target Price of KRW 1,300,000 (4)

Hanwha Aerospace Statement of Cash Flows (Unit: KRW billions)

Dec Year-End	2023	2024	2025	2026E	2027E
Operating CF	1,390	1,393	4,050	678	3,536
Net Income	977	2,540	2,202	3,335	3,823
Depreciation	339	331	873	968	1,092
Chg in Working Cap	462	-903	571	-3,302	-1,012
Accounts Receivable Decrease (Increase)	-583	-1,097	-1,029	-352	-446
Inventory Decrease (Increase)	-714	-882	-1,108	-162	-1,017
Accounts Payable Increase (Decrease)	244	-78	2,099	-2,480	631
Investing CF	-3,029	-1,367	-4,021	-4,675	-4,246
Tangible Asset Disposal (Acquisition)	-450	-577	-1,691	-1,788	-3,180
Intangible Asset Decrease (Increase)	-118	-149	-263	-153	-735
Investment Asset Decrease (Increase)	-10	-372	-552	-2,285	-66
Financing CF	368	1,066	4,682	3,966	1,620
Borrowings Increase (Decrease)	445	1,233	2,035	2,731	2,032
Capital Increase (Decrease)	-51	-139	4,014	-360	-412
Dividends Paid	-51	-120	-195	-360	-412
Gross Cash Flow	909	2,702	4,353	4,211	4,548
(-) Working Capital Increase (Decrease)	-1,047	2,556	-765	3,706	1,012
(-) Capex	453	578	1,711	1,790	3,180
(+) Asset Sales	-115	-147	-244	-151	-735
Free Cash Flow	1,388	-580	3,163	-1,436	-380
(-) Other Investment	3,036	-1,384	1,708	46	265
Excess Cash	-1,648	804	1,455	-1,482	-644
NOPLAT	472	1,643	3,078	3,780	4,181
(+) Dep	339	331	873	968	1,092
(-) Working Capital Investment	-1,047	2,556	-765	3,706	1,012
(-) Capex	453	578	1,711	1,790	3,180
OpFCF	1,405	-1,161	3,006	-748	1,081

Hanwha Aerospace Key Indicators (Unit: KRW, x, %)

Dec Year-End	2023	2024	2025	2026E	2027E
Per Share					
EPS	15,886	49,621	27,247	45,986	63,156
BPS	69,689	109,587	187,825	253,802	308,835
DPS	1,771	3,443	7,000	8,000	10,000
CFPS	17,667	58,319	84,412	81,673	88,196
ROA (%)	4.7	7.3	2.9	4.1	5.1
ROE (%)	25.6	53.9	19.1	20.8	22.4
ROIC (%)	13.9	16.3	17.8	18.9	17.1
Multiples (x, %)					
PER	8.0	16.0	29.1	17.2	12.5
PBR	1.8	7.2	4.2	3.1	2.6
PSR	0.8	3.3	1.5	1.3	1.2
PCR	7.2	13.6	9.4	9.7	9.0
EV/EBITDA	9.3	23.5	11.5	8.5	7.4
Dividend Yield	1.4	0.4	0.9	1.0	1.3
Stability (%)					
Debt Ratio	317.2	279.1	221.4	182.5	174.9
Net Debt/Equity	50.2	65.6	28.1	32.4	33.0
Net Debt/EBITDA	251.8	365.3	119.0	123.7	120.9
Current Ratio	76.1	90.1	102.4	109.2	115.0
Interest Coverage (x)	4.0	7.1	6.3	8.3	10.1
Asset Structure (%)					
Invested Capital	32.4	67.8	54.6	56.1	58.6
Cash + Investment Assets	67.6	32.2	45.4	43.9	41.4
Capital Structure (%)					
Borrowings	47.6	49.3	43.7	43.4	43.2
Equity	52.4	50.7	56.3	56.6	56.8

Note: IFRS consolidated basis

Hanwha Systems (272210): Earnings Burden Eases; Focus on Air Defense and Space Businesses

Buy (Initiation)

Target Price (New)

100,000 KRW

Current Price (10/23)

66,300 KRW

Upside

▲50.8%

Market Cap	**12,525.4 bn KRW**
Shares Outstanding	**188,919 thousand**
52-Week High / Low	**162,700 / 45,350**
90-Day Avg. Daily Value	**49.168 bn KRW**
Foreign Ownership	**9.2%**
Hanwha Aerospace (+3)	**59.5%**
Natl. Pension Service (+1)	**6.2%**
Treasury shares (+1)	**1.0%**

Share Price Returns (%)

Return	1M	3M	6M	12M
Absolute	-10.3	-4.9	-49.0	18.8
Rel. (KOSPI)	11.3	16.9	-34.7	-25.5

▪ With the sale of Philly Shipyard easing the earnings burden, Hanwha Systems' growth direction is expected to become clearer around air defense systems and the space business. This is the time to note, in the short term, the expansion of the air defense systems business area and, over the medium to long term, the growth potential of the space business, which extends beyond satellite manufacturing to data sales.

Hanwha Systems Financial Summary (Unit: KRW billions, KRW, %, x)

Category	2024	2025	2026E	2027E
Revenue	2,804	3,664	4,372	4,520
Operating Profit	219	120	299	479
EBITDA	333	300	532	867
Net Income (Ctrl)	454	242	112	338
EPS	2,405	1,282	661	1,808
Net Debt	152	1,128	1,182	2,358
PER	27.6	51.7	100.2	36.7
PBR	5.0	2.6	2.6	2.4
EV/EBITDA	38.0	45.5	25.8	17.2
Dividend Yield	0.5	0.8	0.8	0.8
ROE	19.6	6.6	2.5	6.7

Sale of Philly Shipyard: The Start of a New Growth Direction

- The company's core business is defense, but since the acquisition of Philly Shipyard, expectations for entry into the US naval vessel market and growth of the space business rose, and the share price climbed sharply. In September 2026, it decided to sell its entire stake in Philly Shipyard, bringing a change in business structure. While Philly Shipyard was a major trigger for the earlier share price rise, it is also true that continued losses weighed on consolidated earnings. With the sale easing the earnings burden going forward, the company's growth drivers are expected to become clearer, centered on air defense systems and the space business.

Air Defense Systems and the Space Business: Central Pillars of Growth

- Going forward, the company's business capabilities are expected to concentrate on air defense systems and the space business. In the short term, air defense systems are expected to drive earnings growth, while over the medium to long term the space business's profit contribution is expected to expand gradually. In particular, considering the recent cooperation with the UAE's EDGE, we note the possibility that the business scope could expand beyond supplying Cheongung-II radars to building an integrated air defense network. Business opportunities for the Cheongwang laser anti-air weapon are expected to expand through higher output and linkage with various weapon systems. In the short term, the space business will center on development and manufacturing of satellites such as military reconnaissance satellites, but in the long term it is expected to expand into a business providing military and private customers with imagery secured from its own satellites and AI-based analysis information.

Buy Rating, Coverage Initiated With a Target Price of KRW 100,000 (1)

- The sale of Philly Shipyard is expected to be an opportunity to reduce consolidated earnings uncertainty in the short term. Stable earnings growth is expected on the basis of the order backlog of the existing defense business, and expansion of the air defense systems business scope is expected to act as an additional growth driver. Over the medium to long term, we note the possibility that the space business will expand from satellite manufacturing to data services and establish itself as a new source of revenue. We initiate coverage of Hanwha Systems with a Buy rating and a target price of KRW 100,000. The target price was calculated by applying a target PER to expected EPS.

[Table 100] Hanwha Systems Valuation

Category	Value	Remarks
Expected EPS (KRW)	2,458	2028 expected EPS
Target P/E (x)	40.6	Applied 80% space premium to the average 2027 PER of 22.5x for three overseas defense companies (Leonardo, Thales, RTX)
Fair Price (KRW)	99,693	Expected EPS * Target P/E
Target Price (KRW)	100,000	
Current Price (KRW)	66,300	
Upside (%)	50.8	

Source: Hanwha Research Center

[Table 101] Hanwha Systems 3Q26 Earnings Estimates (Unit: KRW billions)

Category	3Q25	4Q25	1Q26	2Q26	3Q26E Estimate	3Q26E Consensus	YoY	QoQ	Difference vs. Consensus
Revenue	808	1,398	807	1,118	983	1,076	21.7%	-12.0%	-8.6%
Operating Profit	48	6	34	104	69	62	90.1%	-12.0%	11.2%
Net Income (Ctrl)	162	12	-55	67	40	58	-68.1%	-22.9%	-31.9%
Operating Margin	5.9	0.4	4.2	9.3	7.0	5.8	3.3%p	0.0%p	1.2%p
Net Margin	20.1	0.8	-6.8	6.0	4.0	5.4	-14.8%p	-0.7%p	0.1%p

Source: FnGuide, Hanwha Research Center

Buy Rating, Coverage Initiated With a Target Price of KRW 100,000 (2)

[Table 102] Hanwha Systems Quarterly and Annual Earnings Trend (Unit: KRW billions, %)

Category	2022	2023	2024	2025	2026(E)	2027(E)	2028(E)	2026 YoY	2027 YoY
Revenue	2,188	2,452	2,804	3,664	4,372	4,520	5,308	19.3	3.4
Defense	1,641	1,817	2,099	2,439	3,025	3,689	4,422	24.0	22.0
ICT	547	636	695	653	763	820	877	16.9	7.5
Other	1	-	10	573	585	11	9	2.1	-98.2
Operating Profit	39	123	219	144	299	479	629	107.2	60.0
Pretax Income	-42	444	581	140	96	449	611	-31.1	367.7
Net Income	-77	349	454	252	112	338	460	-55.8	202.9
Operating Margin	1.8	5.0	7.8	3.9	6.8	10.6	11.8	-	-
Pretax Margin	-1.9	18.1	20.7	3.8	2.2	9.9	11.5	-	-
Net Margin	-3.5	14.2	16.2	6.9	2.6	7.5	8.7	-	-

Source: Hanwha Systems, Hanwha Research Center

Buy Rating, Coverage Initiated With a Target Price of KRW 100,000 (3)

Hanwha Systems Income Statement (Unit: KRW billions)

Dec Year-End	2023	2024	2025	2026E	2027E
Revenue	2,452	2,804	3,664	4,372	4,520
Gross Profit	304	416	392	626	794
Operating Profit	123	219	120	299	479
EBITDA	244	333	300	532	867
Net Interest	13	-8	4	-48	-53
FX Gains/Losses	1	16	3	30	30
Equity Method	-46	329	24	-12	-13
Pretax Income	444	581	115	119	449
Net Income	343	445	209	77	341
Net Income (Ctrl)	349	454	242	112	338
Growth (%)					
Revenue	12.1	14.3	30.7	19.3	3.4
Operating Profit	213.6	78.9	-45.3	168.5	48.8
EBITDA	69.7	36.4	-9.9	77.0	63.1
Net Income	Turn to profit	29.8	-53.0	-55.1	263.7
Margins (%)					
Gross Margin	12.4	14.8	10.7	14.3	17.6
Operating Margin	5.0	7.8	3.3	7.4	10.6
EBITDA Margin	10.0	11.9	8.2	12.2	19.2
Pretax Margin	18.1	20.7	3.1	2.7	9.9
Net Margin	14.0	15.9	5.7	2.1	7.6

Hanwha Systems Balance Sheet (Unit: KRW billions)

Dec Year-End	2023	2024	2025	2026E	2027E
Current Assets	1,890	2,286	2,929	5,303	5,253
Cash & Equiv.	509	321	488	2,216	1,894
Accounts Receivable	204	313	452	534	674
Inventory	501	610	811	1,129	1,203
Non-current Assets	2,593	3,524	7,399	7,276	8,789
Investment Assets	1,687	2,154	5,866	5,559	5,572
Tangible Assets	348	636	823	1,022	2,055
Intangible Assets	559	734	710	695	1,162
Total Assets	**4,483**	**5,810**	**10,327**	**12,580**	**14,042**
Current Liabilities	1,951	2,677	3,335	5,680	6,016
Accounts Payable	301	436	722	874	1,105
Current Debt	191	109	426	2,196	2,196
Non-current Liab.	378	678	1,991	1,935	2,816
Long-term Debt	139	364	1,190	1,202	2,055
Total Liabilities	**2,329**	**3,355**	**5,326**	**7,615**	**8,832**
Capital Stock	945	945	945	945	945
Capital Surplus	1,100	1,092	1,123	1,127	1,127
Retained Earnings	212	589	709	742	986
Capital Adjustments	-117	-136	2,073	2,073	2,073
Treasury Stock	-34	-34	-34	-34	-34
Total Equity	**2,154**	**2,456**	**5,001**	**4,965**	**5,209**

Buy Rating, Coverage Initiated With a Target Price of KRW 100,000 (4)

Hanwha Systems Statement of Cash Flows (Unit: KRW billions)

Dec Year-End	2023	2024	2025	2026E	2027E
Operating CF	83	133	79	662	832
Net Income	343	445	209	94	341
Depreciation	122	114	180	210	388
Chg in Working Cap	-168	-218	-211	-17	64
Accounts Receivable Decrease (Increase)	6	-92	-143	-71	-141
Inventory Decrease (Increase)	-128	-106	-188	-318	-74
Accounts Payable Increase (Decrease)	42	2	234	175	230
Investing CF	-864	-442	-1,170	-728	-1,925
Tangible Asset Disposal (Acquisition)	-132	-240	-297	-349	-1,369
Intangible Asset Decrease (Increase)	-27	-25	-19	-18	-519
Investment Asset Decrease (Increase)	-32	-129	-227	-202	-11
Financing CF	27	35	1,237	1,664	760
Borrowings Increase (Decrease)	72	84	1,109	1,753	853
Capital Increase (Decrease)	-47	-52	-65	-93	-93
Dividends Paid	-47	-52	-65	-93	-93
Gross Cash Flow	260	386	353	750	768
(-) Working Capital Increase (Decrease)	55	-224	134	72	-64
(-) Capex	132	240	299	349	1,369
(+) Asset Sales	-27	-25	-18	-18	-519
Free Cash Flow	45	345	-97	310	-1,056
(-) Other Investment	785	490	704	103	26
Excess Cash	-740	-146	-801	206	-1,082
NOPLAT	107	180	87	270	364
(+) Dep	122	114	180	210	388
(-) Working Capital Investment	55	-224	134	72	-64
(-) Capex	132	240	299	349	1,369
OpFCF	41	279	-165	58	-553

Hanwha Systems Key Indicators (Unit: KRW, x, %)

Dec Year-End	2023	2024	2025	2026E	2027E
Per Share					
EPS	1,849	2,405	1,282	661	1,808
BPS	11,327	13,181	25,668	25,863	27,157
DPS	280	350	500	500	500
CFPS	1,376	2,041	1,871	3,969	4,064
ROA (%)	8.3	8.8	3.0	1.1	2.6
ROE (%)	16.9	19.6	6.6	2.5	6.7
ROIC (%)	22.9	27.8	12.9	36.5	22.5
Multiples (x, %)					
PER	9.4	27.6	51.7	100.2	36.7
PBR	1.5	5.0	2.6	2.6	2.4
PSR	1.3	4.5	3.4	2.9	2.8
PCR	12.6	32.5	35.4	16.7	16.3
EV/EBITDA	12.7	38.0	45.5	25.8	17.2
Dividend Yield	1.6	0.5	0.8	0.8	0.8
Stability (%)					
Debt Ratio	108.1	136.6	106.5	153.4	169.6
Net Debt/Equity	-8.3	6.2	22.6	23.8	45.3
Net Debt/EBITDA	-72.9	45.6	375.4	222.3	271.9
Current Ratio	96.9	85.4	87.8	93.4	87.3
Interest Coverage (x)	8.8	9.8	3.2	3.5	5.7
Asset Structure (%)					
Invested Capital	19.1	23.9	8.2	10.5	23.7
Cash + Investment Assets	80.9	76.1	91.8	89.5	76.3
Capital Structure (%)					
Borrowings	13.3	16.2	24.4	40.6	44.9
Equity	86.7	83.8	75.6	59.4	55.1

Note: IFRS consolidated basis

LIG Defense & Aerospace (079550): Supply Constraints on Ballistic Missile Interception Systems Persist Despite Global Production Increases



Buy (Initiation)

Target Price (New)

670,000 KRW

Current Price (10/23)
475,500 KRW

Upside
▲40.9%

Market Cap	**10,461.0 bn KRW**
Shares Outstanding	**22,000 thousand**
52-Week High / Low	**1,020,000 / 364,500**
90-Day Avg. Daily Value	**85.56 bn KRW**
Foreign Ownership	**26.0%**
LIG (+9)	**38.2%**
Natl. Pension Service (+1)	**9.9%**
Treasury shares (+1)	**0.7%**

Share Price Returns (%)

Return	1M	3M	6M	12M
Absolute	-31.3	-35.4	-50.8	3.3
Rel. (KOSPI)	-9.7	-13.6	-36.5	-41.1

- Although the new orders that were expected have not yet come, the global supply shortage of ballistic missile defense systems is expected to continue for a considerable period going forward. The existing high-margin order backlog alone is expected to support stable earnings growth through 2029, and we judge that additional exports of Cheongung-II and development of Cheongung-III will further widen medium- to long-term growth opportunities.

LIG Defense & Aerospace Financial Summary (Unit: KRW billions, KRW, %, x)

Category	2024	2025	2026E	2027E
Revenue	3,276	4,307	4,968	5,825
Operating Profit	223	319	440	602
EBITDA	309	444	592	749
Net Income (Ctrl)	219	253	400	567
EPS	9,974	11,516	18,325	25,955
Net Debt	-75	896	1,624	1,521
PER	47.7	41.3	25.9	18.3
PBR	8.7	7.3	5.9	4.7
EV/EBITDA	33.6	25.6	20.4	16.0
Dividend Yield	0.5	0.6	0.9	1.3
ROE	19.4	19.2	25.0	28.3

Expectations Raised by the War, but Orders Not Yet Visible

- The biggest topic in the global defense industry in 2026 was the war between the US-Israel and Iran that broke out at the end of February. In particular, the shortage of Patriot interceptor missiles stood out, and the company's Cheongung-II drew attention as an alternative that could complement it. The share price also rose sharply for a time, reflecting expectations of expanded exports. However, about 8 months since the outbreak of war, new order results as large as the market expected have not materialized, and the share price has been sluggish.

Ballistic Missile Defense Systems: Supply Constraints Expected to Persist Despite Production Increases

- Not only the Patriot but European defense companies are also pursuing expansion of air defense system and interceptor missile production capacity. Even with increased supply volume, Western air defense systems with ballistic missile interception capability are expected to remain insufficient. For the Patriot, a considerable portion of the increased output is expected to go to rebuilding inventories of existing operating countries. We judge that it will not be easy for the European SAMP/T NG to secure sufficient supply capacity until around 2030. The long delivery times of global ballistic missile defense systems are likely to persist for a considerable period. Additional export opportunities for Cheongung-II are expected to continue. If the Cheongung-III under development is also fielded, we expect the company's medium- to long-term export competitiveness to be strengthened further.

Buy Rating, Coverage Initiated With a Target Price of KRW 670,000 (1)



- Despite the short-term gap in new orders, the company's medium- to long-term earnings growth potential remains valid. Based on the high-margin defense order backlog secured mainly from three Middle Eastern countries, stable profit growth is expected not only in 2027 but also in 2028-2029. If additional export contracts are secured, visibility of earnings growth will extend into the 2030s. Short-term share price momentum is expected to focus on the progress of additional air defense system orders in the Middle East. We initiate coverage of LIG Defense & Aerospace with a Buy rating and a target price of KRW 670,000. The target price was calculated by applying a target PER to expected EPS.

[Table 103] LIG Defense & Aerospace Valuation

Category	Value	Remarks
Expected EPS (KRW)	25,955	2027 expected EPS
Target P/E (x)	26.0	Applied average 2027 PER of 25.5x for two overseas air defense system companies (RTX, Kongsberg Gruppen)
Fair Price (KRW)	674,830	Expected EPS * Target P/E
Target Price (KRW)	670,000	
Current Price (KRW)	475,500	
Upside (%)	40.9	

Source: Hanwha Research Center

[Table 104] LIG Defense & Aerospace 3Q26 Earnings Estimates (Unit: KRW billions)

Category	3Q25	4Q25	1Q26	2Q26	3Q26E Estimate	3Q26E Consensus	YoY	QoQ	Difference vs. Consensus
Revenue	1,049	1,405	1,168	1,110	1,151	1,241	9.7%	3.6%	-7.3%
Operating Profit	90	39	171	106	91	104	1.7%	-13.8%	-12.4%
Net Income (Ctrl)	59	5	140	83	78	86	31.8%	-5.4%	-9.1%
Operating Margin	8.5	2.8	14.7	9.5	7.9	8.4	-0.6%p	-1.6%p	-0.5%p
Net Margin	5.7	0.3	12.0	7.4	6.8	6.9	1.1%p	-0.6%p	-0.1%p

Source: FnGuide, Hanwha Research Center

Buy Rating, Coverage Initiated With a Target Price of KRW 670,000 (2)

[Table 105] LIG Defense & Aerospace Quarterly and Annual Earnings Trend (Unit: KRW billions, %)

Category	2022	2023	2024	2025	2026(E)	2027(E)	2028(E)	2026 YoY	2027 YoY
Revenue	2,221	2,309	3,276	4,307	4,968	5,825	7,006	15.4	17.2
Domestic	1,815	1,950	2,503	3,451	3,561	3,305	4,148	3.2	-7.2
Exports	406	358	774	856	1,407	2,519	2,858	64.5	79.0
Operating Profit	179	186	223	323	440	602	768	35.9	36.9
Pretax Income	158	185	203	282	455	646	801	61.1	42.1
Net Income	123	175	219	255	400	567	720	56.8	41.6
Operating Margin	8.1	8.1	6.8	7.5	8.8	10.3	11.0	-	-
Pretax Margin	7.1	8.0	6.2	6.6	9.2	11.1	11.4	-	-
Net Margin	5.5	7.6	6.7	5.9	8.1	9.7	10.3	-	-

Source: LIG Defense & Aerospace, Hanwha Research Center

Buy Rating, Coverage Initiated With a Target Price of KRW 670,000 (3)

LIG Defense & Aerospace Income Statement (Unit: KRW billions)

Dec Year-End	2023	2024	2025	2026E	2027E
Revenue	2,309	3,276	4,307	4,968	5,825
Gross Profit	347	453	663	876	1,111
Operating Profit	186	223	319	440	602
EBITDA	255	309	444	592	749
Net Interest	-6	-2	-8	-30	-28
FX Gains/Losses	0	0	0	13	32
Equity Method	0	0	1	2	2
Pretax Income	185	203	278	455	646
Net Income	175	213	237	383	539
Net Income (Ctrl)	175	219	253	400	567
Growth (%)					
Revenue	4.0	41.9	31.5	15.4	17.2
Operating Profit	4.1	19.9	43.0	37.6	36.9
EBITDA	9.6	21.1	43.5	33.5	26.5
Net Income	42.3	21.6	11.6	61.2	40.8
Margins (%)					
Gross Margin	15.0	13.8	15.4	17.6	19.1
Operating Margin	8.1	6.8	7.4	8.8	10.3
EBITDA Margin	11.1	9.4	10.3	11.9	12.9
Pretax Margin	8.0	6.2	6.5	9.2	11.1
Net Margin	7.6	6.5	5.5	7.7	9.3

LIG Defense & Aerospace Balance Sheet (Unit: KRW billions)

Dec Year-End	2023	2024	2025	2026E	2027E
Current Assets	2,536	3,895	5,622	6,819	7,067
Cash & Equiv.	466	569	157	129	42
Accounts Receivable	309	425	553	737	800
Inventory	229	364	593	679	737
Non-current Assets	1,280	2,297	2,443	2,591	2,732
Investment Assets	198	230	262	223	229
Tangible Assets	894	1,371	1,533	1,687	1,807
Intangible Assets	188	696	647	682	696
Total Assets	**3,816**	**6,192**	**8,065**	**9,410**	**9,798**
Current Liabilities	2,651	4,617	6,216	6,994	7,061
Accounts Payable	460	727	1,418	1,623	1,561
Current Debt	245	276	733	1,205	1,165
Non-current Liab.	113	311	373	605	457
Long-term Debt	54	218	320	549	399
Total Liabilities	**2,763**	**4,928**	**6,589**	**7,598**	**7,517**
Capital Stock	110	110	110	110	110
Capital Surplus	142	142	153	153	153
Retained Earnings	728	905	1,106	1,442	1,910
Capital Adjustments	72	52	61	61	61
Treasury Stock	-5	-5	-4	-4	-4
Total Equity	**1,052**	**1,264**	**1,476**	**1,812**	**2,281**

Buy Rating, Coverage Initiated With a Target Price of KRW 670,000 (4)

LIG Defense & Aerospace Statement of Cash Flows (Unit: KRW billions)

Dec Year-End	2023	2024	2025	2026E	2027E
Operating CF	467	952	-584	-310	487
Net Income	175	213	237	383	539
Depreciation	69	86	124	153	147
Chg in Working Cap	213	605	-1,105	-848	-227
Accounts Receivable Decrease (Increase)	-72	-83	-759	-699	-63
Inventory Decrease (Increase)	-16	-106	-234	-84	-58
Accounts Payable Increase (Decrease)	107	256	162	211	-62
Investing CF	-97	-975	-276	-315	-287
Tangible Asset Disposal (Acquisition)	-59	-497	-166	-233	-230
Intangible Asset Decrease (Increase)	-24	-70	-74	-54	-52
Investment Asset Decrease (Increase)	0	0	-10	-1	-1
Financing CF	-177	120	439	595	-288
Borrowings Increase (Decrease)	-144	162	491	659	-190
Capital Increase (Decrease)	-33	-42	-52	-64	-98
Dividends Paid	-33	-42	-52	-64	-98
Gross Cash Flow	295	396	572	613	715
(-) Working Capital Increase (Decrease)	-232	-679	998	918	227
(-) Capex	59	497	170	234	230
(+) Asset Sales	-23	-70	-70	-53	-52
Free Cash Flow	444	508	-666	-593	205
(-) Other Investment	33	482	134	-44	4
Excess Cash	411	26	-799	-549	201
NOPLAT	176	162	272	370	502
(+) Dep	69	86	124	153	147
(-) Working Capital Investment	-232	-679	998	918	227
(-) Capex	59	497	170	234	230
OpFCF	418	430	-771	-630	192

LIG Defense & Aerospace Key Indicators (Unit: KRW, x, %)

Dec Year-End	2023	2024	2025	2026E	2027E
Per Share					
EPS	7,953	9,974	11,516	18,325	25,955
BPS	47,835	54,970	65,005	80,302	101,606
DPS	1,950	2,400	2,950	4,500	6,000
CFPS	13,392	18,010	25,994	27,847	32,488
ROA (%)	5.1	4.4	3.6	4.6	5.9
ROE (%)	17.6	19.4	19.2	25.0	28.3
ROIC (%)	19.4	16.6	15.9	13.1	14.2
Multiples (x, %)					
PER	16.4	47.7	41.3	25.9	18.3
PBR	2.7	8.7	7.3	5.9	4.7
PSR	1.2	3.2	2.4	2.1	1.8
PCR	9.7	26.4	18.3	17.1	14.6
EV/EBITDA	10.6	33.6	25.6	20.4	16.0
Dividend Yield	1.5	0.5	0.6	0.9	1.3
Stability (%)					
Debt Ratio	262.6	389.9	446.4	419.3	329.6
Net Debt/Equity	-15.9	-5.9	60.7	89.6	66.7
Net Debt/EBITDA	-65.6	-24.3	202.0	274.3	203.0
Current Ratio	95.7	84.4	90.5	97.5	100.1
Interest Coverage (x)	19.3	18.7	25.3	12.4	17.6
Asset Structure (%)					
Invested Capital	55.7	58.4	84.6	90.5	93.2
Cash + Investment Assets	44.3	41.6	15.4	9.5	6.8
Capital Structure (%)					
Borrowings	22.1	28.1	41.6	49.2	40.7
Equity	77.9	71.9	58.4	50.8	59.3

Note: IFRS consolidated basis

Korea Aerospace Industries (047810): KF-21 Exports Are Ultimately a Matter of Timing



Buy (Initiation)

Target Price (New)

140,000 KRW

Current Price (10/23)
102,200 KRW

Upside
▲37.0%

Market Cap	**9,962.0 bn KRW**
Shares Outstanding	**97,475 thousand**
52-Week High / Low	**202,000 / 97,000**
90-Day Avg. Daily Value	**48.328 bn KRW**
Foreign Ownership	**23.7%**
The Export-Import Bank of Korea (+1)	**26.4%**
Hanwha Aerospace (+3)	**15.9%**
Natl. Pension Service (+1)	**7.1%**

Share Price Returns (%)

Return	1M	3M	6M	12M
Absolute	-19.8	-37.7	-44.0	0.9
Rel. (KOSPI)	1.8	-15.9	-29.7	-43.4

- Although the share price fell on weak earnings this year, medium- to long-term earnings growth potential has not changed. This is the time to note that, amid the global fighter supply shortage, KF-21 export opportunities are actually growing.

Korea Aerospace Industries Financial Summary (Unit: KRW billions, KRW, %, x)

Category	2024	2025	2026E	2027E
Revenue	3,634	3,696	5,261	6,564
Operating Profit	241	269	352	574
EBITDA	345	380	480	729
Net Income (Ctrl)	172	186	255	486
EPS	1,765	1,907	2,612	4,986
Net Debt	950	2,116	1,846	2,038
PER	57.9	53.6	39.1	20.5
PBR	5.8	5.4	4.8	4.0
EV/EBITDA	31.7	31.8	24.6	16.5
Dividend Yield	0.5	0.5	0.6	1.0
ROE	10.4	10.5	13.0	21.2

An Eventful 2026, but Medium- to Long-Term Growth Unchanged

- 2026 was an eventful year for the company. Delivery deferrals due to the light armed helicopter engine issue and worsening profitability of the complete-aircraft export project lowered annual earnings expectations, and the share price and PER fell together. However, the KF-21 program, the core of medium- to long-term growth, is proceeding as planned. KF-21 deliveries to the Korean Air Force are set to increase sequentially every year. We judge there is no change in the direction of stepwise earnings growth that will continue through at least 2029.

KF-21: Exports Are Ultimately a Matter of Timing

- 4.5-generation fighter makers have long pursued production capacity expansion, but actual production increases are proceeding more slowly than expected. Even if target production capacity is achieved, considering the cumulative order backlog, it is not easy to shorten delivery times for new order volumes to within 4 years. The export competitiveness of the KF-21, which has price competitiveness and fast delivery, is expected to stand out. Over the long term, we expect export competitiveness to be strengthened further as weapons integration, domestic missile development and engine localization proceed. Results of the related programs will become visible in the early-to-mid 2030s, and in the 2040s it is expected to evolve into a next-generation fighter platform. We judge that KF-21 exports are a matter of timing. From the buyers' perspective as well, the optimal ordering window is until the late 2020s, when fast delivery can be guaranteed.

Buy Rating, Coverage Initiated With a Target Price of KRW 140,000 (1)

- 2027 is the year when the direction of earnings growth becomes clearer as mass production of complete aircraft expands at home and abroad. Short-term share price momentum is expected to focus on whether a KF-21 export contract with Indonesia is concluded. If an export contract is concluded before the Korean Air Force's full-scale operational fielding, it would be highly significant in that the KF-21's competitiveness was recognized early in overseas markets. We initiate coverage of Korea Aerospace Industries with a Buy rating and a target price of KRW 140,000. The target price was calculated by applying a target PER to expected EPS.

[Table 106] Korea Aerospace Industries Valuation

Category	Value	Remarks
Expected EPS (KRW)	4,986	2027 expected EPS
Target P/E (x)	30.6	Applied 30% premium to the average 2027 PER of 21.3x for four overseas fighter companies (Boeing, Lockheed Martin, Saab, Dassault Aviation)
Fair Price (KRW)	138,231	Expected EPS * Target P/E
Target Price (KRW)	140,000	
Current Price (KRW)	102,200	
Upside (%)	37.0	

Source: Hanwha Research Center

[Table 107] Korea Aerospace Industries 3Q26 Earnings Estimates (Unit: KRW billions)

Category	3Q25	4Q25	1Q26	2Q26	3Q26E Estimate	3Q26E Consensus	YoY	QoQ	Difference vs. Consensus
Revenue	702	1,467	1,093	1,168	1,133	1,220	61.3%	-3.0%	-7.2%
Operating Profit	60	77	67	48	72	82	20.0%	49.1%	-11.9%
Net Income (Ctrl)	39	61	42	36	50	49	28.7%	39.3%	2.0%
Operating Margin	8.6	5.2	6.1	4.1	6.4	6.7	-2.2%p	2.2%p	-0.3%p
Net Margin	5.5	4.1	3.8	3.1	4.4	4.0	-1.1%p	1.3%p	0.4%p

Source: FnGuide, Hanwha Research Center

Buy Rating, Coverage Initiated With a Target Price of KRW 140,000 (2)

[Table 108] Korea Aerospace Industries Quarterly and Annual Earnings Trend (Unit: KRW billions, %)

Category	2022	2023	2024	2025	2026(E)	2027(E)	2028(E)	2026 YoY	2027 YoY
Revenue	2,787	3,819	3,634	3,633	5,261	6,564	7,452	44.8	24.7
Domestic Business	1,897	1,945	2,107	1,710	2,716	3,647	4,202	58.8	34.3
Complete-Aircraft Exports	127	1,047	640	927	1,345	1,483	1,707	45.1	10.3
Aerostructures	748	806	851	924	1,106	1,326	1,411	19.6	19.9
Consolidated Subsidiaries	15	21	35	71	95	108	132	33.0	13.9
Operating Profit	142	247	241	269	352	574	755	30.8	63.1
Pretax Income	122	269	197	202	284	564	739	40.4	98.7
Net Income	118	224	172	186	255	486	637	37.0	90.9
Operating Margin	5.1	6.5	6.6	7.4	6.7	8.7	10.1	-	-
Pretax Margin	4.4	7.0	5.4	5.6	5.4	8.6	9.9	-	-
Net Margin	4.3	5.9	4.7	5.1	4.8	7.4	8.6	-	-

Source: Korea Aerospace Industries, Hanwha Research Center

Buy Rating, Coverage Initiated With a Target Price of KRW 140,000 (3)



Korea Aerospace Industries Income Statement (Unit: KRW billions)

Dec Year-End	2023	2024	2025	2026E	2027E
Revenue	3,819	3,634	3,696	5,261	6,564
Gross Profit	464	481	608	703	1,066
Operating Profit	248	241	269	352	574
EBITDA	369	345	380	480	729
Net Interest	16	-23	-61	-86	-18
FX Gains/Losses	8	-4	4	30	20
Equity Method	-2	-1	-6	-4	-4
Pretax Income	269	197	202	284	564
Net Income	221	171	187	258	491
Net Income (Ctrl)	224	172	186	255	486
Growth (%)					
Revenue	37.0	-4.9	1.7	42.3	24.8
Operating Profit	74.8	-2.7	11.8	30.8	63.0
EBITDA	33.1	-6.5	10.3	26.3	51.9
Net Income	91.0	-22.8	9.6	37.5	90.5
Margins (%)					
Gross Margin	12.1	13.3	16.5	13.4	16.2
Operating Margin	6.5	6.6	7.3	6.7	8.7
EBITDA Margin	9.6	9.5	10.3	9.1	11.1
Pretax Margin	7.0	5.4	5.5	5.4	8.6
Net Margin	5.8	4.7	5.1	4.9	7.5

Korea Aerospace Industries Balance Sheet (Unit: KRW billions)

Dec Year-End	2023	2024	2025	2026E	2027E
Current Assets	4,681	5,439	7,595	8,213	7,737
Cash & Equiv.	771	125	68	1,013	621
Accounts Receivable	422	428	750	762	836
Inventory	1,736	2,359	3,637	3,441	3,163
Non-current Assets	2,459	2,587	2,775	3,555	4,633
Investment Assets	840	828	837	819	848
Tangible Assets	815	913	994	1,067	1,182
Intangible Assets	804	846	944	1,669	2,603
Total Assets	**7,140**	**8,026**	**10,370**	**11,767**	**12,371**
Current Liabilities	4,400	4,915	6,351	7,013	7,368
Accounts Payable	688	982	1,029	1,202	1,363
Current Debt	404	448	820	1,013	1,013
Non-current Liab.	1,119	1,384	2,122	2,595	2,421
Long-term Debt	235	626	1,364	1,846	1,646
Total Liabilities	**5,520**	**6,298**	**8,473**	**9,608**	**9,789**
Capital Stock	487	487	487	487	487
Capital Surplus	128	128	128	169	169
Retained Earnings	1,003	1,111	1,235	1,441	1,864
Capital Adjustments	-21	-21	-21	-21	-21
Treasury Stock	0	0	0	0	0
Total Equity	**1,620**	**1,727**	**1,897**	**2,159**	**2,582**

Buy Rating, Coverage Initiated With a Target Price of KRW 140,000 (4)

Korea Aerospace Industries Statement of Cash Flows (Unit: KRW billions)

Dec Year-End	2023	2024	2025	2026E	2027E
Operating CF	-700	-728	-903	1,270	1,110
Net Income	221	171	187	258	491
Depreciation	121	104	111	128	155
Chg in Working Cap	-1,218	-1,166	-1,315	859	439
Accounts Receivable Decrease (Increase)	-136	-6	-306	-9	-73
Inventory Decrease (Increase)	-153	-629	-1,266	190	278
Accounts Payable Increase (Decrease)	115	150	102	101	160
Investing CF	-108	-189	-186	-1,007	-1,289
Tangible Asset Disposal (Acquisition)	-78	-161	-128	-171	-228
Intangible Asset Decrease (Increase)	-98	-86	-61	-744	-977
Investment Asset Decrease (Increase)	72	64	18	-68	-50
Financing CF	-546	369	1,030	679	-263
Borrowings Increase (Decrease)	-522	418	1,079	727	-200
Capital Increase (Decrease)	-24	-49	-49	-49	-63
Dividends Paid	-24	-49	-49	-49	-63
Gross Cash Flow	492	442	509	483	671
(-) Working Capital Increase (Decrease)	1,209	934	1,149	-797	-439
(-) Capex	78	161	129	171	228
(+) Asset Sales	-98	-86	-61	-744	-977
Free Cash Flow	-892	-739	-829	365	-95
(-) Other Investment	13	238	180	-38	83
Excess Cash	-906	-977	-1,010	403	-128
NOPLAT	204	209	249	320	499
(+) Dep	121	104	111	128	155
(-) Working Capital Investment	1,209	934	1,149	-797	-439
(-) Capex	78	161	129	171	228
OpFCF	-962	-783	-917	1,074	865

Korea Aerospace Industries Key Indicators (Unit: KRW, x, %)

Dec Year-End	2023	2024	2025	2026E	2027E
Per Share					
EPS	2,298	1,765	1,907	2,612	4,986
BPS	16,388	17,500	18,776	21,303	25,640
DPS	500	500	500	650	1,000
CFPS	5,048	4,531	5,222	4,953	6,885
ROA (%)	3.0	2.3	2.0	2.3	4.0
ROE (%)	14.8	10.4	10.5	13.0	21.2
ROIC (%)	23.3	10.5	7.7	8.2	11.9
Multiples (x, %)					
PER	21.8	57.9	53.6	39.1	20.5
PBR	3.1	5.8	5.4	4.8	4.0
PSR	1.3	2.7	2.7	1.9	1.5
PCR	9.9	22.6	19.6	20.6	14.8
EV/EBITDA	12.9	31.7	31.8	24.6	16.5
Dividend Yield	1.0	0.5	0.5	0.6	1.0
Stability (%)					
Debt Ratio	340.7	364.7	446.6	445.0	379.1
Net Debt/Equity	-8.2	55.0	111.5	85.5	78.9
Net Debt/EBITDA	-36.0	275.7	556.6	384.5	279.4
Current Ratio	106.4	110.7	119.6	117.1	105.0
Interest Coverage (x)	8.5	7.2	4.1	3.9	26.1
Asset Structure (%)					
Invested Capital	46.6	73.0	81.2	68.1	75.5
Cash + Investment Assets	53.4	27.0	18.8	31.9	24.5
Capital Structure (%)					
Borrowings	28.3	38.4	53.5	57.0	50.7
Equity	71.7	61.6	46.5	43.0	49.3

Note: IFRS consolidated basis

Hyundai Rotem (064350): Valuation Too Low to Worry About Further Declines

Buy (Initiation)

Target Price (New)

110,000 KRW

Current Price (10/23)	Upside
82,700 KRW	**▲33.0%**

Market Cap	**9,026.1 bn KRW**
Shares Outstanding	**109,142 thousand**
52-Week High / Low	**269,000 / 82,700**
90-Day Avg. Daily Value	**53.559 bn KRW**
Foreign Ownership	**39.0%**
Hyundai Motor Company (+4)	**33.8%**
Natl. Pension Service (+1)	**7.1%**
BlackRock Fund Advisors (+13)	**5.0%**

Share Price Returns (%)

Return	1M	3M	6M	12M
Absolute	-28.6	-50.3	-64.9	-65.8
Rel. (KOSPI)	-7.0	-28.5	-50.6	-110.1

- Although the share price has been sluggish owing to earnings that fell short of market expectations and order delays, the need for tanks and the company's medium- to long-term competitiveness have not changed. With global competitors' long delivery times continuing, the company's fast delivery remains a strength, and we judge that the share price does not sufficiently reflect this growth potential.

Hyundai Rotem Financial Summary (Unit: KRW billions, KRW, %, x)

Category	2024	2025	2026E	2027E
Revenue	4,377	5,839	6,562	6,367
Operating Profit	457	1,006	1,027	967
EBITDA	504	1,065	1,096	1,044
Net Income (Ctrl)	407	770	800	712
EPS	3,728	7,055	7,326	6,527
Net Debt	-419	-1,145	-1,792	-2,668
PER	22.2	11.7	11.3	12.7
PBR	4.4	2.9	2.4	2.0
EV/EBITDA	17.1	7.4	6.6	6.1
Dividend Yield	0.2	0.7	0.7	0.7
ROE	21.8	30.1	23.2	17.2

High Market Expectations, Disappointing Results

- In 2026, expectations for the company centered on earnings growth from the full-scale mass production of the second Poland implementation contract and the materialization of a large Iraq order. We judge that both fell short of market expectations. The second Poland implementation contract is proceeding normally, but the pace at which it was reflected in earnings did not meet market expectations. The Iraq order also appears delayed as the budget burden from intensifying geopolitical conflict in the Middle East and the blockade of the Strait of Hormuz has come to the fore. Since the start of the year, the share price has been the weakest among the five large domestic defense companies.

The Role of Tanks Remains; Fast Delivery Builds Competitiveness

- Through the Russia-Ukraine war and the US-Iran war, concerns have been raised that the role of tanks could shrink in a drone-centered battlefield environment. The core role of tanks in seizing and defending territory in ground warfare remains valid. Through integration with counter-drone defense systems, major tanks are adapting to the changing battlefield environment. Meanwhile, the Western tank market is expected to maintain a supply structure centered on the US, Germany and Korea. Because competitors are pursuing next-generation tank development and a transition to mass production around 2030, the possibility of a sharp expansion of production capacity in a short period is low. With competitors' long delivery times continuing, relatively fast delivery and cost-effectiveness are expected to support the company's medium- to long-term order competitiveness.

Buy Rating, Coverage Initiated With a Target Price of KRW 110,000 (1)

- The company's current PER is 11x, which we judge to be a level at which the growth premium of the defense sector is virtually not reflected, considering its medium- to long-term growth potential. Since the valuation discount to domestic defense companies was already large, the recent share price decline has made the investment appeal even more prominent. In the short term, expanded earnings reflection of the second Poland implementation contract and progress in the Iraq export project are expected to act as share price momentum. We initiate coverage of Hyundai Rotem with a Buy rating and a target price of KRW 110,000. The target price was calculated by applying a target PER to expected EPS.

[Table 109] Hyundai Rotem Valuation

Category	Value	Remarks
Expected EPS (KRW)	6,527	2027 expected EPS
Target P/E (x)	17.0	Applied 17x, an 18% discount to the average 2027 PER of 20.7x for two overseas defense companies (General Dynamics, Rheinmetall)
Fair Price (KRW)	110,775	Expected EPS * Target P/E
Target Price (KRW)	110,000	
Current Price (KRW)	82,700	
Upside (%)	33.0	

Source: Hanwha Research Center

[Table 110] Hyundai Rotem 3Q26 Preliminary Results

Category	3Q25	4Q25	1Q26	2Q26	3Q26P Preliminary	3Q26P Consensus	YoY	QoQ	Difference vs. Consensus
Revenue	1,620	1,626	1,457	1,606	1,621	1,801	0.1%	0.9%	-10.0%
Operating Profit	278	267	224	232	247	276	-11.0%	6.3%	-10.5%
Net Income (Ctrl)	198	222	201	185	177	216	-11.0%	-4.7%	-18.2%
Operating Margin	17.1	16.5	15.4	14.5	15.2	15.3	-1.9%p	0.8%p	-0.1%p
Net Margin	12.3	13.7	13.8	11.5	10.9	12.0	-1.4%p	-0.6%p	-1.1%p

Source: FnGuide, Hanwha Research Center

Buy Rating, Coverage Initiated With a Target Price of KRW 110,000 (2)

[Table 111] Hyundai Rotem Quarterly and Annual Earnings Trend (Unit: KRW billions, %)

Category	2022	2023	2024	2025	2026(E)	2027(E)	2028(E)	2026 YoY	2027 YoY
Revenue	3,163	3,587	4,377	5,839	6,562	6,367	9,402	12.4	-3.0
AD&RH	1,059	1,578	2,365	3,215	3,644	3,264	5,887	13.3	-10.4
RS	1,779	1,554	1,496	2,090	2,495	2,714	3,097	19.4	8.8
EP	325	456	516	534	423	389	419	-20.8	-8.0
Operating Profit	147	210	457	1,006	1,027	967	1,591	2.1	-5.9
Pretax Income	104	181	510	1,004	1,075	938	1,556	7.1	-12.8
Net Income	198	161	407	770	800	712	1,181	3.8	-10.9
Operating Margin	4.7	5.9	10.4	17.2	15.6	15.2	16.9	-	-
Pretax Margin	3.3	5.0	11.6	17.2	16.4	14.7	16.6	-	-
Net Margin	6.3	4.5	9.3	13.2	12.2	11.2	12.6	-	-

Source: Hyundai Rotem, Hanwha Research Center

Buy Rating, Coverage Initiated With a Target Price of KRW 110,000 (3)

Hyundai Rotem Income Statement (Unit: KRW billions)

Dec Year-End	2023	2024	2025	2026E	2027E
Revenue	3,587	4,377	5,839	6,562	6,367
Gross Profit	464	829	1,392	1,498	1,442
Operating Profit	210	457	1,006	1,027	967
EBITDA	250	504	1,065	1,096	1,044
Net Interest	0	12	25	73	56
FX Gains/Losses	0	0	0	32	90
Equity Method	1	0	0	0	0
Pretax Income	181	510	1,004	1,075	938
Net Income	157	405	770	803	713
Net Income (Ctrl)	161	407	770	800	712
Growth (%)					
Revenue	13.4	22.0	33.4	12.4	-3.0
Operating Profit	42.4	117.4	120.3	2.1	-5.9
EBITDA	34.5	101.4	111.5	2.9	-4.7
Net Income	-19.4	158.5	90.1	4.2	-11.2
Margins (%)					
Gross Margin	12.9	18.9	23.8	22.8	22.6
Operating Margin	5.9	10.4	17.2	15.6	15.2
EBITDA Margin	7.0	11.5	18.2	16.7	16.4
Pretax Margin	5.0	11.6	17.2	16.4	14.7
Net Margin	4.4	9.3	13.2	12.2	11.2

Hyundai Rotem Balance Sheet (Unit: KRW billions)

Dec Year-End	2023	2024	2025	2026E	2027E
Current Assets	3,690	3,686	7,163	7,442	8,301
Cash & Equiv.	1,008	770	1,277	1,949	2,825
Accounts Receivable	405	965	3,438	1,467	1,229
Inventory	269	316	503	611	693
Non-current Assets	1,551	1,599	2,155	2,328	2,449
Investment Assets	225	203	224	311	318
Tangible Assets	1,234	1,292	1,820	1,886	1,977
Intangible Assets	92	104	111	131	154
Total Assets	**5,241**	**5,285**	**9,318**	**9,770**	**10,750**
Current Liabilities	3,205	3,020	5,776	5,467	5,783
Accounts Payable	1,012	670	988	1,049	1,189
Current Debt	339	329	56	81	81
Non-current Liab.	390	256	500	526	544
Long-term Debt	256	22	76	75	75
Total Liabilities	**3,595**	**3,276**	**6,277**	**5,994**	**6,327**
Capital Stock	546	546	546	546	546
Capital Surplus	520	520	520	520	520
Retained Earnings	317	689	1,434	2,168	2,815
Capital Adjustments	297	291	579	579	579
Treasury Stock	0	0	0	0	0
Total Equity	**1,647**	**2,009**	**3,041**	**3,776**	**4,423**

Buy Rating, Coverage Initiated With a Target Price of KRW 110,000 (4)

Hyundai Rotem Statement of Cash Flows (Unit: KRW billions)

Dec Year-End	2023	2024	2025	2026E	2027E
Operating CF	734	142	904	986	1,140
Net Income	157	405	770	803	713
Depreciation	40	47	60	69	77
Chg in Working Cap	437	-541	-120	50	334
Accounts Receivable Decrease (Increase)	-93	-552	-2,453	1,975	238
Inventory Decrease (Increase)	131	-153	-196	-132	-82
Accounts Payable Increase (Decrease)	212	77	316	107	140
Investing CF	-270	233	-232	11	-202
Tangible Asset Disposal (Acquisition)	-54	-81	-129	-110	-141
Intangible Asset Decrease (Increase)	-26	-30	-27	-43	-51
Investment Asset Decrease (Increase)	-201	325	-94	285	-4
Financing CF	-576	-300	-237	-48	-55
Borrowings Increase (Decrease)	-580	-291	-221	18	10
Capital Increase (Decrease)	0	-11	-22	-65	-65
Dividends Paid	0	-11	-22	-65	-65
Gross Cash Flow	313	687	1,140	1,044	807
(-) Working Capital Increase (Decrease)	-722	409	-59	-58	-334
(-) Capex	55	81	129	110	141
(+) Asset Sales	-26	-30	-27	-43	-51
Free Cash Flow	954	168	1,042	950	948
(-) Other Investment	274	114	161	129	7
Excess Cash	681	54	881	821	941
NOPLAT	182	363	772	767	735
(+) Dep	40	47	60	69	77
(-) Working Capital Investment	-722	409	-59	-58	-334
(-) Capex	55	81	129	110	141
OpFCF	890	-80	761	784	1,004

Hyundai Rotem Key Indicators (Unit: KRW, x, %)

Dec Year-End	2023	2024	2025	2026E	2027E
Per Share					
EPS	1,475	3,728	7,055	7,326	6,527
BPS	15,391	18,737	28,211	34,937	40,864
DPS	100	200	600	600	550
CFPS	2,864	6,298	10,442	9,570	7,390
ROA (%)	3.2	7.7	10.5	8.4	6.9
ROE (%)	10.1	21.8	30.1	23.2	17.2
ROIC (%)	12.6	26.6	45.5	41.6	42.6
Multiples (x, %)					
PER	18.0	22.2	11.7	11.3	12.7
PBR	1.7	4.4	2.9	2.4	2.0
PSR	0.8	2.1	1.5	1.4	1.4
PCR	9.3	13.1	7.9	8.6	11.2
EV/EBITDA	10.0	17.1	7.4	6.6	6.1
Dividend Yield	0.4	0.2	0.7	0.7	0.7
Stability (%)					
Debt Ratio	218.2	163.1	206.4	158.7	143.0
Net Debt/Equity	-25.1	-20.9	-37.6	-47.5	-60.3
Net Debt/EBITDA	-165.4	-83.3	-107.4	-163.6	-255.6
Current Ratio	115.1	122.1	124.0	136.1	143.5
Interest Coverage (x)	7.3	27.5	127.7	199.5	81.7
Asset Structure (%)					
Invested Capital	49.0	61.3	55.2	44.9	33.9
Cash + Investment Assets	51.0	38.7	44.8	55.1	66.1
Capital Structure (%)					
Borrowings	26.5	14.9	4.2	4.0	3.4
Equity	73.5	85.1	95.8	96.0	96.6

Note: IFRS consolidated basis

Target Price and Investment Opinion History (1)

Hanwha Aerospace Target Price-Price Difference

Date	Investment Opinion	Target Price (KRW)	vs. Avg Price (%)	vs. High/Low (%)
2024.11.04	Buy	462,812	-22.58	5.96
2025.02.12	Buy	640,059	2.05	17.54
2025.04.09	Buy	925,623	-11.80	-2.23
2025.06.10	Buy	1,100,000	-18.68	-9.27
2025.08.01	Buy	1,350,000	-25.89	-1.48
2026.02.10	Buy	1,500,000	-8.62	2.47
2026.05.04	Buy	1,800,000	-40.70	-18.61
2026.10.26	Buy	1,300,000	-	-

Hanwha Systems Target Price-Price Difference

Date	Investment Opinion	Target Price (KRW)	vs. Avg Price (%)	vs. High/Low (%)
2025.02.10	Buy	31,000	10.72	36.29
2025.04.16	Buy	46,000	-9.27	2.93
2025.05.20	Buy	57,000	-5.05	20.35
2025.07.30	Buy	74,000	0.01	119.86
2026.04.01	Buy	160,000	-44.33	-14.31
2026.10.26	Buy	100,000	-	-

LIG Defense & Aerospace Target Price-Price Difference

Date	Investment Opinion	Target Price (KRW)	vs. Avg Price (%)	vs. High/Low (%)
2024.11.11	Buy	304,000	-26.00	-4.61
2025.02.17	Buy	350,000	-17.84	6.43
2025.05.09	Buy	460,000	8.97	37.17
2025.07.23	Buy	740,000	-24.05	37.84
2026.05.11	Buy	1,050,000	-29.67	-4.57
2026.10.26	Buy	670,000	-	-

Korea Aerospace Industries Target Price-Price Difference

Date	Investment Opinion	Target Price (KRW)	vs. Avg Price (%)	vs. High/Low (%)
2024.10.30	Buy	69,000	-9.56	37.68
2025.04.03	Buy	96,000	-2.42	19.79
2025.11.28	Buy	130,000	3.64	31.69
2026.02.06	Buy	201,000	-23.31	0.50
2026.10.26	Buy	140,000	-	-

Target Price and Investment Opinion History (2)

Hyundai Rotem Target Price-Price Difference

Date	Investment Opinion	Target Price (KRW)	vs. Avg Price (%)	vs. High/Low (%)
2024.10.29	Buy	75,000	-25.16	-4.67
2025.02.07	Buy	95,000	-4.23	18.42
2025.03.24	Buy	135,000	-18.98	1.19
2025.05.27	Buy	170,000	7.28	25.88
2025.07.14	Buy	240,000	-14.62	1.88
2025.11.04	Buy	290,000	-30.33	-14.14
2026.04.27	Buy	320,000	-49.60	-15.94
2026.10.26	Buy	110,000	-	-

Stock Investment Ratings

- 'Buy' is assigned if expected absolute return is over +15% for the next year.
- 'Hold' for -15% to +15%; 'Sell' if less than -15%.
- Evaluation and change cycles depend on industries/stocks and the analyst's estimates.

Industry Investment Ratings

- 'Positive' if sector expected to outperform historic returns over next year.
- 'Neutral' if sector expected to match past returns.
- 'Negative' if underperformance is expected.
- Based on analyst's comprehensive judgment.

Distribution of Investment Ratings in Our Research Reports (As of September 30, 2026)

Investment Rating	Buy	Hold	Sell	Total
Share of Financial Investment Products	96.4%	3.6%	0.0%	100.0%

Compliance Notice

Publication Date: October 26, 2026

This English version was translated from the original Korean-language report using Claude (Anthropic). Please refer to the Korean original for the authoritative version.

This report was prepared by the research analyst based on objective facts, accurately reflecting their own opinions without improper external pressure or interference. The analyst has not provided this report to institutional investors or third parties in advance. (Chae Un-Sam)

Our company does not hold 1% or more of the issued shares of the securities covered in this report as of the publication date.

In connection with the preparation of this research report, we disclose that our financial investment analyst Chae Un-Sam attended a corporate briefing at the expense of Hanwha Aerospace and Hanwha Systems during October 1-2, 2026. The above "Hanwha Aerospace and Hanwha Systems" are corporations in an affiliate relationship with our company as defined in the "Monopoly Regulation and Fair Trade Act."

This report is distributed only to our company's customers to assist investors' securities investment, and the copyright belongs to our company. Illegal reproduction and distribution are prohibited. The content in this report was obtained from sources that our Research Center believes to be reliable, but our company cannot guarantee its accuracy or completeness. Therefore, this report cannot be used as evidence of legal liability related to customers' securities investment results under any circumstances.